

One network, chained to itself across the LHCb magnet: a label-free extrapolator trained by what an error costs

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Abstract

Track reconstruction in LHCb calls an extrapolator, the routine that transports a track state from one detector plane to another, millions of times a second; crossing the spectrometer magnet is its most expensive step. The first paper of this programme [1] built a label-free surrogate for the field-propagation part of the job from the discrete-time construction of Raissi, Perdikaris and Karniadakis [2], and left one failure: inside a network serving every kind of step, the single 5178 mm crossing sat a factor 22 above the scheme it was trained to solve. Here I cut the crossing into N equal steps, train *one* network to take a step of length $\Delta z = L/N$ wherever it starts, and apply it to its own output N times; the network's outputs are written as a correction to the straight line, not as the stage states directly. The construction stays label-free: the training signal is the reconstruction residual of a q -stage Gauss–Legendre collocation step evaluated in the measured field map, and no reference trajectory enters the objective. On 1452 held-out tracks from an official LHCb simulation sample, chaining removes the discretisation: the same scheme solved exactly and chained 64 times sits $0.147\,\mu\text{m}$ from a sixth-order reference, while the trained chains of 64 steps or more sit between 137 and $246\,\mu\text{m}$, limited by their optimisation and by the accumulation. The experiment is a change of objective. An audit of that loss finds it spending 89.8 per cent of itself on 3–8 GeV tracks, 1.23 per cent on the 10–50 GeV region of interest and 0.65 per cent on the quarter of the crossing where an error costs most. Replacing it with a loss that measures every residual by the displacement it would cause at the far side of the magnet, as a fraction of that track's own bend and with the region of interest weighted up — built from the network's inputs and the field map alone — moves those three shares to 38.3, 47.0 and 23.3 per cent. At $N = 64$, $q = 2$, $\Delta z = 80.9\text{ mm}$ the median endpoint error falls from 145.7 to $88.8\,\mu\text{m}$, the error inside the weighted window from 108.1 to $24.8\,\mu\text{m}$, a factor 4.37, and the median per-step slope error from 1.30×10^{-6} to 7.34×10^{-7} . Both pre-registered conditions are met, though the second is weak: the pooled-loss counterpart has since cleared it too. The price is specific: below the crossover between 3–8 and 8–20 GeV the reweighted network is worse in five of six comparisons (these counts run over three matched pairs, and two of the three pooled counterparts had not reached the plateau rule), the median below 3 GeV rises from 353.7 to $615.1\,\mu\text{m}$, the 95th percentile from 898 to $1399\,\mu\text{m}$, and the fraction beyond a millimetre from 4.41 to 7.16 per cent. The whole of the gain is in the bending plane, the vertical plane does not move, and what remains is *more* coherent along the chain in all six matched comparisons, 0.31 to 0.53 against the 0.0625 to 0.125 of independent steps. Shortening the step from 80.9 to 20.2 mm halves each step's error and does not improve the crossing. Against the simulated particle, network and reference are indistinguishable: the three cost-weighted networks sit 12.5 to 25.2 times closer to one another than either is to the truth in every band from < 3 to 20–50 GeV and in the loss window, because neither models the material. In the thirty-track > 50 GeV band the pooled networks instead approach the reference-to-truth gap, the ratio falling to 0.85.

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1 Introduction

This paper reports one experiment and the machinery needed to read it. A single small neural network is trained to take one fixed step in z across the LHCb magnet, and is then applied to its own output as many times as it takes to cross. It is trained without labels: it is never shown where a particle went, only required to satisfy the algebra of an implicit Runge–Kutta step evaluated in the measured magnetic field. The experiment is the loss. The same network, at the same three step lengths, is trained twice, once under the loss the construction comes with and once under a loss that measures every residual by what that residual would cost at the far side of the magnet, and the two are compared on the same held-out tracks by the same statistics.

I have written this section, and the theory section that follows it, for a reader with no prior exposure to LHCb, to track fitting, or to physics-informed neural networks. Every term is defined at first use and every symbol is introduced in a “where ...” clause beside the equation it appears in. Section 1.1 says what an extrapolator is and why its cost matters. Section 1.2 states what the first study of this programme [1] established and the one failure it left behind. Section 1.3 states the design change this paper makes in response. Section 1.4 sets the two questions the paper answers, and Section 1.5 is the road map.

1.1 The extrapolator and why it is the cost in the fit

LHCb is one of the four large experiments at the Large Hadron Collider at CERN, rebuilt for the current data-taking period as a fully software-triggered forward spectrometer [3]. Protons are brought into collision at its centre tens of millions of times a second, each collision sprays out charged particles, and the detector’s job is to work out which particles were produced and where they came from. It does that by placing layers of position-sensitive silicon and scintillating fibre along the beam direction, so that a particle passing through leaves a series of measured points, called **hits**. A **track** is the reconstructed path of one charged particle through the detector, fitted to those hits.

Three groups of layers matter here. The vertex detector surrounds the collision point. The **Upstream Tracker** is a set of four silicon planes immediately in front of the magnet. The **scintillating-fibre tracker**, usually written SciFi, is twelve layers immediately behind it. Between the second and the third sits a large dipole magnet whose field is roughly vertical and about one tesla at its strongest. A charged particle passing through a magnetic field is deflected, and the deflection is inversely proportional to its momentum: fast particles are barely bent, slow particles are bent a great deal. The magnet is therefore the instrument that measures momentum. It is also the reason the geometry of a track is not a straight line, and the reason this paper exists.

Throughout, z runs along the beam direction, x is horizontal and transverse to the beam, and y is vertical. Because LHCb detects particles that go forward at small angles to the beam, every particle of interest moves monotonically in z , and it is convenient to describe a track not as a function of time but as a function of z .

A **track state** is the complete description of a track at one value of z . It is a list of five numbers,

$$S = (x, y, t_x, t_y, qop), \tag{1}$$

where x and y are the transverse positions of the track at that z , in millimetres; $t_x = dx/dz$ and $t_y = dy/dz$ are the local slopes, which are pure numbers with no units; and the fifth component encodes the particle’s charge q , in units of the proton charge, divided by its momentum p . I use the convention of Allen, the GPU high-level trigger of LHCb [4], in which that component is stored as

$$\text{qop} = 0.299792458 \frac{q}{p[\text{GeV}]}.$$
 (2)

Five numbers suffice, because once one knows where a particle is, which way it is pointing, and its charge-to-momentum ratio, the magnetic field determines the rest of its path. Because the slopes are dimensionless, every slope error in this paper is quoted as a dimensionless number in scientific notation. No slope is multiplied by 10^3 or converted to an angular unit anywhere in the text, the tables, the figures or the scripts that make them.

An **extrapolator** is the piece of software that takes a track state at one plane z_0 and returns the track state at another plane z_1 . That is the whole of its job, and it is used constantly. A track fit walks along a candidate track from measurement to measurement, and every walk is an extrapolation. The reconstruction asks whether a track found in front of the magnet and a track found behind it are the same particle, which means transporting one of them to meet the other. A vertex fit asks where several tracks came closest to one another, which requires each of them to be carried back towards the collision point, iteratively. In the high-level trigger all of this happens millions of times per second, on every event, inside the time budget that decides whether the event is written to disk at all.

The standard extrapolator does the job by numerical integration: it steps along z in small increments, reading the magnetic field at every step. It is accurate and it is not cheap. Across the magnet it is the most expensive routine in the fit, and its cost is not even predictable, because the number of steps depends on the track. A method that could take a single large step, at comparable accuracy, using a fixed and known amount of arithmetic, would be worth having twice over: it would be faster, and a fixed arithmetic cost is far easier to schedule on the parallel hardware the trigger runs on than a loop whose trip count varies from track to track.

That is the surrogate this programme is trying to build. One restriction holds throughout and is stated once here: the surrogate replaces **field propagation only**. Real particles also scatter off the detector material and lose energy to it. Those effects stay with the existing extrapolator, they are not in the equations the network is trained on, and none of the errors reported in this paper include them. Section 2.7 says exactly what that means for the interpretation of every number, because the difference between a field-only prediction and a real particle’s path is large and is measured here rather than assumed away.

1.2 What the first study established, and the one failure it left

The first paper of this programme [1] asked whether the label-free construction works at all on LHCb geometry, and answered yes with one exception that this paper exists to attack.

On a single frozen leg — one fixed step from the last Upstream Tracker plane to the first SciFi plane, a distance of 5178 mm — a small network trained purely on the reconstruction residuals of an eight-stage Gauss–Legendre scheme, with no labels anywhere in the loss, reached a median endpoint error of $115\,\mu\text{m}$. That is three and a half orders of magnitude better than ignoring the magnet and continuing along the incoming slopes, and it is delivered in one forward pass against the median of 44 root-finder residual evaluations the same scheme needs when the same equations are solved directly. Where the numerical scheme was the limiting factor, the network reproduced the scheme’s own error to within a percent. Where the network was the limiting factor, the limit was shown to be the optimiser rather than the network’s capacity, by a rank correlation of $+0.965$ between final training loss and held-out error across 96 converged runs. The method also worked in the strongest sense available for a label-free claim: it reached the same accuracy on a field polarity for which no labelled sample exists.

The failure is equally specific. The first paper also trained a single *general* network meant to serve every kind of step — short hops between sensors, backward fetches towards the collision point, and the magnet crossing — by making the start plane and the step length inputs alongside the state. Averaged over the whole test population that network reached $2.1\,\mu\text{m}$ against $16.1\,\mu\text{m}$ for a straight line, and it beat the straight line on every leg type. But the one step the extrapolator exists for, the cross-magnet crossing inside that shared network, sat a factor of 22 above the numerical scheme it was being trained to solve; its supervised twin, trained on reference labels instead of on the equations, sat a factor of 11 above. Chaining two or three legs together, that single leg dominated everything else. The first paper’s own conclusion is blunt about it: the cross-magnet leg inside the shared network is the item everything else waits on.

Two readings of that failure are possible. Either the crossing is intrinsically hard for a network of this kind, or a single step of 5178 mm is simply too much to ask of one forward pass and the crossing should be built out of smaller pieces. This paper takes the second reading seriously and measures what happens.

1.3 One network of step Δz applied N times

The crossing studied here is fixed and is the same one throughout the paper. It runs from $z_0 = 2648.2\text{ mm}$, the last Upstream Tracker plane, to $z_1 = 7826.0\text{ mm}$, the first SciFi plane, a distance $L = 5177.8\text{ mm}$. The crossing is cut into N equal steps of length $\Delta z = L/N$, and **one** network is trained to take a step of that length, wherever in the crossing it starts. A crossing is then performed by applying that same network N times, feeding each step’s output state in as the next step’s input, a procedure I call **chaining**. At $N = 64$ the step is 80.90 mm; at $N = 128$ it is 40.45 mm; at $N = 256$ it is 20.23 mm.¹

Three design points make that possible, and each is a change from what came before.

1. **The network’s input is the state, not the position.** In the original discrete-time construction [2, §3] the network’s input is a spatial coordinate and its output is the solution there, so the network belongs to one initial condition and one step and cannot be reused. Here the input is the track state of Eq. (1), so the network is a *map* from a state to the state one step later, and the same map can be applied to whatever state comes out of the previous application.
2. **The start plane is an input.** The field a step crosses depends on where along the magnet the step begins: on the beam axis the field magnitude runs from about 0.22 to about 1.05 tesla across the crossing. One network can serve all N steps only if it is told where it is, so the start plane z_{start} enters as a sixth input, mapped linearly onto $[-1, 1]$ across the possible start planes.
3. **The momentum is carried, not predicted.** A magnetic field does no work, so $q\mathbf{p}$ is conserved exactly. It is an input to every step and is passed through each application unchanged; only the four remaining components are ever predicted.

The precedent for a single step-network applied repeatedly is Wang and Perdikaris [5], who integrate parametric evolution equations over long horizons by training one operator for a fixed time step and then marching with it, rather than by training a network to represent the solution over the whole horizon at once. The construction used here is the discrete-time one of Raissi, Perdikaris and Karniadakis, and the marching is theirs.

This is also a deliberate departure from the study immediately preceding this one, in which a *separate* network was trained for each step — 128 networks for a 128-step crossing, each trained on the previous one’s outputs. That design follows the source construction literally, because

¹All four geometry numbers, and the step lengths, are read from the track set’s own metadata rather than recomputed; see the data availability statement.

there the network’s input is the coordinate; it stops being necessary the moment the input is the state. One network of step Δz applied N times is both a smaller object to train and a far more plausible thing to deploy, since the trigger would hold one set of weights rather than N of them.

One further departure from the source construction was chosen for the earlier fixed-step study and inherited here without re-examination: the network does not emit the stage states directly but a correction to the straight line, a form carried over from the first study and set out in Section 2.4.

1.4 The two questions of this paper

Two questions were set in advance, and what each answer would have to contain was fixed before the runs were submitted.

1. **Under the loss the construction comes with, is the endpoint error the step or the accumulation, and where does the loss look?** The first half is a decomposition: measure the error of one application of the network in isolation, measure the error of the full chain, and see whether the second is the first multiplied up by the chain or something structurally different. The second half is an audit of the objective itself: take a trained network, draw states evenly over the crossing, and ask what fraction of the loss comes from which tracks and from which part of the crossing, against what those same states’ errors actually cost at the far end.
2. **What does a cost-weighted, still label-free loss buy, and what does it cost?** If the audit shows the loss spending itself somewhere other than where the error is, the natural response is to reweight it. The weight must be built from the network’s inputs and the field map alone, because a weight that reads a reference trajectory would destroy the one property that makes the construction worth having. The question then has two halves that must both be reported: what improves, and what gets worse.

Both halves of the second question have answers, and they point in opposite directions. For the 64-step network the endpoint error falls from $145.7\,\mu\text{m}$ to $88.8\,\mu\text{m}$ overall, and within the 10–50 GeV window that the weight targets it falls from $108.1\,\mu\text{m}$ to $24.8\,\mu\text{m}$, a factor of four. Over the same tracks the 95th percentile rises from $898\,\mu\text{m}$ to $1399\,\mu\text{m}$, and the median for tracks below 3 GeV rises from $354\,\mu\text{m}$ to $615\,\mu\text{m}$. A loss that is told which tracks matter does better on those tracks and worse on the others, and that trade is the subject of Sections 5 and 7.

1.5 Road map

Section 2 builds the method from the equation of motion upwards: the differential equation a track satisfies in z ; what makes a Runge–Kutta method implicit and why the Gauss–Legendre family is the right one; the construction that turns that scheme into a training objective with no labels; what changes when one network is chained to itself; the loss the construction comes with and its two measured consequences; the cost-weighted loss derived term by term; the three things every number in this paper is measured against; and the metrics and the stopping rule.

Section 3 describes the track sample, the crossing, the cuts, the momentum bands, the units and the compute. Section 4 measures the numerical scheme on its own, with no network anywhere, which is what fixes the floor underneath everything else. Section 5 is the result: the grid of step counts and stage counts under the original loss, where each loss puts its weight, the convergence of each run to its own plateau, the headline comparison of three networks under two losses, the per-component and per-band breakdown, the region near 5 GeV, what the remaining

error is made of, what shorter steps do, the comparison against the simulated truth, and the cost per track.

Section 6 is reserved. It is held for the study in which the loss window is moved down onto the 3–8 GeV band; those runs are in training as this is written, and the section states what it will contain rather than reporting numbers that do not yet exist. Section 7 reads the results against the two questions and lists the caveats, Section 8 says what is worth doing next, and Section 9 is the summary. Appendix A carries the provenance: repository, commits, data sets, compute, and the maps from every figure and every table to the file it was computed from.

2 Theory

This section contains everything needed to read the results, and nothing that depends on them. I start from the differential equation a charged particle satisfies in the LHCb field and say where each term comes from (Section 2.1). I then define what a Runge–Kutta method is, what makes one implicit, and why the Gauss–Legendre family is the natural choice for taking very few, very large steps (Section 2.2). Section 2.3 turns that scheme into a training objective that never touches a label. Section 2.4 says what changes when one network is applied to its own output N times. Section 2.5 states the loss the construction comes with and the two consequences of its normalisation, both of which are measured rather than asserted. Section 2.6 derives the replacement loss term by term, with a worked example. Section 2.7 defines the three fixed points every number in this paper is quoted against, and Section 2.8 defines every statistic and the rule that decided when a run had finished.

2.1 The equation of motion in z

A charged particle in a magnetic field $\mathbf{B}$ feels the Lorentz force $q \mathbf{v} \times \mathbf{B}$. Rewriting Newton’s second law with z rather than time as the independent variable, and expressing the result in the state variables of Eq. (1), gives the system

$$\frac{dx}{dz} = t_x, \quad \frac{dy}{dz} = t_y, \quad (3)$$

$$\frac{dt_x}{dz} = \kappa \text{qop} \mathcal{N} [t_x t_y B_x - (1 + t_x^2) B_y + t_y B_z], \quad (4)$$

$$\frac{dt_y}{dz} = \kappa \text{qop} \mathcal{N} [(1 + t_y^2) B_x - t_x t_y B_y - t_x B_z], \quad (5)$$

$$\frac{d \text{qop}}{dz} = 0, \quad (6)$$

where $\kappa = 10^{-3}$ is the unit conversion constant that accompanies the Allen qop convention of Eq. (2); $\mathcal{N} = \sqrt{1 + t_x^2 + t_y^2}$ is the ratio of path length to z length, which appears because the force acts along the particle’s path while the derivative is taken along z ; and B_x , B_y , B_z are the three components of the field in tesla, evaluated at the point (x, y, z) the particle currently occupies. This is the system the reference integrator implements, and it is identical to the one Allen uses.

Three properties of Eqs. (3)–(6) shape everything that follows.

- **The system is non-autonomous.** The right-hand side depends on z explicitly, through the field, and not only on the state. This is unlike the textbook problems on which the technique used here was first demonstrated.
- **qop is conserved**, by Eq. (6). A magnetic field does no work, so restricting the surrogate to field propagation makes the momentum an input that is carried through unchanged, and only four components are ever predicted.

- **The field is not an analytic function.** It is a measured map, stored on a regular grid with 100 mm cells, and the field at an arbitrary point is obtained by trilinear interpolation between the eight surrounding grid values. Such an interpolant is continuous but its gradient jumps at every cell boundary; in the standard classification it is C^0 and not C^1 . Section 4 shows what that costs a high-order method.

Analogy. Equations (3)–(5) say the same thing as a description of a car on a road. The position changes at a rate given by the current heading, and the heading changes at a rate given by how hard the steering wheel is turned. The magnetic field is the steering wheel, and qop is how strongly this particular car responds to it. A heavy, fast car, meaning high momentum and therefore small qop, barely turns; a light, slow one turns sharply.

Worked micro-example. Take a particle of $p = 10$ GeV and charge $+1$, so that $\text{qop} = 0.0300$ by Eq. (2), travelling nearly along the beam, $t_x = t_y = 0$ and hence $\mathcal{N} = 1$, through a field of $B_y = -1.0$ T with $B_x = B_z = 0$. Equation (4) gives $dt_x/dz = \kappa \text{qop} (-1) B_y = 10^{-3} \times 0.0300 \times 1.0 = 3.0 \times 10^{-5}$ per millimetre. Treating the field as constant across the 5177.8 mm crossing — which it is not, but the estimate is only meant to fix the order of magnitude — the slope would grow to about 0.155, and the particle would move sideways by roughly $0.155 \times 5177.8/2 \approx 400$ mm. So the quantity this surrogate has to reproduce is a sideways correction of a few hundred millimetres, and getting it right to $100 \mu\text{m}$ means getting it right to something like one part in five thousand.

2.2 Gauss–Legendre collocation as an implicit Runge–Kutta method

Given a state $S(z_0)$ and the derivative function $f(S, z)$ defined by Eqs. (3)–(6), a numerical scheme advances the solution by a step Δz . The general Runge–Kutta method with q stages does so by evaluating f at q intermediate points inside the step and combining the results:

$$S_j = S_0 + \Delta z \sum_{k=1}^q A_{jk} f(S_k, z_0 + c_k \Delta z), \quad j = 1, \dots, q, \quad (7)$$

$$S_{\text{end}} = S_0 + \Delta z \sum_{k=1}^q b_k f(S_k, z_0 + c_k \Delta z), \quad (8)$$

where S_0 is the state at the start of the step; the S_j are the q **stage states**, the method’s internal estimates of the solution at intermediate planes; the $c_k \in [0, 1]$ are the **nodes**, the fractional positions of those planes within the step; A_{jk} is the $q \times q$ **stage matrix**; and the b_k are the q **weights** that combine the stage rates into the step’s final answer. Together (c, A, b) are called the tableau, and they are the whole of the method.

If A is strictly lower triangular the scheme is **explicit**: stage 1 depends on nothing, stage 2 on stage 1, and so on, so the stages can be evaluated one after another. If A is full the scheme is **implicit**: every stage depends on every other stage, including itself, and Eq. (7) is a coupled nonlinear system that has to be solved by iteration.

The **Gauss–Legendre** family is the implicit family obtained by taking the nodes c_k to be the roots of the degree- q Legendre polynomial mapped onto $[0, 1]$, and then choosing A and b so that the scheme integrates polynomials exactly to the highest attainable degree. The resulting method has formal order $2q$, the maximum any q -stage Runge–Kutta method can have, and its classical theory is set out by Hairer and Wanner [6]. It is also a **collocation** method, which is the picture I use throughout: the q -stage scheme is exactly equivalent to fitting a polynomial of degree q through the step and requiring that the polynomial satisfy the differential equation exactly at the q node positions. The stage states are the values of that polynomial at those nodes. The same family is symplectic, a structure-preserving property discussed at length by Hairer, Lubich and Wanner [7], and the identity that expresses it is used below as a check on the construction.

Analogy. Explicit stepping is like walking across a room in small steps with your eyes on your feet: each step follows from the last, and you always know what to do next. Collocation is like drawing a single smooth curve from one side of the room to the other and then checking, at a handful of chosen points along it, that the curve is heading in the direction the physics says it should be heading at that point. If it passes every check, and the curve is smooth, it is very likely right everywhere in between.

Worked micro-example: the $q = 2$ tableau. The nodes are the roots of the shifted Legendre polynomial of degree two, $c = \left(\frac{1}{2} - \frac{\sqrt{3}}{6}, \frac{1}{2} + \frac{\sqrt{3}}{6}\right) \approx (0.21132, 0.78868)$, the weights are $b = \left(\frac{1}{2}, \frac{1}{2}\right)$, and

$$A = \begin{pmatrix} 1/4 & 1/4 - \sqrt{3}/6 \\ 1/4 + \sqrt{3}/6 & 1/4 \end{pmatrix} \approx \begin{pmatrix} 0.2500 & -0.0387 \\ 0.5387 & 0.2500 \end{pmatrix}. \quad (9)$$

The entry $A_{12} = -0.0387$ is nonzero and sits above the diagonal, so the first stage depends on the second: there is no order in which these two can be evaluated one at a time. That is exactly what implicitness means.

The tableau is not hard-coded in this work; it is built for any q from the Legendre roots, with A obtained from barycentric collocation integrals, and the construction is then checked against every identity it must satisfy. The battery is run for $q = 1, 2, 3, 4, 8, 12$ and 16 and checks that each row of A sums to its own node; that the quadrature (c, b) integrates polynomials exactly to degree $2q - 1$, which is what gives the scheme its order $2q$; that the collocation conditions hold to degree $q - 1$; that the symplecticity identity $b_j A_{jk} + b_k A_{kj} = b_j b_k$ holds, an exact property of Gauss methods that the construction never imposes and therefore has to earn; and, for $q = 1, 2, 3$, agreement with the closed-form tableaux in the literature. The worst residual over all of those checks and all seven stage counts is 2.2×10^{-16} , against a tolerance of 5×10^{-13} : machine zero. Section 4.1 reports the battery in full. Two node sets are worth quoting here for scale, because they show how the stage planes crowd towards the ends of the step as q grows: at $q = 8$ the first and last nodes are 0.01986 and 0.98014 , and at $q = 16$ they are 0.00530 and 0.99470 .

2.3 The discrete-time construction: training with no labels

Ordinarily Eqs. (7) and (8) are a problem to be solved: given S_0 , find the q stage states and the endpoint that satisfy the $q + 1$ equations. Solving them requires a root-finder, which is iterative and whose cost is not known in advance — exactly the property the trigger does not want.

The discrete-time construction of Raissi, Perdikaris and Karniadakis [2, §3] inverts the arithmetic. A network is asked to **output** all q stage states and the endpoint at once, given the starting state:

$$\mathcal{M}_\theta : S_0 \mapsto (S_1, S_2, \dots, S_q, S_{\text{end}}), \quad (10)$$

where θ are the network's weights. Those outputs are then substituted into the scheme's own equations, read backwards. Each of the $q + 1$ equations can be rearranged to express the *starting* state in terms of a proposed stage or endpoint:

$$S_0^{(j)} = S_j - \Delta z \sum_k A_{jk} f(S_k, z_0 + c_k \Delta z), \quad (11)$$

$$S_0^{(\text{end})} = S_{\text{end}} - \Delta z \sum_k b_k f(S_k, z_0 + c_k \Delta z). \quad (12)$$

Each of the $q + 1$ outputs therefore produces its own **reconstruction** of the starting state. If the network's proposal genuinely solves the scheme, all $q + 1$ reconstructions equal the true S_0 , which is known because it is the input. If it does not, they do not. The discrepancies are the

reconstruction residuals, and they are the entire training signal. Writing $d \in \{x, y, t_x, t_y\}$ for the four predicted components and n for the training state,

$$r_{n,j,d} = S_{0,n,d}^{(j)} - S_{0,n,d}, \quad (13)$$

which is a mismatch in millimetres for the two position components and a dimensionless mismatch for the two slope components. A loss is any positive function of those residuals that vanishes when they do; the two used in this paper differ only in how each residual is weighted before squaring, and they are given in Sections 2.5 and 2.6.

Nothing in Eq. (13) is a label. No correct answer ever enters it. What enters it is the starting state, which is the input; the tableau, which is arithmetic; and the magnetic field, which is a measured map. This is what **label-free** means in this paper, and it is a precise statement rather than a slogan: no reference trajectory, from any integrator or any simulation, appears anywhere in the objective or in any quantity the objective is built from. The field enters through f , evaluated at the positions the network itself proposes, which is why the field must be available in a differentiable form — the gradient of the loss with respect to the weights runs through the field lookup.

The practical value of that property is worth stating once. Producing labels for a surrogate of this kind means running the reference integrator over a large sample, and it has to be redone whenever the field map, the detector geometry or the state conventions change. An objective that never touches a label is immune to all three, and can be trained for a configuration for which no labelled sample has ever been produced.

Analogy. Instead of solving a maze by walking it, you hand someone a proposed route and ask them to walk it backwards from every waypoint on it. If every backwards walk lands on the entrance, the route is right. Nobody ever had to be told the correct route.

Worked micro-example, and the cost argument. Take $q = 8$ and one training state. The network emits $9 \times 4 = 36$ numbers. The rates f are evaluated once at each of the eight stage states, requiring eight field lookups. The eight stage reconstructions of Eq. (11) and the one endpoint reconstruction of Eq. (12) are then formed by two small matrix products and compared against the four known components of the input. That is 36 residual numbers from one forward pass and eight field lookups, with no iteration and no branch. Solving the same eight-stage system directly with a root-finder took, on the cross-magnet leg of the first study [1], a median of 44 residual evaluations, each of which is itself eight field lookups plus the root-finder’s internal Jacobian work. The arithmetic advantage of a trained network is not that it is approximate; it is that its cost is fixed and known before the track arrives.

2.4 Chaining one network N times

Everything above describes one step. This section describes what it takes to make the same network serve all N of them, and what the chain does to the errors.

The network of this paper takes six inputs: the five components of the state, each divided by its spread over the training population, and the start plane z_{start} , mapped linearly onto $[-1, 1]$ across the possible start planes $z_0, \dots, z_1 - \Delta z$. It emits $4(q + 1)$ numbers, the four predicted components of each of the q stage states and of the endpoint. The stage planes of a step are then

$$z_j = z_{\text{start}} + c_j \Delta z, \quad j = 1, \dots, q, \quad (14)$$

which is where the field is read when the residuals of Eq. (11) are formed. Knowing z_{start} is what allows one set of weights to serve every step: a step near the entrance of the magnet does much less bending than a step through its middle, and without the start plane the network would have no way to tell the two apart.

The crossing itself is then a loop, and it is worth writing out because every endpoint number in this paper is produced by it.

1. Start from the track’s real state S on the last Upstream Tracker plane, and set $z_{\text{start}} = z_0$.
2. Evaluate the network on (S, z_{start}) and keep only its endpoint output, the four predicted components at $z_{\text{start}} + \Delta z$. The q stage states are used in training and discarded here.
3. Form the next input state by appending the particle’s qop, which is carried through unchanged by Eq. (6) and is never predicted.
4. Advance z_{start} by Δz and return to step 2, until N applications have been made and the state sits on z_1 .

One further design point belongs here, because it is what makes a single network behave sensibly over a whole crossing. The network does not emit the state directly. It emits the **correction to the straight line**, scaled by a per-track estimate of how large that correction ought to be:

$$\text{output}_j = \text{straight}_j + \text{scale}_j \odot \text{raw}_j, \quad \text{straight}_j = (x + t_x(z_j - z_{\text{start}}), y + t_y(z_j - z_{\text{start}}), t_x, t_y), \quad (15)$$

where $\odot$ is the componentwise product and raw_j is the network’s unscaled output, a pure number. The scale is built from the inputs and the field map alone, and therefore keeps the construction label-free. To first order in the field, Eqs. (4) and (5) give a slope change of $\kappa |\text{qop}| I_B$ over the step and a position change of about half that ramped over the step, so

$$\text{scale}^{\text{slope}} = \kappa |\text{qop}| I_B, \quad \text{scale}^{\text{pos}} = \frac{1}{2} \kappa |\text{qop}| I_B |\Delta z|, \quad I_B = \int |\mathbf{B}| dz, \quad (16)$$

where I_B is the field magnitude integrated along the straight line through the input state, evaluated with a 16-point midpoint rule, and small floors keep the scale away from zero where there is no field. The y components carry their own scale, built the same way from a term-by-term bound on the y rate of Eq. (5) rather than from $|\mathbf{B}|$, because the bending in y is far smaller than the bending in x and sharing one scale between them makes the y output far too coarse. Two properties recommend this parameterisation: what the network learns is the bending, which is physics, rather than where the plane happens to be, which is geometry; and it returns the straight line exactly when there is no field, so the trivial baseline is built into the design.

What the network learns: the correction to the straight line. The outputs are the q stage states and the endpoint, but none of them is emitted as it is. Each is written as the straight-line extrapolation of the input state to its own plane plus a network term scaled per track, as Eq. (15) states and Eq. (17) of Section 2.5 writes out in full. What the network learns is therefore the **correction to the straight line** — the magnet’s bending over one step — and not the stage states themselves. This is a departure from the source construction: in the discrete-time construction of Raissi, Perdikaris and Karniadakis [2, §3] the network emits the stage states and the endpoint directly, which is how Eq. (10) is written. The loss is unaffected. It sees only the stage states that result, through the reconstruction residual of Eq. (13), and it is formed from them in the same way whichever form produced them. I introduced the form on 5 September 2026 for a network that had to serve step lengths from 70 mm to 8.6 m [1, §3.10], and chose it explicitly for the fixed-step study of 14 September 2026 that preceded this one; the single-network chain of this paper then inherited it without re-examination, and the theory as first written did not state that the learned quantity is the correction rather than the stage states. Over that range of step lengths the bending over one step spans five orders of magnitude and a single population-wide output scale cannot cover it; that motivation — a step length that varies from one training state to the next — does not apply when every step has the same Δz .

The two properties above hold here as they did there, but whether the form is load-bearing in this design has not been measured, and measuring it is the first item of future work in Section 8.

The chain is the reason this paper needs a section on weighting at all. Each application’s output is the next application’s input, so an error made at step k is not simply added to the total: the remaining $N - k$ applications propagate it. In particular a *slope* error made early is multiplied by the distance the track still has to travel before it reaches z_1 . That distance is what I call the **lever arm**, and it is the single most important quantity in Section 2.6.

Analogy. Aiming a rifle. Being a millimetre off at the muzzle costs a millimetre at the target. Being a thousandth of a radian off in *angle* at the muzzle costs a metre at a target a kilometre away. The two mistakes are the same size on the rifle and completely different sizes where it matters, and an objective that scores them equally at the muzzle is scoring the wrong thing.

2.5 The pooled loss and its two consequences

The two losses in full. Both losses of this paper are built on the same reconstruction residual, and both are stated below in the form in which they were coded: one self-contained expression each, with the network’s output and the reconstruction substituted, so that nothing about either objective is left implicit. Nothing in this paragraph is new — every piece of it has appeared in Sections 2.3 and 2.4 — and Figure 1 draws the construction the two share.

First, what the network emits. Writing $S_n = (x, y, t_x, t_y, q/p)_n$ for the input state on the start plane z_n , the j -th output of Eq. (10) is

$$\hat{S}_{n,j} = (x + t_x c_j \Delta z, y + t_y c_j \Delta z, t_x, t_y)_n + \sigma(S_n) \odot \text{NN}_\theta(S_n/s_{\text{in}}, \zeta(z_n))_j, \quad j = 1, \dots, q, \text{end}, \quad (17)$$

where $c_{\text{end}} = 1$, so that the last output sits on $z_n + \Delta z$; $\odot$ is the componentwise product; NN_θ is the network itself, six inputs to $4(q + 1)$ outputs, and $(\cdot)_j$ selects the four numbers belonging to output j ; $\sigma(S_n)$ are the per-track output scales of Eq. (16), built from $\kappa |q/p|_n \int |\mathbf{B}| dz$ taken along the straight line through S_n , with a separate integral I_y serving the two y components; s_{in} is the spread of each input component over the first round’s training states; and ζ maps the start plane linearly onto $[-1, 1]$ across the possible start planes $z_0, \dots, z_1 - \Delta z$. The first bracket is the straight line of Eq. (15), and the second is the quantity the network actually learns: the correction to the straight line, not the stage state itself, as the paragraph *What the network learns* of Section 2.4 sets out. This is the parameterisation the network module documents, and it returns the straight line exactly when the network’s output is zero.

Second, what is done with those outputs. Equations (11) and (12) become one expression once the endpoint row of the tableau is written as $A_{\text{end},k} \equiv b_k$:

$$R_{n,j} = \hat{S}_{n,j} - \Delta z \sum_{k=1}^q A_{jk} f(\hat{S}_{n,k}, z_n + c_k \Delta z), \quad r_{n,j,d} = R_{n,j,d} - S_{n,d}, \quad (18)$$

with j now running over the q stages and the endpoint alike, f the rate of Eqs. (3)–(6) evaluated at the network’s own proposed states and carrying that track’s conserved $(q/p)_n$, and $r_{n,j,d}$ the reconstruction residual of Eq. (13). Panel (b) of Figure 1 is this equation drawn.

The pooled loss is then, in full,

$$\mathcal{L}_{\text{pooled}}(\theta) = \frac{1}{4(q+1)N_s} \sum_{n=1}^{N_s} \sum_{j=1}^{q+1} \sum_{d \in \{x, y, t_x, t_y\}} \left(\frac{\hat{S}_{n,j,d} - \Delta z \sum_k A_{jk} f_d(\hat{S}_{n,k}, z_n + c_k \Delta z) - S_{n,d}}{s_d} \right)^2 \quad (19)$$

where $N_s = 32,000$ is the number of collocation states in a training round, redrawn every round, and the four s_d are fixed constants of the run, measured on the first round’s states and unchanged

thereafter. This is the discrete-time loss of Raissi, Perdikaris and Karniadakis [2, §3] with two changes: a per-component normalisation, without which x would dominate the sum entirely, and a mean in place of a sum, which only rescales it. There is no separate regulariser and no second term. The normalisation is the loss.

The cost-weighted loss is the same bracket, multiplied before squaring by a weight that depends on the track, on the output plane and on the component:

$$\mathcal{L}_{\text{cw}}(\theta) = \frac{1}{4(q+1)N_s} \sum_{n=1}^{N_s} \sum_{j=1}^{q+1} \sum_{d \in \{x, y, t_x, t_y\}} \left(\frac{a_n g_{n,j,d}}{D_{\text{ref}}} \left[\hat{S}_{n,j,d} - \Delta z \sum_k A_{jk} f_d(\hat{S}_{n,k}, z_n + c_k \Delta z) - S_{n,d} \right] \right)^2 \quad (20)$$

where $g_{n,j,d} = (1, 1, \ell_{n,j}, \ell_{n,j})$ for $d = x, y, t_x, t_y$ respectively, with the lever arm $\ell_{n,j}$ of Eq. (24); D_{ref} is the fixed reference bend of the run; and

$$a_n = \text{clamp} \left(\sqrt{W(p_n)} \frac{D_{\text{ref}}}{D_n}, \left[\frac{1}{5}m, 5m \right] \right),$$

with the track bend $D_n = \kappa |q/p|_n \bar{I} L$ of Eq. (25), the momentum window W of Eq. (26), and m the median of the unclamped a_n over the round's own batch of states. The clamp is written here because Eq. (25) gives a_n before it is applied, and the loss that was trained is the clamped one. Figure 2 draws the two weight maps against each other.

Set side by side, the two losses are one formula with two readings of a single divisor:

$$\mathcal{L}(\theta) = \text{mean}_{n,j,d} \left(\frac{r_{n,j,d}}{s} \right)^2, \quad s = \begin{cases} s_d, & \text{pooled: one constant per component;} \\ \frac{D_{\text{ref}}}{a_n g_{n,j,d}}, & \text{cost-weighted: one value per track,} \\ & \text{per output plane and per component.} \end{cases} \quad (21)$$

The residual is the same in both and so is the minimiser: both vanish exactly when $r = 0$, and no positive weight can move that. What the weight changes is the shape of the surface around the minimum, which is what the rest of this paper measures. The ablations reported in Section 5.2 switch one factor of $a_n g_{n,j,d}$ off at a time by replacing the quantity that varies with a constant — the lever arm by its mean over the planes, D_n by D_{ref} , or W by one — so that the overall size of the loss is unchanged and only its distribution over states moves.

One implementation detail belongs beside these formulae rather than with the results. At each L-BFGS restart the trainer multiplies whichever loss it is using by the constant $1/\mathcal{L}_0$, where $\mathcal{L}_0$ is that loss's value at the start of the restart, so that the optimiser always begins from a number of order one. A constant multiplier rescales the gradient without changing its direction and moves no minimum; it is a conditioning device and nothing more.

The loss the construction comes with treats every residual alike, once each component has been put on a comparable footing. The state components have very different units and sizes — x is hundreds of millimetres while t_x is a few tenths and dimensionless — so squaring and summing them directly would let x dominate entirely. Each residual is therefore divided by a fixed scale s_d before squaring:

$$\mathcal{L}_{\text{pooled}}(\theta) = \text{mean}_{n,j,d} \left(\frac{r_{n,j,d}}{s_d} \right)^2, \quad (22)$$

where n runs over the training states, j over the $q+1$ outputs, d over the four predicted components, and s_d is the **pooled** spread of component d over the training states, measured from the training population itself and therefore label-free like everything else. This is the

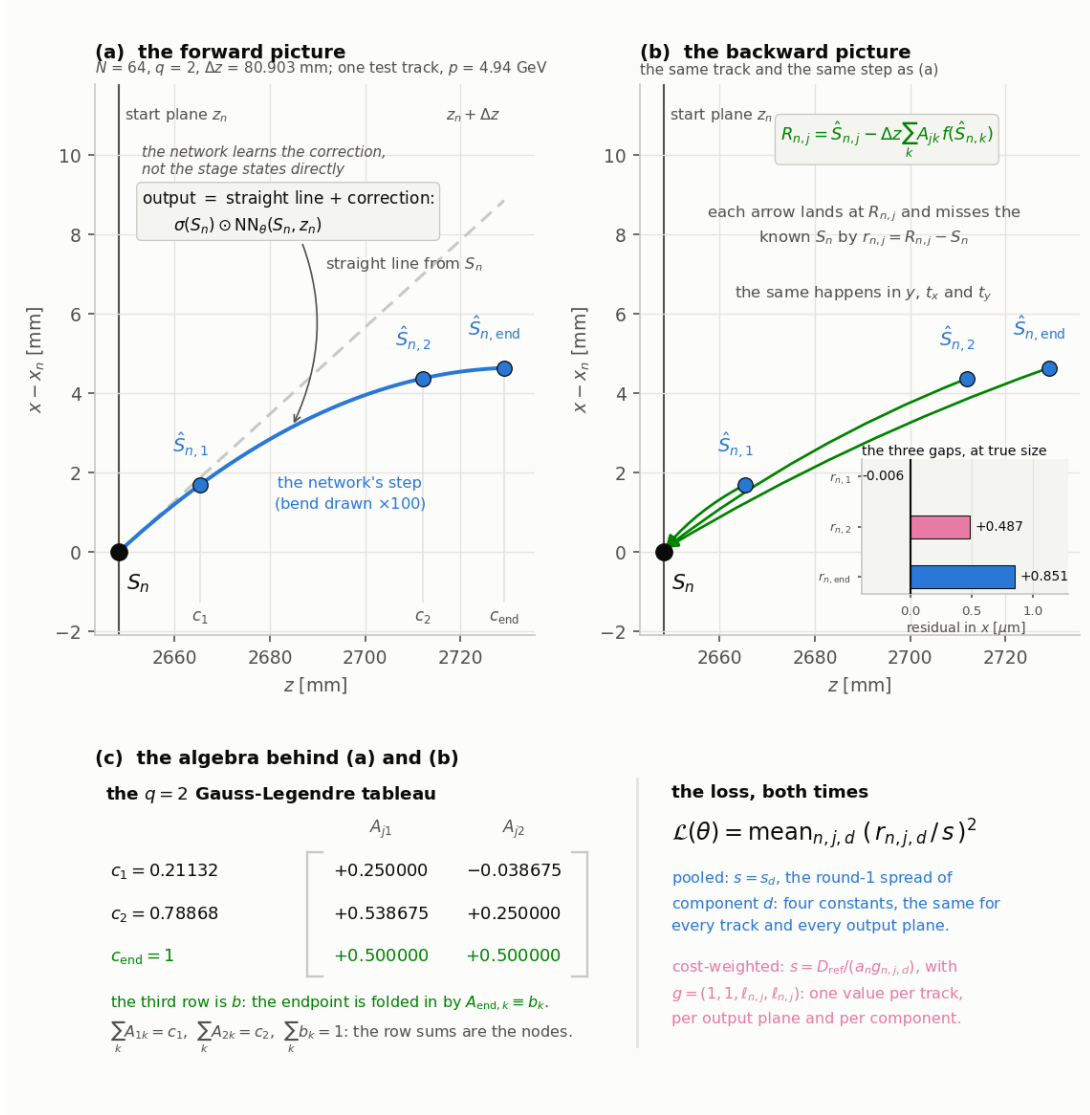


Figure 1: How one training step produces its residuals, drawn on the real geometry of the $N = 64$, $q = 2$, $\Delta z = 80.903 \text{ mm}$ network and on one real test track of 4.94 GeV . (a) The forward picture: the input state S_n on the start plane z_n (black dot), the straight line it would follow with no field (dashed), and the network's three outputs at $z_n + c_j \Delta z$ for $c_1 = 0.21132$, $c_2 = 0.78868$ and $c_{\text{end}} = 1$, each of them the straight line plus a correction $\sigma(S_n) \odot \text{NN}_\theta$ as in Eq. (17), so that what the network learns is the correction to the straight line, not the stage states directly. That correction is real but small — $42 \mu\text{m}$ over an 80.9 mm step for this track — so the bend is drawn exaggerated by a factor of 100; the straight line itself and the plane positions are to scale. (b) The backward picture: from each output the reconstruction of Eq. (18) runs the step backwards to the start plane, and each landing point misses the known S_n by that output's reconstruction residual. On the scale of the panel the three landings are indistinguishable from S_n , so the inset draws the three gaps in x at their true size, in micrometres; y , t_x and t_y behave the same way. No reference trajectory enters either panel: only the input, the tableau, and the field at the network's own proposed positions. (c) The algebra behind both: the $q = 2$ Gauss-Legendre tableau with its actual numbers and with the endpoint row $A_{\text{end},k} \equiv b_k$ appended, and the one-line form of the loss with the two choices of divisor that separate Eq. (19) from Eq. (20). Source: scripts/fig16_loss_schematic.py.

mean squared normalised reconstruction residual, and it is what the first study and the study immediately preceding this one both used. Equation (19) at the head of this section is the same expression written out, with the residual of Eq. (13) substituted and the sums over states, outputs and components shown explicitly.

Equation (22) asks for equal *absolute* accuracy in every component, on every track, at every plane — and it gets it: the measured per-step slope error is flat along the crossing. Two consequences follow, and both were measured before any reweighted network was trained, by drawing 8000 (track, plane) states evenly over the 64 start planes of a trained $N = 64$, $q = 2$ network and asking what share of the loss each group of states carries.

Consequence one: the loss is spent on the softest tracks. A soft track bends more, so its residuals are larger in absolute terms, so it takes over an objective that counts absolute residuals. Of the pooled loss over those 8000 states, 89.8 per cent comes from tracks of 3–8 GeV, which are 45.8 per cent of the states; 8.8 per cent comes from below 3 GeV; and 1.23 per cent from the 10–50 GeV window that the momentum region of interest sits in, although that window holds 32.0 per cent of the states. The concentration is extreme even within that: the top one per cent of states by contribution carry 98.3 per cent of the whole loss.

Consequence two: the loss does not know where along the crossing it is. A slope residual and a position residual count the same in Eq. (22), although a slope error made on the first plane is multiplied by more than five metres before the track reaches z_1 and a position error is not. Measured the same way, the first quarter of the crossing — where an error has the longest lever arm and therefore costs the most — carries 0.65 per cent of the loss, and the last quarter, where an error costs least, carries 6.2 per cent. The rank correlation between a state’s share of the loss and what that state’s error actually costs at z_1 is +0.68 overall and +0.53 within the 10–50 GeV window.

Neither distribution is something anyone asked for. They are what a pooled normalisation produces, and together they set the question of this paper: if the loss is asked for the right thing instead, how much of the endpoint error goes away?

2.6 The cost-weighted loss, derived

The replacement loss measures every residual as **the displacement it would cause at the first SciFi plane, as a fraction of that track’s own total bend**, with the momentum region of interest weighted up. It changes the weights inside the loss and nothing else: the same network code, the same residuals of Eq. (13), the same tracks, the same optimiser.

The form. Write the weighted loss as

$$\mathcal{L} = \text{mean}_{n,j,d} \left(\frac{a_n e_{n,j,d}}{D_{\text{ref}}} \right)^2, \quad \begin{array}{ll} e_x = r_x, & e_y = r_y, \\ e_{t_x} = r_{t_x} \ell_{n,j}, & e_{t_y} = r_{t_y} \ell_{n,j}, \end{array} \quad (23)$$

where every e is now in millimetres; $\ell_{n,j}$ is the lever arm defined below; a_n is a factor that depends only on the track; and D_{ref} is a fixed constant of the run that makes the whole expression dimensionless. The two positions are weighted equally, in millimetres, because the endpoint measure is radial; the separate y scale of Eq. (16) stays where it belongs, in the network’s output parameterisation, and no longer doubles as a loss weight. Equation (20) in Section 2.5 writes this out in full, with the residual substituted and the clamp on a_n included.

The lever arm.

$$\ell_{n,j} = z_1 - (z_{\text{start},n} + c_j \Delta z) + \Delta z, \quad (24)$$

where $z_{\text{start},n}$ is the start plane of state n , $c_j \in [0, 1]$ is the position of output j inside the step (with $c_{q+1} = 1$ for the endpoint), and Δz is the step length. It is the distance that output still has to travel before the track reaches z_1 , *plus one step*. Multiplying a slope residual by it converts a slope into the millimetres of displacement that slope will have produced by the time the track arrives, which is what finally puts a slope residual and a position residual into the same units so that they can be added.

The extra Δz is **additive, not a floor**, and the choice matters. Without it the lever arm would be exactly zero at the last plane of the crossing, and the loss would place no weight at all on the final step — which is wrong, because an error made there still lands in the fit at full size as a position error. A floor, $\max(z_1 - z, \Delta z)$, would fix the last plane but leave every other plane untouched and introduce a kink. Adding one step everywhere makes the last plane’s weight small but finite and leaves every other plane essentially unchanged, because Δz is 1.6 per cent of L at $N = 64$ and less at the shorter steps. At $N = 64$ the lever arm runs from 80.9 mm to 5258.7 mm, a range of a factor 65.

Figure 2 draws what that range does. Its first panel is the lever arm seen geometrically, as the displacement a fixed slope error opens up by the time the track reaches z_1 ; the remaining three are weight maps over the plane z and the momentum p , one for each loss, and they are the shortest statement of what this section is about.

The track factor.

$$a_n = \sqrt{W(p_n)} \frac{D_{\text{ref}}}{D_n}, \quad D_n = \kappa |q/p|_n \bar{I} L, \quad (25)$$

where D_n is the track’s **total transverse bend** across the whole crossing; $\kappa = 10^{-3}$ is the unit conversion of Eqs. (4)–(5); $\bar{I} = 3752.05 \text{ T}\cdot\text{mm}$ is the field magnitude integrated along the z axis over the crossing, evaluated once with a 1024-point midpoint rule and thereafter a pure constant; $L = 5177.8 \text{ mm}$; and $W(p)$ is the momentum window defined next. D_{ref} is the median of D_n over the run’s first-round states, 861.56 mm for the $N = 64$, $q = 2$ run. Dividing by D_n converts an absolute residual into a *relative* one: an error of a given size counts for more on a track that barely bends than on one that bends a great deal. Since $D_n \propto 1/p$, that is precisely what removes the momentum skew of Section 2.5.

D_n is deliberately the bend over the *whole* crossing, a constant of the track, and not the field integral of the particular step the residual sits in. Dividing by the per-step integral would ask for constant relative accuracy at every z , which in absolute terms would make the high-field middle of the magnet worse — and the middle is exactly where the bending happens. D_{ref} is likewise fixed from the first round so that the objective does not drift between rounds; it only sets the overall size of the loss, which the trainer rescales anyway.

The momentum window. $W(p)$ is a smooth top hat: equal to one on 10–50 GeV and falling off as a Gaussian in $\log p$ outside it, with a floor,

$$W(p) = \max\left(0.05, \exp\left[-\left(\frac{\log(p/p_{\text{edge}})}{\log 2}\right)^2\right]\right), \quad (26)$$

where $p_{\text{edge}} = 10 \text{ GeV}$ below the band and 50 GeV above it, and $W = 1$ inside. The roll-off constant $\log 2$ means the window falls by a factor $1/e$ one factor of two outside the band. It is a window and not a cut because one network still has to work everywhere: W is 0.05 at 1 and 2 GeV, 0.368 at 5 GeV, 1 at 10, 20 and 50 GeV, 0.368 at 100 GeV and 0.05 at 200 GeV. The square root in Eq. (25) is there because a_n is squared inside Eq. (23), so $\sqrt{W}$ makes W itself the multiplier on the squared residual. Throughout this paper 10–50 GeV is called the **loss window**, because that is what it is: a parameter of the objective. It is not one of the paper’s reporting bands, which are < 3 , 3–8, 8–20, 20–50 and $> 50 \text{ GeV}$; where the loss window appears in a table it appears as an extra row and is labelled as such.

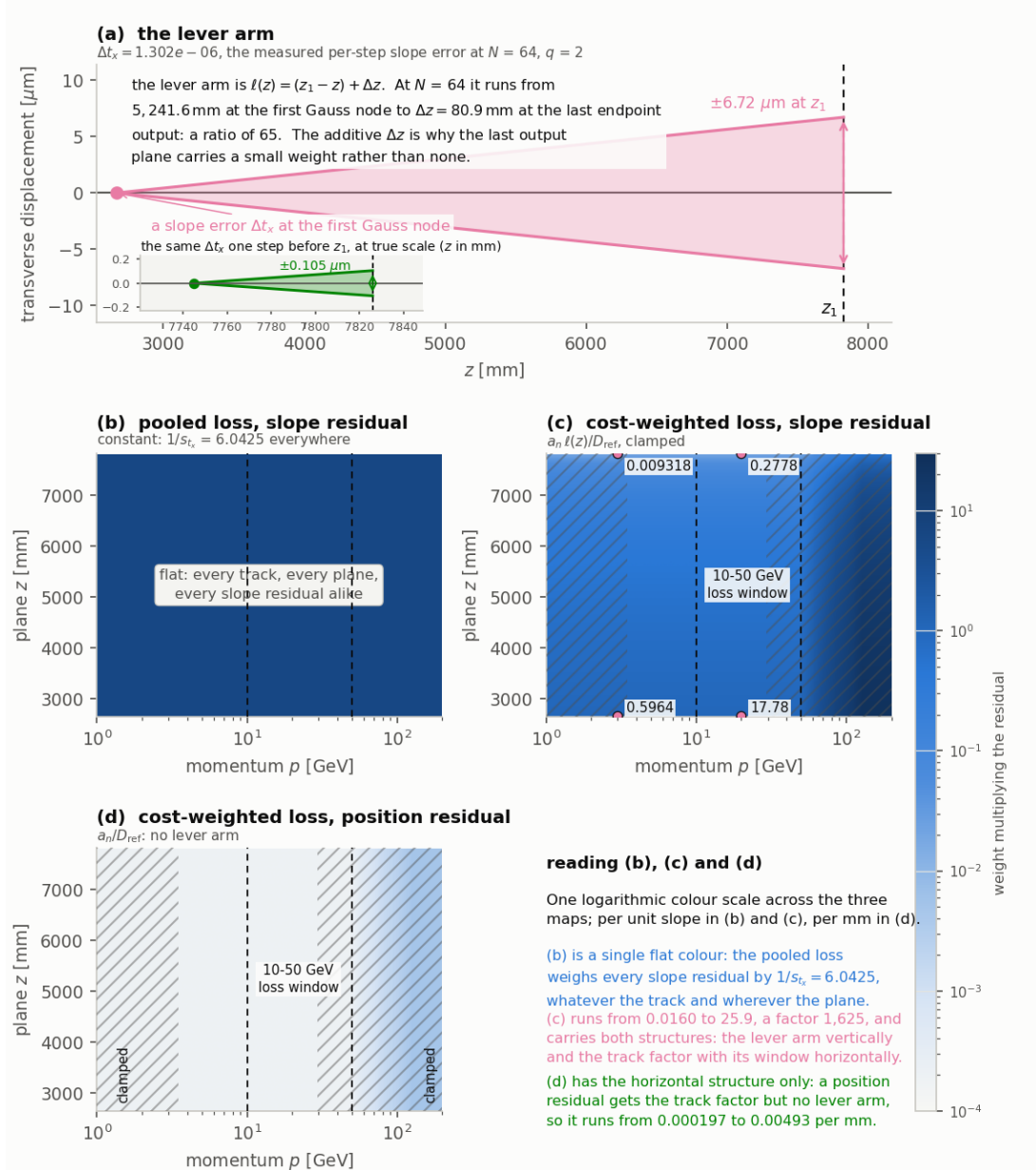


Figure 2: What each loss does with the residuals, for the $N = 64$, $q = 2$ geometry. (a) The lever arm, geometrically: a slope error Δt_x made at the first Gauss node of the first step fans out into a displacement of $\pm 6.72 \mu\text{m}$ by the time the track reaches z_1 , while the same error made one step before z_1 fans into $\pm 0.105 \mu\text{m}$, drawn at true scale in the inset. Throughout the panel $\Delta t_x = 1.30 \times 10^{-6}$, the measured per-step slope error of the pooled-loss network of this step length. The lever arm of Eq. (24) adds one step to that geometric distance, so it runs from 5241.6 mm at the first Gauss node — the largest lever arm the loss actually weighs — down to $\Delta z = 80.9$ mm at the last endpoint output, a ratio of 65. The range quoted in the text above and in Figure 3, 80.9 to 5258.7 mm, is the weighting module’s own reported range, whose upper end is the value of Eq. (24) at $c = 0$, a plane on which no output sits; the two ends differ by 0.3 per cent and give the same factor 65. (b) The pooled loss as a weight map over the plane z and the momentum p : one flat colour, the constant $1/s_{t_x} = 6.0425$, on every track and at every plane. (c) The cost-weighted loss on the same axes, $a_n \ell(z)/D_{\text{ref}}$ with the clamp applied. The hatched momenta are those at which the clamp binds, the dashed lines are the 10–50 GeV loss window, and the four marked points are the worked example at the end of this section, annotated with its unclamped weights 0.596, 0.0093, 17.8 and 0.278. (d) The same for a position residual, a_n/D_{ref} , which carries the track factor but no lever arm and so has no vertical structure at all. Panels (b) to (d) share one logarithmic colour scale, so the flat map and the 1625-fold range of the drawn cost-weighted map can be read against one another directly. Source: `scripts/fig17_loss_weights_schematic.py`.

The clamp. The factor a_n scales as $|q/p|$, which spans about a factor 200 across this sample and 40 000 once squared. Left alone, the handful of tracks above 100 GeV would take a fifth of the loss. So a_n is clamped to $[1/5, 5]$ of its median over the round’s own batch of training states. The threshold moves slightly between rounds because the batch does; D_{ref} does not. The factor five was chosen by a scan over the clamp before any training, reported with the rest of the pre-flight in Section 5.2.

The loss is still label-free. Both factors are functions of the input state and the field map alone. $\ell_{n,j}$ is geometry: the start plane, the nodes and the step length. a_n needs $|q/p|$, which is the fifth component of the input, and $\bar{I}$, which is an integral of the measured map. Neither reads a reference trajectory, from an integrator or from a simulation, so Eq. (23) preserves the property that makes the whole construction worth having. That is not a detail: a weight built from the reference error — which would be the obvious way to weight by cost — would have made the objective as expensive to prepare as supervised training and as fragile to a change of field map.

The reweighting must not move the minimum. A weight changes how much each residual counts. It must not change *what a zero residual is*, or the network would be trained towards a different solution rather than towards the same solution with a different emphasis. That is a checkable statement and it was checked before any training: collocation states solved exactly by a root-finder, with no network anywhere, give machine zero under every weighting mode, between 6×10^{-30} and 2×10^{-29} of the straight line’s loss in the same units. The minimiser of Eq. (23) is the minimiser of Eq. (22); only the shape of the surface around it differs.

Why not weight by the field. The obvious alternative is to weight up the strong-field region, on the grounds that that is where the bending happens and therefore where the work is. It is wrong, and measurably so. Taking the step-by-step error anatomy of a trained $N = 64$, $q = 2$ chain, over the 64 steps of the crossing, with the field magnitude on the axis running from 0.221 to 1.048 T:

- the per-step slope error against the field magnitude: Spearman -0.575 , Pearson -0.623 . The per-step error is *worst where the field is weakest*, at the two low-field ends, not in the magnet’s core;
- a step’s cost at z_1 against the distance it still has to travel: Spearman $+0.996$, Pearson $+0.982$;
- a step’s cost at z_1 against the field magnitude: Spearman $+0.124$, Pearson -0.018 .

What a step’s error costs at the far side follows the lever arm almost perfectly and the field hardly at all. The lever arm is therefore the right variable and the field is not, and the same measurement is the reason D_n uses the bend over the whole crossing rather than the field integral of its own step.

A worked example. Take the $N = 64$, $q = 2$ run, where $\Delta z = 80.90$ mm and $D_{\text{ref}} = 861.56$ mm, and compare a 3 GeV track with a 20 GeV one. The 3 GeV track has $qop = 0.0999$ and bends $D_n = 1941$ mm across the crossing; it sits outside the loss window, so $W = 0.05$, the floor, and $a_n = \sqrt{0.05} \times 861.56/1941 = 0.0992$. The 20 GeV track has $qop = 0.0150$ and bends $D_n = 291$ mm; it sits inside the window, so $W = 1$ and $a_n = 861.56/291 = 2.959$, a factor 29.8 larger.

Now place each of them at a plane, and take the lever arm at the two ends of the crossing in round terms, $\ell = L$ for an output near the start and $\ell = \Delta z$ for the last output plane, whose

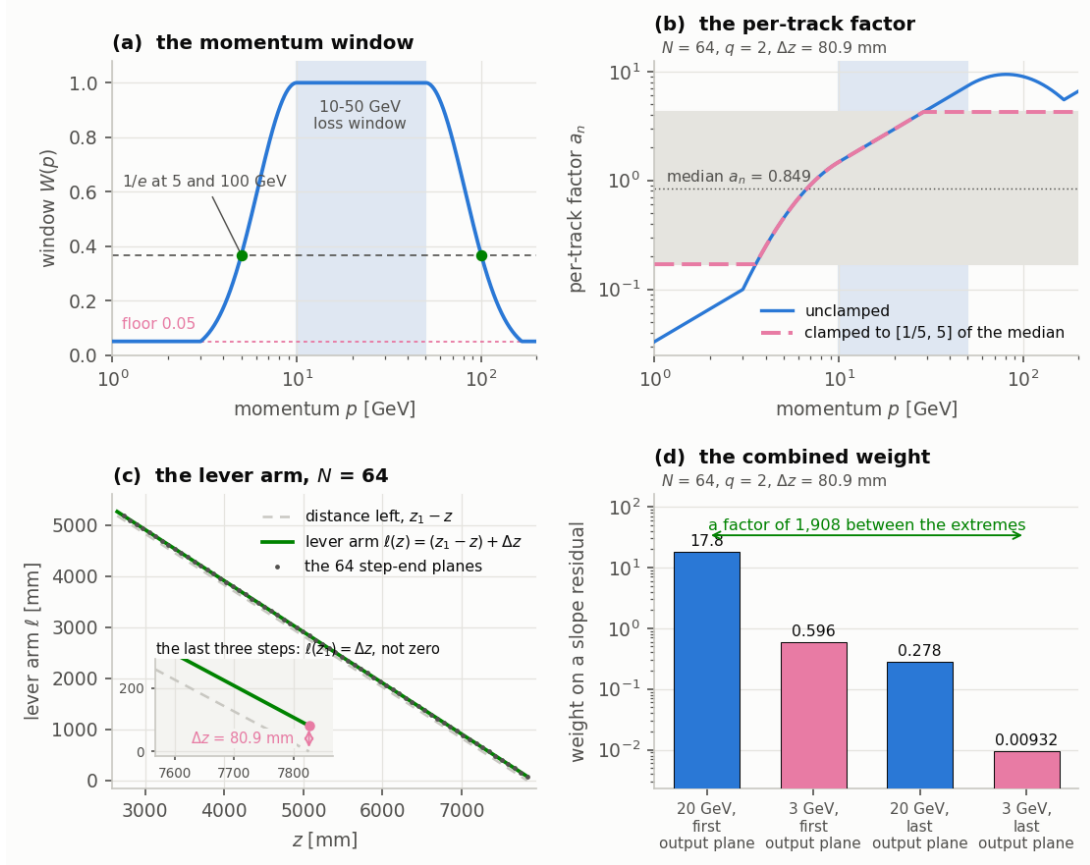


Figure 3: The four pieces of the cost-weighted loss of Eq. (23). (a) The momentum window $W(p)$ of Eq. (26): flat at one over the 10–50 GeV loss window, falling as a Gaussian in $\log p$ outside it and floored at 0.05, so that no track is ever ignored. (b) The track factor $a_n(p)$ of Eq. (25), unclamped and clamped to $[1/5, 5]$ of the batch median, showing which tracks the clamp removes weight from at each end. (c) The lever arm $\ell(z)$ of Eq. (24) across the crossing for the $N = 64$, $q = 2$ network, from 5258.7 mm at the start to 80.9 mm at the last output plane, with the additive Δz visible as the nonzero value at z_1 . (d) The four weights of the worked example — a 3 GeV and a 20 GeV track, each at the first and at the last output plane — whose extremes differ by a factor 1908. Source: `scripts/fig07_weights.py`, from the constants of the loss module.

ratio is exactly $N = 64$. A slope residual on the 20 GeV track near the start carries a weight $a_n \ell / D_{\text{ref}} = 17.8$ per unit of slope. The same residual on the 3 GeV track at the same place carries 0.596 — the factor 29.8 again. The same residual on the 20 GeV track at the *last* plane carries 0.278, a factor 64 down from the lever arm alone. And on the 3 GeV track at the last plane it carries 0.0093. The extremes differ by a factor **1908**. A position residual, which has no lever arm, carries 1.15×10^{-4} per millimetre on the 3 GeV track and 3.43×10^{-3} per millimetre on the 20 GeV one. Under Eq. (22) all four of those slope residuals would have counted the same. These are the unclamped values; the clamp of $[1/5, 5]$ then trims the tails of the a_n distribution, which is why the realised spread is smaller than 1908 and why the clamp is a knob worth scanning.

Figure 3 shows the four factors of the weight separately.

2.7 The three reference points

An error in micrometres means nothing on its own. Every number in this paper is quoted against three fixed points, which answer three different questions and must not be confused with one another. The numbers themselves are in Section 4; what follows is what each one can and cannot say.

1. **Exact collocation: the ceiling.** The same q -stage Gauss–Legendre equations, on the same states, at the same step length, chained the same N times, solved directly by a root-finder in double precision with no network anywhere. Because the training loss is zero exactly when the network’s outputs solve those equations, this is the best a network of that (N, q) could possibly be: the answer the objective is pointing at. It says whether a given network is limited by its own optimisation or by the scheme it is solving. It says nothing about whether the scheme is the right scheme.
2. **Sixth-order Runge–Kutta at 0.1 mm: the reference.** Butcher’s seven-stage explicit method of order six, integrating Eqs. (3)–(6) through the same measured field map in 0.1 mm steps from the same starting state. Written RK6 throughout. Every error in this paper that is not explicitly against something else is against this: it is the same physics the network is trained on, integrated far more finely than the network’s step. It says how well the network solves the problem it was given. It cannot say whether that problem is the particle’s real trajectory, because it is a *field-only* reference: it knows the magnetic field and nothing about the material the particle crosses.
3. **The Geant4-true SciFi state: the truth.** The simulated particle’s actual state on its own first SciFi plane, taken from the simulation that produced the sample, including multiple Coulomb scattering off the material in the way and any energy it lost there. Comparing the reference against this measures what the field-only description leaves out; the classical parameterisation of the scattering width is that of Lynch and Dahl [8]. This is the only one of the three that is a physical statement about a particle. It says what accuracy is worth chasing: driving a field-only surrogate far below the material term buys nothing in a fit that has to model the material anyway. It cannot be used to score the network directly, because the network was never trained on it and by construction cannot reproduce it.

The ordering matters and is easy to get wrong. A network can be far above its ceiling and still far below the material term, in which case the honest reading is that the surrogate is limited by its optimiser but the limitation does not yet matter physically. Section 5.9 makes that comparison explicitly rather than leaving it to the reader.

2.8 Metrics and the stopping rule

Two different errors are reported throughout and they must be kept apart.

- **Endpoint error.** The network is applied N times, by the loop of Section 2.4, starting from a test track’s real state on the last Upstream Tracker plane, and its state on the first SciFi plane is compared with the RK6 reference carried to that same plane. This is the quantity the track fit would actually experience, and it contains every accumulation effect the chain produces.
- **Single-step error.** The network is applied *once*, starting from an RK6 reference state on some intermediate plane, and its output is compared with RK6 taken from that same state over that same step. This measures the network in isolation, with no accumulation whatever. Where a single-step number appears it is always said to be a single step; where it does not, the number is an endpoint.

The position error of a prediction is quoted in two ways. The **radial** metric is $\sqrt{\Delta x^2 + \Delta y^2}$, and it is the headline measure of this paper, because a track fit cares how far away the prediction is and not in which direction. The **max metric** is $\max(|\Delta x|, |\Delta y|)$, which is what the first study used and what several of the stored result files carry, so it is reported wherever a comparison

with earlier work is being made. Slope errors are reported per component and, as stated in Section 1.1, as dimensionless numbers.

Over a set of tracks, five summaries are used.

- The **median** of the absolute deviation, which is the headline for every table. It is used rather than the mean because the distribution has a long tail and a mean would report the tail rather than the typical track.
- The **95th percentile**, written p95, of the same distribution. This is where the tail shows, and a real-time system schedules against the tail rather than against the median.
- The **signed median**: the median of the signed deviation rather than of its magnitude. A signed median much smaller than the median magnitude means the typical track is missed in either direction, which is scatter; a signed median close to the median magnitude means the tracks are being bent systematically one way, which is a bias and a different problem.
- The **68 per cent half-width**: half the distance between the 16th and 84th percentiles of the signed deviation. For a Gaussian it equals the standard deviation, but unlike the standard deviation it is not dragged by a few outliers.
- The **RMS**, the root mean square of the signed deviation, quoted beside the half-width wherever the two disagree, because the size of the disagreement is itself the statement about how heavy the tail is.

Two further quantities are used in the decomposition of Section 5.7 and are defined here.

Coherence asks whether a chain’s per-step errors add or cancel. For one track, form the magnitude of the *sum* of the per-step slope errors along the chain and divide it by the sum of their magnitudes. If the N steps erred independently and at random, that ratio would be about $1/\sqrt{N}$; if every step erred in the same direction it would be one. Quoting the measured ratio beside its $1/\sqrt{N}$ reference says directly by what factor the steps are more aligned than chance.

The **explained fraction** asks whether the endpoint error really is accumulated slope error. Build a model of the endpoint error as the sum over steps of each step’s own position error plus that step’s slope error multiplied by the distance it still has to travel, and take the ratio of that model to the measured endpoint error. A fraction near one means the decomposition is complete and the endpoint error is exactly what the lever-arm picture says it is; a fraction well below one means something else is happening.

Training, and the vocabulary of stopping. The networks are trained with L-BFGS, a quasi-Newton optimiser that builds an approximation to the curvature of the loss surface from the last few gradient differences and takes a step using it [9]. It converges quickly to its own idea of a minimum and then stops, so training does not proceed as one long descent but as a sequence of fresh starts. Three terms follow from that and are used throughout.

1. A **restart** is one run of the optimiser, from the current weights, to its own convergence. Its curvature history is discarded at the start of each one.
2. A **round** is 25 consecutive restarts, after which a fresh batch of 32 000 collocation states is drawn and the network is scored on the validation tracks. The round is the unit in which progress is measured, because a single restart’s validation number wanders by 10 to 20 per cent.
3. The **validation headline** of a run is the median validation endpoint error of its last ten rounds, quoted with the round-to-round spread beside it. It is not the final checkpoint’s number, which is one draw from that wandering.

The plateau rule. A run has **plateaued** once the median validation error of its last ten rounds is no more than 5 per cent below the median of the ten rounds before it, and that holds at each of the last three rounds. The rule was written into the folder specification before the runs were submitted, and it was applied to the earlier pooled-loss runs’ own logs as well as to the new ones, so that neither side of the comparison could be favoured by a rule chosen afterwards.

It is one-sided on purpose. An earlier two-sided version — the last ten rounds within 5 per cent of the ten before, in either direction — was tried and discarded, because the round-to-round error wanders by 8 to 23 per cent and ten-round medians therefore rarely land within 5 per cent of each other even long after a run has stopped improving. The question that matters for stopping is not whether a run is steady but whether it is *still getting better*, so the rule fires only when the improvement has gone.

Convergence is never called from the loss. This deserves its own paragraph because ignoring it cost this programme a week of farm time and one set of published numbers. The training loss and the held-out error are not the same signal. Applying the rule above to the earlier pooled-loss runs’ round logs showed that *none* of the sixteen had plateaued when training was stopped by hand on the strength of a flat loss: every one of the eleven runs with twenty or more rounds was still improving, by 5 to 24 per cent per ten rounds. Eleven falls out of eleven has probability 0.0005 if the round-to-round movement were pure noise, so that is a trend and not scatter. Those runs’ numbers were therefore where training stopped, not where it converges, and every one of them had to be extended to its own plateau before any comparison could mean anything. All comparisons in Section 5 use the extended runs; the one table that uses the stopped snapshot says so in its caption.

The pre-registered criterion. For completeness, the criterion by which the reweighting was to be judged was also fixed in advance, and it is the one place in this paper where the loss window appears as a reporting band. Success required the $N = 64$, $q = 2$ network’s test radial error to clear the ± 13 per cent run-to-run band downward, *and* its median per-step slope error to fall below its then-current value of 1.6×10^{-6} , both reported overall and in the 10–50 GeV window. A partial result was defined as an in-band improvement without an overall one, to be read as the window having been made too hard, with the clamp as the knob. A null result was defined as nothing clearing the band, to be read as the residual not being the binding constraint. Section 5.4 reports the outcome against that criterion and nothing else.

3 Data and machinery

This section describes the material a chained network is trained on and scored against, and the machinery that turns raw simulation into that material. I begin with the simulated sample and where it came from (Section 3.1), then the piece of the detector the paper is about, the magnet crossing, and the field that acts inside it (Section 3.2). I then give the selection as a numbered cascade with the row counts at every stage (Section 3.3), the momentum spectrum and the five bands every table in this paper is reported in (Section 3.4), and the three checks the tracks had to pass before any network saw them (Section 3.5). Two short subsections close the section: the units, stated once and properly, because the slope convention in this paper differs from the labels on some of the files it reads (Section 3.6); and the training protocol and its cost (Section 3.7). A reader who wants only the numbers can read Tables 1 to 3 and skip the prose.

3.1 The sample

Every track in this paper comes from official LHCb simulation. None of it was generated here. That is a policy decision taken on 21 July 2026 and it has a plain justification: a small

configuration error in home-grown event generation is easy to make and hard to see, whereas a centrally produced sample carries validated conditions with it.

The sample is the LHCb *TestFileDB* entry `expected_2024_minbias_xdigi`. *TestFileDB* is LHCb's registry of reference data files: each entry names a set of files together with the exact software conditions they were produced under, so that quoting the entry name fixes the data and its provenance in one string. The entry points at production 00212966. Its files are in **XDIGI** format, the extended digitised format, which holds the digitised detector banks *and* the packed simulation truth, so that for every simulated particle I can read both what the detector would have recorded and what the particle actually did. The conditions are detector description `dddb-20231017` and conditions tag `sim-20231017-vc-mu100` for 2024 data taking. **DDDB** is the detector description database, which says where every piece of the detector is and what it is made of; **CondDB** is the conditions database, which says what state the detector was in, including the magnet current and polarity. Five of the entry's eight files, 19 GB, are mirrored on the local CVMFS filesystem, and 200 events were read from them. No grid access was needed at any point.

The `mu100` in the conditions tag is the magnet polarity: it is the **magnet-up** configuration, meaning the dipole's vertical field component points upward and positive particles are deflected in the $-x$ direction. This matters here for one reason, and I state it plainly because it is the kind of detail that silently invalidates work.

Polarity. The cross-magnet labels stored in the harvested training file were integrated with the *magnet-down* map, `field.v8r1.down.bin`, md5 `af284c69...58e`: that was the default in force when the file was written, before the sample's own polarity was established. The sample is a magnet-up sample. **Every track in this paper is integrated on the magnet-up map, `field.v8r1.up.bin`, md5 `9e49ddc4313b589f273540e7e0bb513b`, and the stored labels are not used anywhere:** I take from that file only the particles, their real states on real detector planes and their momenta, all of which are field-free quantities, and I regenerate every trajectory myself. The two maps are exact negatives of one another, so the mistake would have been a sign error on the whole bend, of order a metre for a soft track, not a subtle one.

3.2 The crossing and the field

The extrapolation step this paper is about is the crossing of the dipole magnet: from the last plane of the Upstream Tracker, at $z_0 = 2648.2$ mm, to the first plane of the scintillating-fibre tracker, at $z_1 = 7826.0$ mm. The span is $L = z_1 - z_0 = 5177.8$ mm. It is the longest routine step in the track fit and the only one in which the trajectory is strongly curved, so it is where a cheap surrogate would be worth most and where it is hardest to build.

I store each track's state on 257 planes,

$$z_k = z_0 + k \frac{L}{256}, \quad k = 0, 1, \dots, 256, \quad (27)$$

spaced 20.2258 mm apart. The number 256 is chosen so that every step count used in this paper divides it: a chain of N steps starts its step k on plane index $k \cdot (256/N)$ and ends on plane 256, so the same stored track serves $N = 2, 64, 128$ and 256 without interpolation. The corresponding step lengths are $\Delta z = L/N = 2588.9, 80.9031, 40.4516$ and 20.2258 mm.

The field is read from the canonical LHCb field map, version `v8r1`, magnet-up polarity, file `/cvmfs/lhcb.cern.ch/lib/lhcb/DBASE/FieldMap/v8r1/cdf/field.v8r1.up.bin`, md5 `9e49ddc4313b589f273540e7e0bb513b`. The map is a table of the three field components on a regular Cartesian grid of 100 mm cells, and a field value at an arbitrary point is obtained by **trilinear interpolation**: the value is a weighted average of the eight corner values of the cell the point falls in, linear in each coordinate. The consequence matters for everything in Section 4

and I flag it here. Trilinear interpolation is continuous but its derivative is not: crossing a cell face, the field is unchanged but its slope jumps. A crossing of the magnet passes about 52 such faces. The field a high-order integrator sees is therefore piecewise smooth with kinks, which is written C^0 , and not the smooth function such an integrator's order was derived for.

On the beam axis the vertical component B_y is 0.207 T at z_0 , rises to a peak of 1.048 T near $z = 4700$ mm, and has fallen back to 0.304 T at z_1 . Almost all of the deflection therefore happens in the middle third of the crossing and almost none at either end, which is worth holding on to: a chain of equal steps in z is not a chain of equally difficult steps. A real 2.9 GeV track in this set is displaced 1204.8 mm in x relative to where a straight line would have taken it, and that displacement falls as the inverse of the momentum, so a 45 GeV track bends about fifteen times less over the same distance. Figure 4 draws these three things together: the field on the beam axis, the plane grids of the three chains, and $x(z)$ for six real test tracks.

3.3 The cut cascade and the split

The tracks are built from the harvested simulation in nine selection steps followed by one fiducial requirement. I give them as a protocol, with the row counts after each step in Table 1.

1. Take every cross-magnet row in the harvested set, both directions of travel. A *row* here is one pair of planes to be extrapolated between, for one particle.
2. Require that the plane before the magnet really is an Upstream Tracker plane, that is $z_{\text{pre}} > 1500$ mm. Below that the particle had no hit in the Upstream Tracker at all and the row would start on a vertex-detector plane, which is a different and much longer extrapolation. This removes 2476 rows.
3. Require that both directions survived for the particle, since the forward and backward legs are built as a matched pair and I do not want a half-pair in the set. This removes 2388 rows.
4. Require $2 < \eta < 5$, where the **pseudorapidity** $\eta = -\ln \tan(\theta/2)$ is a dimensionless measure of how forward a particle goes, with θ the angle to the beam axis; $\eta = 2$ is about 15.4° from the beam and $\eta = 5$ about 0.8° . This window is LHCb's nominal acceptance. It is evaluated on the truth pseudorapidity at the particle's origin and removes 1240 rows.
5. Require $1 < p < 200$ GeV. This is already the harvested set's own domain cut, so it removes nothing here, and I list it so that the domain is on the record.
6. Require that the particle is not an electron, $|\text{PID}| \neq 11$. Electrons radiate in material and their trajectories are not describable by an equation of motion that conserves momentum. This removes 4780 rows, the largest single cut.
7. Re-check that both directions are still present after the cuts. Nothing is removed.
8. Keep the forward row only, one row per particle.
9. Require that the row's own planes sit within 60 mm of the nominal z_0 and z_1 , so that every track in the set really is crossing the same gap. In the surviving set z_{pre} runs from 2593.2 to 2663.3 mm and z_{post} from 7818.7 to 7833.5 mm. This removes 775 particles and leaves **14,482** crossings.
10. **Fiducial.** A track is *fiducial* if it stays inside the region the field map covers for its whole path: I require x and y to lie inside the map at all 257 stored planes and every stored number to be finite. On the magnet-up polarity, which is the sample's own, this removes nothing. It is not a formality: on the reversed polarity soft backward legs bend out to the edge of the map, where the loader clamps the field and the trajectory stops meaning anything.

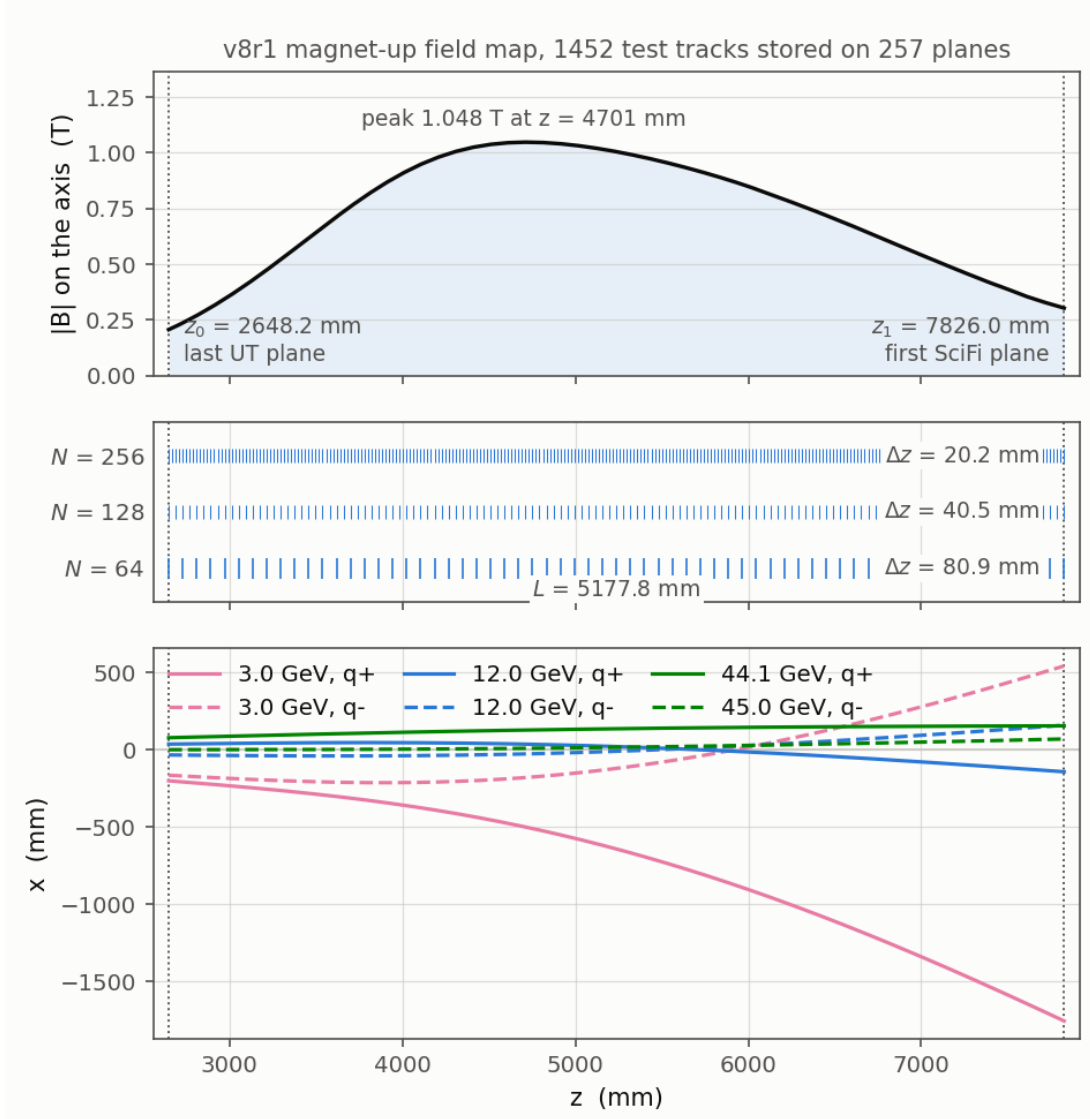


Figure 4: The magnet crossing this paper is about, in three registers. Top: the field magnitude on the beam axis ($x = y = 0$) between $z_0 = 2648.2$ mm and $z_1 = 7826.0$ mm, read from the v8r1 magnet-up map: 0.207 T at the entry plane, a peak of 1.048 T near $z = 4700$ mm, and 0.304 T at the exit plane, so almost all of the deflection happens in the middle third. Middle: the plane grids of the three chains used in this paper, $N = 64, 128$ and 256 , drawn as ticks at $z_0 + k \Delta z$ with Δz labelled; the $N = 256$ grid is the 257-plane grid every stored trajectory lives on. Bottom: $x(z)$ for six real test tracks, the nearest track to 3, 12 and 45 GeV in each charge, taken from the stored sixth-order trajectories and drawn to scale in millimetres. The $1/p$ bend is the whole reason a single step across the magnet cannot work. Source: `scripts/fig01_crossing.py`, `results/fig01_crossing.csv`.

The 14,482 crossings are split **by particle**, inherited from the harvested set with seed 20260718: 11,567 training, 1463 validation and 1452 test. Splitting by particle rather than by row matters because the rows of one particle are strongly correlated; splitting by row would leak information from training into test. The validation split is used for the stopping rule and the test split is used for every number reported as a result.

3.4 The momentum spectrum and the five bands

Everything in this paper that varies with momentum is reported in five bands, < 3 , $3-8$, $8-20$, $20-50$ and > 50 GeV, with the band population always shown. Where a loss or a design choice singles out the $10-50$ GeV window I report that window as an extra row, labelled “10–50 (loss

Table 1: The cut cascade that produced the track set. A row is one plane pair for one particle; the particle column counts distinct particles surviving. The last two lines are the two requirements that define the set used throughout this paper: a common crossing and a fiducial trajectory. Source: `results/tab_dataset_cuts.csv`.

	requirement	rows in	removed	rows out	particles
1	cross-magnet rows, both directions	41,398	0	41,398	21,932
2	the pre-magnet plane is an Upstream Tracker plane	41,398	2476	38,922	20,655
3	both directions kept for the particle	38,922	2388	36,534	18,267
4	$2 < \eta < 5$	36,534	1240	35,294	17,647
5	$1 < p < 200$ GeV	35,294	0	35,294	17,647
6	not an electron	35,294	4780	30,514	15,257
7	both directions still present	30,514	0	30,514	15,257
8	forward row only	30,514	15,257	15,257	15,257
9	planes within 60 mm of z_0 and z_1	15,257	775	14,482	14,482
10	fiducial: inside the map at all 257 planes	14,482	0	14,482	14,482

Table 2: The three splits in the paper’s five momentum bands, with the 10–50 GeV loss window reported alongside. The window overlaps the 8–20 and 20–50 bands on purpose and is not part of the partition, so its count is not added into the split total. The > 50 GeV band is thin everywhere and holds **30 test tracks**; every statement made about it in this paper is read with that in mind. Source: `results/tab_dataset_bands.csv`.

	train (11,567)		validation (1463)		test (1452)	
band [GeV]	n	median p	n	median p	n	median p
< 3	1339	2.62	181	2.63	164	2.63
3–8	5283	4.83	668	4.89	655	4.94
8–20	3391	11.91	441	11.82	433	12.48
20–50	1326	27.34	154	27.97	170	26.88
> 50	228	65.34	19	61.04	30	70.64
10–50 (loss window)	3756	16.50	470	15.85	487	16.18

window)”; it deliberately overlaps two of the five bands and is not part of the partition. Band membership is taken from each track’s own momentum, not from any stored band index.

Table 2 gives the populations. The three splits carry the same spectrum: median momentum 6.75, 6.78 and 6.86 GeV in train, validation and test, and band fractions that agree to about one percentage point in every band. The set is dominated by soft tracks, as a minimum-bias sample must be: 56 to 58 per cent of every split sits below 8 GeV and the 20–50 GeV band holds about 11 per cent. The extremes are 1.50 and 178.6 GeV in training, and 1.73 and 143.4 GeV in test. The pseudorapidity fills the selection window, 2.00 to 5.00, with a median of 3.49 in training and 3.53 in test.

Two further properties of the population are used later. The charge is close to balanced, 51.2 per cent positive in training and 50.9 per cent in test, which is what lets Section 4.4 separate a charge-antisymmetric systematic from a charge-blind width by splitting the set in two. And the particle composition is dominated by pions: in the test split, 1120 pions (77.1 per cent), 163 kaons (11.2), 152 protons (10.5) and 17 muons (1.2); the training split is the same to within a percentage point. Electrons are absent by construction. Figure 5 shows the sample in four registers: the momentum spectrum of the three splits with the band edges drawn, the composition and charge balance of the test split, the starting footprint of the test tracks on the plane z_0 , and the pseudorapidity distribution.

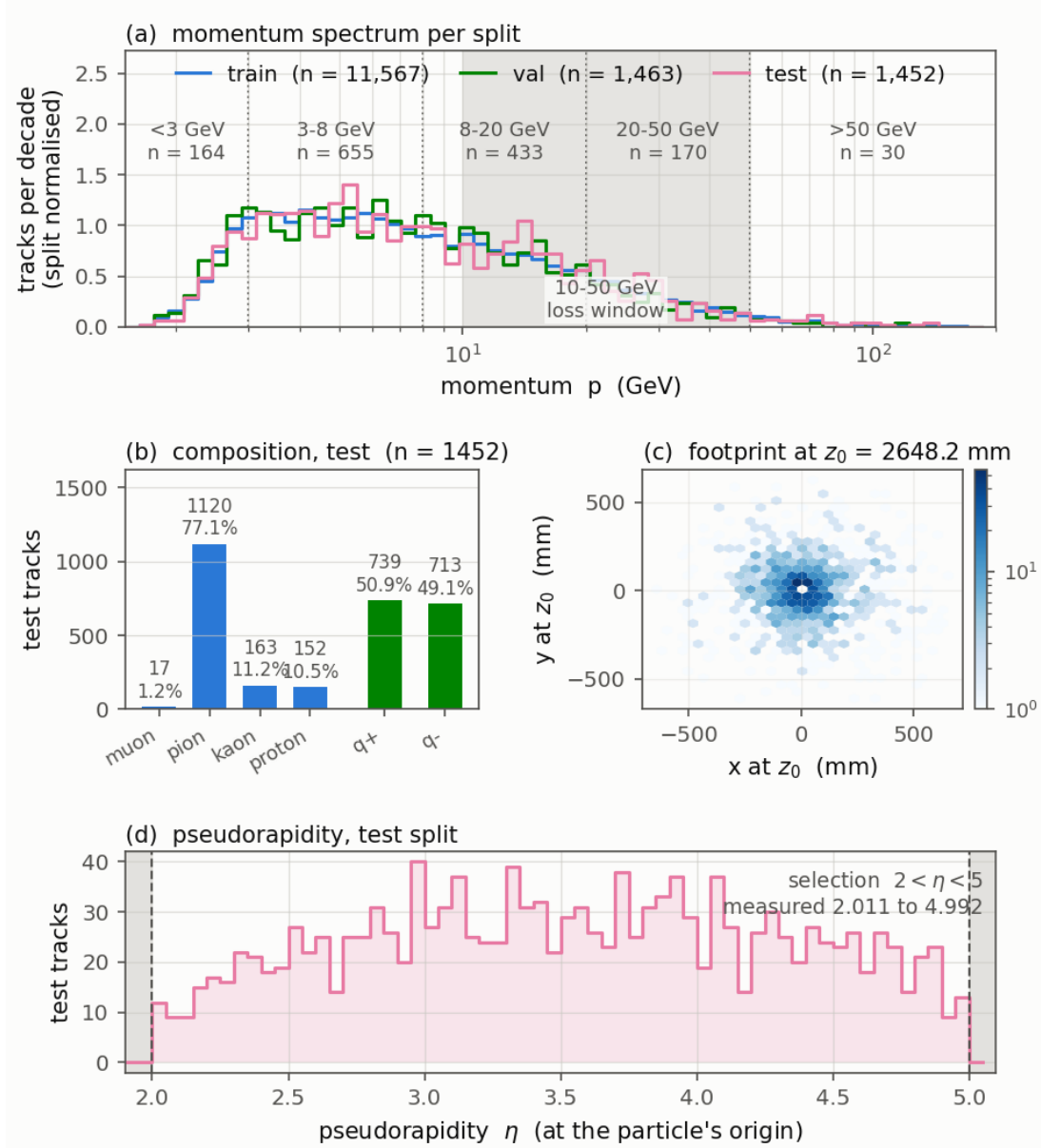


Figure 5: What the track sample is made of. Panel (a): the momentum spectrum of the three splits, each normalised so that the shapes can be compared, on a logarithmic axis from 1.5 to 200 GeV, with the four interior band edges drawn, each band labelled with its test-split count, and the 10–50 GeV loss window shaded. Panel (b): the composition of the test split, the four particle types that survive the non-electron cut (1120 pions, 163 kaons, 152 protons, 17 muons) and the charge balance (50.9 per cent positive), with counts and percentages printed on the bars. Panel (c): the starting footprint of the test tracks, (x, y) on the plane $z_0 = 2648.2$ mm, as a hexagonal-bin density on a logarithmic colour scale. Panel (d): the pseudorapidity distribution of the test split with the $2 < \eta < 5$ selection drawn. Every count on the figure is asserted against `results/paper_numbers.json` before the figure is drawn. Source: `scripts/fig02_sample.py`, `results/fig02_sample.csv`.

3.5 The gates on the tracks

Three checks stand between the raw selection and any use of these tracks. I call them gates, and all three pass.

Gate 1: the track set reproduces its predecessor bit for bit. The same particles, selected the same way, were used in the preceding study of this programme, on a coarser plane grid of 129 planes. The new file is required to carry the same validation and test particles,

Table 3: The material floor: the particle’s real simulated state on its own first fibre-tracker plane, minus the field-only sixth-order trajectory carried to that same plane, over all three splits concatenated (14,482 crossings). Positions are the max metric $\max(|\Delta x|, |\Delta y|)$ in micrometres; the slope column is the median of $\max(|\Delta t_x|, |\Delta t_y|)$ and is dimensionless. The last column is the fraction of tracks missing by more than one millimetre. The 10–50 GeV row overlaps two bands and is not part of the partition. Source: `results/tab_material_floor.csv`.

band [GeV]	n	median [μm]	p95 [μm]	median slope	> 1 mm
all	14,482	1736	12,628	4.94×10^{-4}	64.6%
< 3	1684	8683	24,293	3.34×10^{-3}	98.8%
3–8	6606	2806	10,849	8.81×10^{-4}	87.9%
8–20	4265	793	4059	2.17×10^{-4}	38.5%
20–50	1650	324	2048	9.36×10^{-5}	13.9%
> 50	277	153	850	4.06×10^{-5}	4.7%
10–50 (loss window)	4713	532	2973	1.45×10^{-4}	25.1%

in the same order, with bit identical start states, real detector states, planes, momenta and charges, and to agree with the old trajectories on the 129 shared planes. It does. The largest position disagreement anywhere is $1.5 \times 10^{-4} \mu\text{m}$; the largest slope disagreement is 6.3×10^{-11} , dimensionless (the gate file records it as 6.3×10^{-8} in its own units of slope $\times 10^3$, and I convert); and q/p agrees exactly. Those residuals are not zero for a reason worth stating: the integrator restarts its 0.1 mm steps at every stored plane, and the new file stores twice as many planes, so the same trajectory is cut into twice as many pieces and the last, shortened step of each piece falls differently. A tenth of a nanometre is the size of that effect.

Gate 2: fiducial. Every one of the 14,482 trajectories stays inside the field map at all 257 planes, with every stored number finite. This is step 10 of the cascade and it removes nothing on this polarity, which is itself the evidence that the polarity is right.

Gate 3: the reference closes. The reference integrator is required to undo itself: integrate a state forward across the magnet and back again, and compare with where it started. Over 200 momentum-stratified legs the round trip closes to a median of $3.7 \times 10^{-6} \mu\text{m}$ and a worst case of $4.4 \times 10^{-5} \mu\text{m}$ on the real map at a 0.1 mm step. This is not a test of accuracy, because both halves carry the same truncation error, but it is a floor: nothing built on this reference can be trusted below the level at which it fails to undo itself. Section 4.3 measures the reference properly.

What no gate can remove. The reference integrates the equation of motion in the magnetic field and nothing else, while the simulated particle also scatters off material and loses energy. The distance between the two is the **material floor**: the gap between the particle’s own real state on its first fibre-tracker plane and the field-only trajectory carried to that same plane. Table 3 gives it in the paper’s bands. It is $1736 \mu\text{m}$ in the median over the whole set and $8683 \mu\text{m}$ below 3 GeV. It is not an error in the tracks and no better integrator and no better network can reduce it. Section 4.4 takes it apart into its physical pieces and says which pieces are fixable.

3.6 Units

This subsection is short and it is binding on everything that follows.

Slopes are dimensionless. A track state is $(x, y, t_x, t_y, q/p)$, and the two slopes are

$$t_x = \frac{dx}{dz}, \quad t_y = \frac{dy}{dz}, \quad (28)$$

Table 4: What is held fixed across every training run in this paper. The parameter count depends on the stage count q because the network’s output layer carries q stage states and one end state; the three step counts used for the headline comparison have 18,956, 22,052 and 26,180 parameters at $q = 2, 8$ and 16. Source: `results/tab.compute.csv`, `results/paper_numbers.json` (compute), and `single_network_chain_discrete_approach/Block_F_reweighted_loss/F4_Writeup/page.md`.

held fixed	value
tracks	14,482 crossings from the official simulated sample, split by particle into 11,567 train, 1463 validation, 1452 test
network	two hidden layers of 128 units, tanh activation; the module is <i>imported</i> from the shared package rather than copied, so it cannot drift between runs
seed	0, the same one for every run
optimiser	L-BFGS, 200 iterations per restart, history 120, strong Wolfe line search; 25 restarts per round
states per round	32,000 collocation states, redrawn every round
field map	v8r1, magnet up, md5 <code>9e49ddc4...513b</code>
reference	seven-stage, sixth-order Runge–Kutta at a 0.1 mm step, from the same start state

built from a simulated hit’s exit minus entry displacement divided by that hit’s own Δz . No arctangent is applied anywhere in the chain from the simulation to these pages, so t_x is a ratio of two lengths and not an angle. I quote slopes and slope errors as dimensionless numbers in scientific notation. I never write “mrad” and I never scale a slope by 10^3 .

This differs from the labels on some of the files this paper reads. Several earlier result files in this programme write “mrad” for a slope difference multiplied by 10^3 , which is not the same quantity as a milliradian. The relation between the two is

$$\Delta\theta \approx \frac{\Delta t_x}{1 + t_x^2}, \quad (29)$$

where $\theta = \arctan t_x$ is the true angle. Inside the acceptance, $2 < \eta < 5$ gives $|t_x| \lesssim 0.27$, so the two agree to within 7 per cent at the very edge of the acceptance and within 1 per cent for most tracks. Whenever this paper reads a file whose slope column is scaled that way, the scaling is divided out and the label discarded; the figure and table generators do this at the point of reading, so no scaled slope reaches a table.

Positions are quoted in micrometres (μm) for errors and millimetres (mm) for geometry, and I say which. **Radial** error means $\sqrt{\Delta x^2 + \Delta y^2}$ and **max metric** means $\max(|\Delta x|, |\Delta y|)$; the max metric is the more conservative of the two and is never larger than the radial value. The metric is named beside every number in this paper. Finally, I distinguish **endpoint error**, the network applied N times in succession from the particle’s real Upstream Tracker state and compared at the far end, from **single-step error**, one application of the network from a state taken off the reference trajectory. These differ by more than two orders of magnitude and I say which one every number is.

3.7 Compute and the training protocol

Table 4 lists what is held fixed across every run in this paper. The point of fixing them is that a difference between two runs is then a difference in the loss and in nothing else.

The optimiser is **L-BFGS** [9], the limited-memory Broyden–Fletcher–Goldfarb–Shanno method: a quasi-Newton method that builds an implicit approximation to the inverse Hessian from the last few dozen gradient differences rather than storing a full matrix, and takes a line-searched step along the resulting direction. It is a deterministic full-batch method, which is why the training protocol is organised in **restarts** rather than epochs: one restart is 200

L-BFGS iterations from the current parameters with a fresh optimiser state, and restarting periodically is what lets the method escape the flat regions where its curvature estimate has gone stale. Restarts are grouped into **rounds** of 25. At the start of every round a fresh set of 32,000 (track, plane) states is drawn, the same number from every plane, so that no network can overfit a fixed draw; the loss is rescaled to 1 at the start of each round and again whenever it has fallen tenfold, each time with a fresh optimiser, so that the line search always works on numbers of order one.

All training ran on the Nikhef HTCondor farm, one job per run, single-threaded and in double precision. Single-threaded is not a preference: multithreaded PyTorch spin-waits on these nodes and is dramatically slower. The median wall time per restart, over the de-duplicated training history of each run, is 77.2 s for the 80.9 mm-step network under the pooled loss and 77.7 s under the cost-weighted loss; 117.9 and 118.3 s for the 40.5 mm-step network; and 219.8 and 184.1 s for the 20.2 mm-step network. Each restart runs its full 200 iterations; none of the six runs stopped early.

One trainer produced both losses. The cost-weighted trainer is the pooled-loss trainer with the loss function swapped and a mode flag that restores the old weights, and it is required to reproduce the old training *bit for bit* in that mode. It does: over three restarts on the same seed and the same states, the two trainers give identical losses to the last digit, 1.0011×10^{-4} , then 8.6855×10^{-8} , 3.2594×10^{-8} and 1.9375×10^{-8} . That gate is what licenses every comparison in Section 5 to be read as a comparison of two losses rather than of two pieces of software.

4 The scheme, measured on its own

Before any network appears, three things have to be measured. The first is the integration scheme the network is being asked to satisfy: its coefficients are constructed rather than copied, and a constructed tableau is worth nothing until it has been checked on the machine that will use it (Section 4.1). The second is how well that scheme does when it is solved *exactly*, with no network in it, and chained across the magnet. Because the training loss is zero precisely when the network solves those same equations, the exact solution is the **ceiling**: no network of that step count and stage count can do better, and a network’s score only means something beside it (Section 4.2). The third is the reference the whole programme is scored against, a sixth-order Runge–Kutta integration at a 0.1 mm step, whose own accuracy I measure rather than assume (Section 4.3). Section 4.4 then asks the question behind all three: the reference solves its equation, but is the equation what the simulated particle did? The answer sets the ceiling of the programme as a whole, and it does not enter any network number in this paper.

4.1 The tableau

Section 2.2 defined the tableau (c, A, b) of a q -stage Gauss–Legendre scheme and said that it is built here rather than looked up. This subsection reports the construction and the battery of identities that checks it, because a constructed tableau is worth nothing until it has been checked on the machine that will use it.

The construction is three lines, and it is problem-independent:

1. take the q -point Gauss–Legendre nodes and weights on $[-1, 1]$ from the standard Legendre routine, and map them affinely onto $[0, 1]$: $c = (x + 1)/2$ and $b = w/2$;
2. form the Lagrange basis polynomials ℓ_k on those nodes, written in **barycentric** form, $\ell_k(x) \propto w_k/(x - c_k)$ normalised to sum to one, which is numerically well conditioned where a Vandermonde solve would not be;
3. set $a_{jk} = \int_0^{c_j} \ell_k(s) ds$, the collocation integrals, each evaluated with the same Gauss rule rescaled onto $[0, c_j]$, which is exact because ℓ_k has degree $q - 1$.

Table 5: The Gauss–Legendre tableau verification, for every stage count this paper uses and three more. Columns three to seven are the maximum absolute residual of each identity; the tolerance is 5×10^{-13} and every entry is at the level of double-precision roundoff, 2.2×10^{-16} or below. The literature column is empty above $q = 3$ because no closed form is tabulated there. The last two columns show the outermost nodes closing in on the ends of the step as q grows, and the weights summing to one. Source: `results/tab_tableau.csv`.

q	order	row sums	quadrature	collocation	symplectic	literature	c_1	c_q	$\sum b$
1	2	0	0	0	0	0	0.5000	0.5000	1.0000000
2	4	2.8×10^{-17}	5.6×10^{-17}	5.6×10^{-17}	5.6×10^{-17}	5.6×10^{-17}	0.2113	0.7887	1.0000000
3	6	1.1×10^{-16}	2.2×10^{-16}	1.1×10^{-16}	2.8×10^{-17}	5.6×10^{-17}	0.1127	0.8873	1.0000000
4	8	5.6×10^{-17}	1.1×10^{-16}	1.1×10^{-16}	4.2×10^{-17}	—	0.0694	0.9306	1.0000000
8	16	2.2×10^{-16}	2.2×10^{-16}	1.1×10^{-16}	7.3×10^{-17}	—	0.0199	0.9801	1.0000000
12	24	2.2×10^{-16}	1.9×10^{-16}	2.2×10^{-16}	5.7×10^{-17}	—	0.0092	0.9908	1.0000000
16	32	1.1×10^{-16}	1.4×10^{-16}	1.4×10^{-16}	2.7×10^{-17}	—	0.0053	0.9947	1.0000000

Five identities then have to hold, and each one catches a different mistake.

1. **Row sums.** $\sum_k a_{jk} = c_j$: a stage really sits at that fraction of the step. A transposed or mis-ordered A fails here.
2. **Quadrature.** $\sum_j b_j c_j^{m-1} = 1/m$ for $m = 1, \dots, 2q$: the rule integrates every polynomial up to degree $2q - 1$ exactly, which is the property that gives the method its order $2q$.
3. **Collocation.** $Ac^{m-1} = c^m/m$ for $m = 1, \dots, q$: the stage matrix really is the integral of the interpolating polynomial. A wrong node vector fails here even when the weights are right.
4. **Symplecticity.** $b_j a_{jk} + b_k a_{kj} = b_j b_k$, an exact property of Gauss methods [7] that the construction never imposes. Finding it satisfied to roundoff is an independent check that these are the Gauss coefficients and not merely a consistent tableau.
5. **Literature.** For $q = 1, 2$ and 3 , comparison element by element with the closed-form tableaus of Hairer and Wanner [6, table IV.5.2]. For $q = 1$ this is the implicit midpoint rule, $c = (1/2)$, $A = (1/2)$, $b = (1)$.

Table 5 gives the result. Every residual, for every stage count from 1 to 16, sits at or below 2.2×10^{-16} , which is one unit in the last place of a double-precision number near one, against a tolerance of 5×10^{-13} . The construction is therefore exact to the arithmetic, and nothing in this paper is limited by the coefficients. The same routine has been run to $q = 50$ with the same outcome.

4.2 Exact collocation across the magnet: the ceiling

Equation (7) is implicit, so a solution of it exists independently of any network. If I solve the stage equations directly with a root finder, to a residual of 10^{-9} in millimetres and in slope units scaled to millimetres, and chain the resulting step N times across the magnet, I obtain the answer a network with zero training loss would produce. That answer’s distance from the reference is the **ceiling** for that (N, q) : the best a perfectly trained network of that geometry could possibly do. A network at $170 \mu\text{m}$ whose ceiling is $0.15 \mu\text{m}$ is limited by its own optimisation; a network at $9250 \mu\text{m}$ whose ceiling is $9271 \mu\text{m}$ is limited by the scheme, and no amount of further training will move it.

Table 6 gives all sixteen cells, and three readings follow from it.

First, the ceiling collapses the moment the step falls below one field cell. At $N = 2$, a step of 2588.9 mm covering half the magnet, the two-stage scheme is $9271 \mu\text{m}$ from the reference: the polynomial of degree two simply cannot follow a metre-scale bend through a varying field, and the error is the scheme’s, not anybody’s implementation. Adding stages helps enormously there,

Table 6: The exact collocation scheme chained across the magnet, and the trained networks beside it, on the 1452 test tracks. Upper block: the endpoint radial median of the exact scheme, $\sqrt{\Delta x^2 + \Delta y^2}$ in micrometres, in all sixteen cells. Middle block: the endpoint max-metric 95th percentile of the exact scheme, which exists only for $N = 2$ and $N = 256$, whose solves were run for this paper and stored per component; the $N = 64$ and 128 cells are read from the earlier chained comparator table, which stores the radial median alone. Lower block: the endpoint radial median of the network trained at each cell under the pooled loss. **The lower block is the 18 September 2026 snapshot** and is quoted for the grid only; the extended runs of Section 5 are better. Source: `results/tab_exact_ceiling.csv` and `results/tab_pooled_grid.csv`.

N	Δz [mm]	$q = 2$	$q = 4$	$q = 8$	$q = 16$
<i>exact scheme, endpoint radial median</i> [μm]					
2	2588.9	9270.9	195.4	7.47	5.17
64	80.903	0.147	0.478	0.103	0.0719
128	40.452	0.511	0.116	0.0115	1.76×10^{-4}
256	20.226	0.228	0.0295	4.26×10^{-4}	1.05×10^{-3}
<i>exact scheme, endpoint max-metric p95</i> [μm], where stored					
2	2588.9	38,521	1009	82.0	45.9
64	80.903	—	—	—	—
128	40.452	—	—	—	—
256	20.226	0.708	0.0989	5.68×10^{-3}	4.50×10^{-3}
<i>trained network, pooled loss, endpoint radial median</i> [μm] (snapshot)					
2	2588.9	9250.0	283.8	188.1	223.6
64	80.903	166.3	172.4	204.2	194.2
128	40.452	207.0	195.6	158.6	246.2
256	20.226	185.3	195.6	193.6	136.8

$195 \mu\text{m}$ at $q = 4$ and $7.5 \mu\text{m}$ at $q = 8$. At $N = 64$ and shorter, where a step is 80.9 mm or less and therefore shorter than the 100 mm field cell, the ceiling at $q = 2$ is already sub-micrometre, 0.147, 0.511 and $0.228 \mu\text{m}$ at $N = 64$, 128 and 256, and the extra stages buy anything between a factor of two and a factor of three thousand on top of that.

Second, the ceiling is not monotone in q at short steps. At $N = 64$ the $q = 4$ cell, $0.478 \mu\text{m}$, is worse than the $q = 2$ cell, $0.147 \mu\text{m}$; at $N = 256$ the $q = 16$ cell, $1.05 \times 10^{-3} \mu\text{m}$, is worse than the $q = 8$ cell, $4.26 \times 10^{-4} \mu\text{m}$. That is not a defect of the solver. It is the kinks. A method's order is a statement about a smooth right-hand side, and the trilinear map's derivative jumps at every cell face. Once the truncation error has fallen to the level set by those jumps, the ordering by classical order stops meaning anything and the remaining differences are the accidents of where the nodes happen to fall relative to the faces. A worked micro-example makes the scale concrete: a 20.2 mm step inside a 100 mm cell sees a perfectly smooth field and the scheme is exact to the arithmetic; a 2589 mm step crosses about 26 faces, and no polynomial of any degree interpolates across a corner.

Third, and this is the reading the rest of the paper rests on, the trained networks are nowhere near their ceilings except at the half-magnet step, where the ceiling is what limits them. At $N = 2$, $q = 2$ the network sits at $9250 \mu\text{m}$ against a ceiling of $9271 \mu\text{m}$: it has solved the scheme, and the scheme is wrong. The other three cells at $N = 2$ sit within a factor of 1.5 to 43 of their own ceilings, so they too are at least partly scheme-limited. At every shorter step the picture is completely different: the networks sit between 137 and $246 \mu\text{m}$ against ceilings between 1.8×10^{-4} and $0.511 \mu\text{m}$, a factor of 360 to 1.4×10^6 above what the scheme permits. Figure 6 draws all sixteen cells twice, against the stage count and against the step length, with each trained network joined to its own ceiling. The chained networks of this paper are therefore limited by optimisation and by the chaining, and not at all by the collocation scheme. Choosing more stages cannot help them; that is the finding that motivates everything in Section 5.

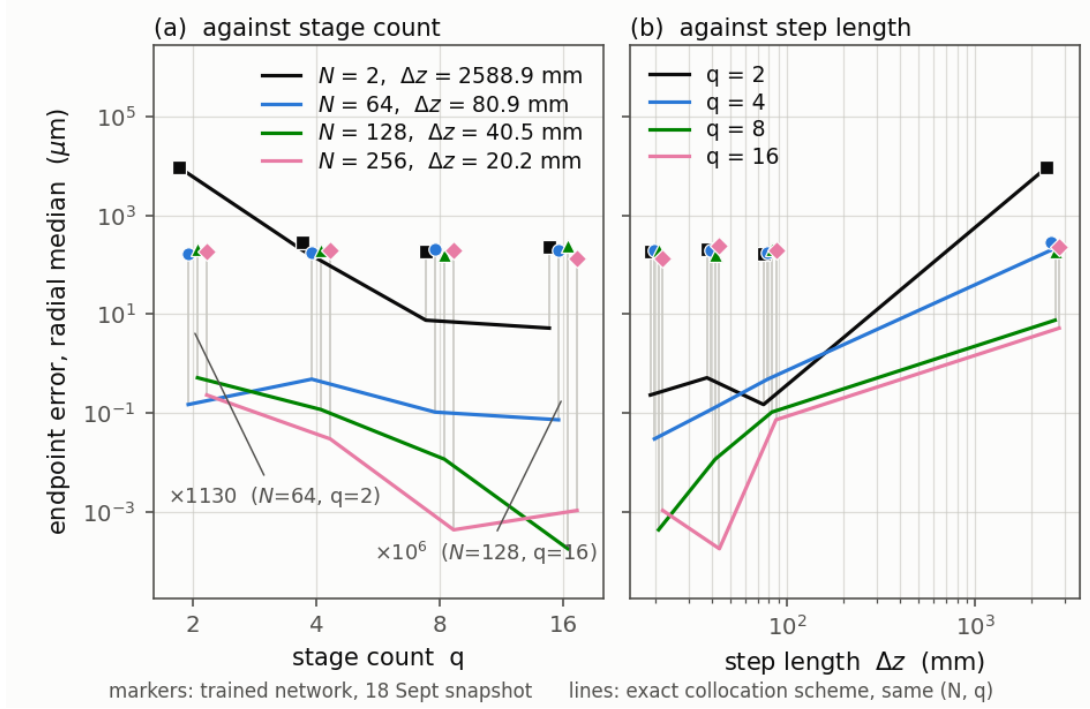


Figure 6: The ceiling and the networks. Both panels carry the same sixteen (N, q) points twice. The lines are the exact collocation scheme chained N times and solved without a network, that is the endpoint radial error a network of that step count and stage count could reach if its residual were driven to zero; the markers are the trained pooled-loss networks at the 18 September 2026 checkpoint, scored the same way on the same 1452 test tracks, and a thin connector joins each network to its own ceiling. Panel (a): against the stage count q , one line per step count N . Panel (b): the same sixteen points against the step length $\Delta z = L/N$, one line per q . The vertical distance is the point of the figure: for every chain of 64 steps or more the trained network sits two to six orders of magnitude above the scheme it is approximating. Source: `scripts/fig03_ceiling.py`, `results/fig03_ceiling.csv`.

4.3 The sixth-order reference

Everything in this paper is scored against a numerical solution of the equation of motion, and that solution has to be a number with a bar on it rather than a convention. The reference is Butcher’s seven-stage *explicit* Runge–Kutta method of order six, in double precision, at a fixed step of 0.1 mm, marched plane to plane with the last step of each leg shortened so that the trajectory lands exactly on the requested plane.

Provenance and identities. The seven coefficient arrays were copied from an earlier study’s reference machinery and are compared with the original element by element, with no tolerance: exactly equal, for the nodes, the stage matrix, the weights, the stated order and the stage count. Each row of A sums to its node to within 2.2×10^{-16} , b sums to one with an error of exactly zero, and A is strictly lower triangular, which is what makes the method explicit. Four mechanical properties were checked on a leg of 5177.8 mm, chosen deliberately not to be a whole number of steps: a forward-then-back integration on a smooth field closes to 1.8×10^{-12} mm; one leg integrated alone agrees with the same leg inside a mixed batch of five different legs *exactly*, difference 0.0, so batching cannot change an answer; a leg shorter than one step and a zero-length leg both behave, the latter returning the start state bit for bit; and q/p passes through unchanged.

Order, where order is visible. The scheme’s order was checked on three problems with exact solutions, chosen so that no single mistake can hide: a linear one, $dy/dt = -y$, a nonlinear one, $dy/dt = -y^2$, and a non-autonomous one, $dy/dt = y \cos t$, which is the only one that can

Table 7: The step ladder of the sixth-order reference on the real magnet-up map, over 200 momentum-stratified cross-magnet legs of median span 5174 mm. Left block: how far the endpoint moves when the step is halved. Right block: the distance from the finest run of the ladder, at 0.05 mm. Errors are $\max(|\Delta x|, |\Delta y|)$ in micrometres. The last row’s two blocks coincide because halving 0.1 mm *is* the comparison with the finest run. Source: `results/tab_rk6_convergence.csv`.

h [mm]	$ S(h) - S(h/2) $ [μm]			$ S(h) - S(0.05 \text{ mm}) $ [μm]		
	median	p95	max	median	p95	max
0.8	9.42×10^{-5}	6.41×10^{-4}	1.13×10^{-3}	1.00×10^{-4}	6.79×10^{-4}	1.10×10^{-3}
0.4	1.18×10^{-5}	1.51×10^{-4}	2.20×10^{-4}	9.78×10^{-6}	1.44×10^{-4}	2.48×10^{-4}
0.2	7.81×10^{-6}	5.02×10^{-5}	8.17×10^{-5}	6.85×10^{-6}	3.54×10^{-5}	5.91×10^{-5}
0.1	2.08×10^{-6}	1.64×10^{-5}	2.44×10^{-5}	2.08×10^{-6}	1.64×10^{-5}	2.44×10^{-5}

catch a wrong node vector. The fitted orders are 6.06, 6.27 and 5.75. Those problems cannot be pushed through the LHCb integrator, which is hard-wired to the equation of motion, so the equation was kept and the field replaced by a smooth analytic stand-in with the same call signature, a Gaussian blob peaking at 1.05 T near $z = 4700$ mm. On that field the maximum absolute error against a fine run is 1.66×10^{-5} , 2.12×10^{-7} , 2.93×10^{-9} and 4.20×10^{-11} mm at steps of 640, 320, 160 and 80 mm, each halving dividing the error by 70 to 78 against the ideal 64. The **fitted order is 6.23**, from the three coarsest points, which are the ones safely above the arithmetic floor; including the 80 mm point as well gives 6.20. The ladder has to be that coarse because on a smooth field the integrator is already at the double-precision floor by an 80 mm step.

Order, on the real map, is not visible. The real map is trilinear on a 100 mm grid and therefore C^0 : continuous, with a derivative that jumps at every cell face. A sixth-order scheme integrating across a kink behaves like a low-order one, and it does.

The successive ratios of the medians in Table 7 are **8.0**, **1.5** and **3.8**, against the 64 an order-six method would give. The first of those is the informative one: $8.0 = 2^3$ is exactly what an **order-three** method gives, and that is what integrating across kinks turns a sixth-order method into. The size of the penalty can be read off directly by running the same code on the smooth stand-in field: at $h = 0.8$ mm the real map moves the endpoint by $9.4 \times 10^{-5} \mu\text{m}$ per halving while the smooth field moves it by $9.9 \times 10^{-7} \mu\text{m}$, a factor of 95. The trilinear grid costs about two orders of magnitude of accuracy, and no choice of integrator recovers it; only a smoother field map would.

The remaining two ratios, 1.5 and 3.8, are not convergence at all. The smooth control is the tell: its $0.8 \rightarrow 0.2$ mm difference is $9.9 \times 10^{-7} \mu\text{m}$ but its $0.2 \rightarrow 0.05$ mm difference is *larger*, $4.0 \times 10^{-6} \mu\text{m}$. On a smooth field this scheme is exact to double precision at any of these steps, so a rising difference can only be rounding accumulating over more steps, of which there are 103,500 at 0.05 mm. That sets an arithmetic floor of a few times $10^{-6} \mu\text{m}$, which the real map’s series reaches at $h = 0.1$ mm. The practical statement is therefore: the endpoint has stopped moving by $h = 0.2$ mm, and at $h = 0.1$ mm the series has reached the floor.

Closure. Integrating each leg forward and back at 0.1 mm returns the start state to a median of $3.66 \times 10^{-6} \mu\text{m}$, a 95th percentile of $2.88 \times 10^{-5} \mu\text{m}$ and a worst case of $4.36 \times 10^{-5} \mu\text{m}$ over the same 200 legs. **Closure** is not an accuracy test, because both halves of the round trip carry the same truncation error and it cancels; it is a floor, and it agrees with the ladder, both saying a few times $10^{-6} \mu\text{m}$ in the median and a few times $10^{-5} \mu\text{m}$ in the tail.

The incumbent engine. Every label in the harvested set and every experiment before this reference existed was built with double-precision fourth-order Runge–Kutta at a 5 mm step. Against the sixth-order reference at 0.05 mm on these same legs, that engine sits at a median

of **0.0127** μm , a 95th percentile of $0.0766 \mu\text{m}$ and a worst case of $0.291 \mu\text{m}$; at a 1 mm step it sits at $8.2 \times 10^{-4} \mu\text{m}$. Thirteen nanometres in the median is orders of magnitude below every ceiling and every network error this paper discusses, so the old engine was never what limited anything and no earlier result is disturbed. What the fine reference buys is headroom.

The number to quote. The reference’s own floor is $5 \times 10^{-5} \mu\text{m}$, that is 0.05 nm: the worst case over the 200 legs of both the step-halving and the closure measurements, rounded up. It is a floor and not an error bar on any one number. It sits a factor 250 below the $0.0127 \mu\text{m}$ at which the old engine differs from it, and six to seven orders of magnitude below anything this paper is asked to resolve. Figure 7 draws the three measurements behind these paragraphs: the order fit on the smooth stand-in field, the step ladder on the real map, and the levels themselves on one logarithmic axis.

Cost. One thread, double precision. The field call is essentially the whole cost and it amortises over the batch, so the per-track figure falls with batch size until the map’s working set stops fitting in cache: 0.87 s per track in a batch of 200, 0.26 s at 1000, 0.134 s at 5000 and **0.12 s at 20,000**. Measured directly on the 200-leg batch it is 1.04 s per track. Peak resident memory for the whole measurement was 146 MB; the cost of this reference is time, not memory.

4.4 What the field-only reference shares with the truth, and what it cannot

The three preceding subsections establish that the scheme is what I think it is, that the exact scheme is far below the networks, and that the reference solves its equation to within 0.05 nm. None of that answers the question a supervisor asks next: the reference is a solution of an equation, but is the equation what the simulated particle actually did? This subsection measures the answer on all 14,482 crossings, takes the gap apart into physical pieces with two tests rather than assertions, and says which pieces are removable.

I say at the outset what does and does not follow. **The networks of this paper are measured against the sixth-order reference, and the gap measured here does not enter any network number.** The network and the reference solve the same equation with the same q/p ; they share every omission. What this subsection sets is the ceiling of the programme: the level at which a field-only extrapolator, however perfect, stops agreeing with the simulation. The comparison of the networks themselves against the simulated truth belongs in Section 5.9.

The three objects. Three different states exist on the first fibre-tracker plane for one and the same particle.

1. **The simulated truth.** Geant4 records a Monte Carlo hit wherever a particle crosses a sensitive volume, and I build a state from the midpoint of its entry and exit points and from the entry-to-exit displacement over its own Δz . The inverse momentum attached to that state comes not from the hit but from the particle record, that is the momentum the particle had *when it was produced*; that detail is the origin of the largest single piece of what follows. The state contains everything the simulation did to the particle, scattering and ionisation loss included, and no detector effect at all: there is no digitisation and no reconstruction anywhere in this paper.
2. **The field-only reference.** The particle’s real state on its last Upstream Tracker plane, carried to z_0 , integrated across the magnet and on to that particle’s own fibre-tracker plane with the sixth-order integrator at 0.1 mm. The equation holds $d(q/p)/dz = 0$: the momentum is constant along the whole crossing, the field is the only thing in the equation, and there is no material of any kind.

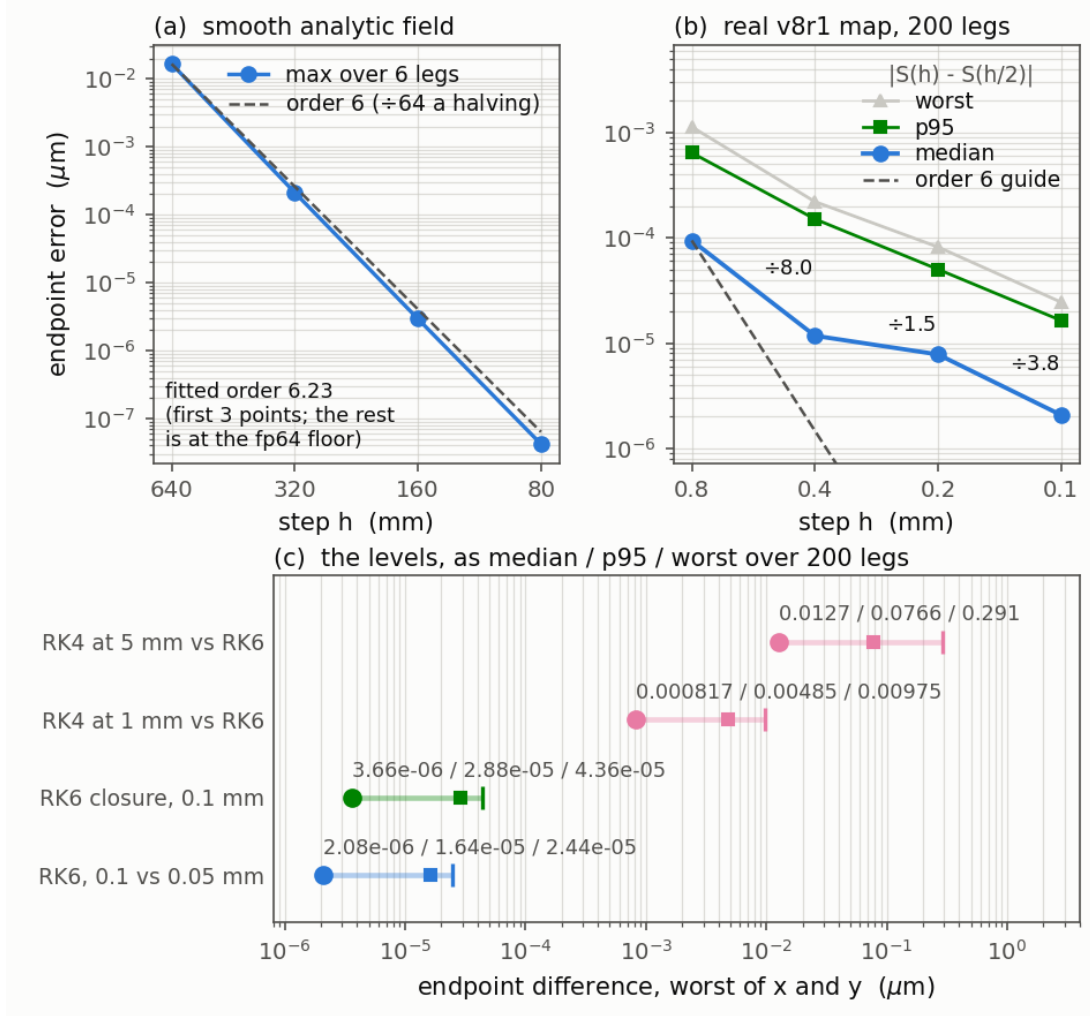


Figure 7: The sixth-order reference measured on itself. Panel (a): the scheme’s own order, measured with the trilinear map replaced by a smooth analytic stand-in so that the method is not asked to integrate across a kink; endpoint error against the same routine at a 0.5 mm step, at steps of 640, 320, 160 and 80 mm, with the fitted slope printed and an order-six guide drawn through the coarsest point. Panel (b): the step ladder on the real magnet-up map, the median, 95th percentile and worst of $|S(h) - S(h/2)|$ over 200 momentum-stratified cross-magnet legs, with the three successive halving ratios 8.0, 1.5 and 3.8 printed beside the median line against the 64 an order-six method would give on a smooth field. Panel (c): the levels that matter on one logarithmic axis, what one further halving moves the endpoint at the chosen 0.1 mm step, what the forward-then-back round trip fails to close by at that step, and how far the incumbent fourth-order engine at 5 mm and at 1 mm sits from the fine reference, each drawn as median, 95th percentile and worst. Source: `scripts/fig04_rk6_reference.py`, `results/fig04_rk6_reference.csv`.

3. The network endpoint. One network applied N times from the same real state, trained on the residual of that same equation. Whatever the equation omits, the network omits by construction.

Since the tracker is not one flat plane, each particle’s own z_{post} differs from z_1 by at most 7.45 mm. Both the reference state and the network state at z_1 are therefore carried from z_1 to that particle’s own plane with the same integrator before any comparison is made.

Why this was not visible earlier. The labels were gated against the simulation when they were built, and the gate looked reassuring: on 100,660 short plane-to-plane legs the label disagreed with the particle’s own next hit by a median of $11.9 \mu\text{m}$, falling to $3.8 \mu\text{m}$ above 5 GeV and rising to $38.8 \mu\text{m}$ below 2 GeV. Those legs are short and the magnet crossing is not.

The difference is the lever arm: a slope error of 10^{-4} displaces a track by $10\text{ }\mu\text{m}$ over a 10 cm gap between two tracker planes and by $518\text{ }\mu\text{m}$ over the 5177.8 mm of the crossing, so the same physics is fifty times larger here.

Test one: what happens under a charge flip. Write $\Delta = S_{\text{true}} - S_{\text{ref}}$ at the particle's own plane. The charge enters the equation of motion only through q/p , and it enters the two slope derivatives linearly. If the momentum handed to the integrator is too large by a relative amount $\epsilon = \Delta p/p$, every slope derivative is scaled by $(1 + \epsilon)$ and, to first order, so is the whole bend. Define the bend

$$b = x_{\text{ref}}(z_{\text{post}}) - [x_0 + t_{x,0}(z_{\text{post}} - z_0)], \quad (30)$$

where x_0 and $t_{x,0}$ are the start position and slope: the horizontal distance between where the reference ends up and where a straight line through the start state would have put it, that is the magnet's whole effect on that track. Its median magnitude over this set is 453 mm, and 14,476 of the 14,482 tracks have $|b| > 20\text{ mm}$, which is the cut applied whenever I divide by it. Then

$$\Delta x \approx \epsilon b \quad \text{flips sign with the charge,} \quad \text{while} \quad \frac{\Delta x}{b} \approx \epsilon \quad \text{does not.} \quad (31)$$

Multiple scattering has no such structure: a scattered particle is as likely to go left as right whatever its charge, so it contributes nothing to the *median* of Δx for either charge and shows up only in the charge-blind *width*. Splitting the set by charge therefore separates a bending systematic from random scattering with no modelling assumption. A worked micro-example: a 5 GeV track that bends by 500 mm and whose reference misses by $2500\text{ }\mu\text{m}$ in x has $\Delta x/b = 5 \times 10^{-3}$, which reads as a momentum defect of $5 \times 10^{-3} \times 5\text{ GeV} = 25\text{ MeV}$.

Reading $\Delta x/b$ as a momentum also distinguishes two candidate causes. If the implied defect is a *fixed number of MeV* independent of momentum, that is what ionisation energy loss looks like: a particle of any momentum loses roughly the same absolute amount crossing the same material. If instead the *ratio* were constant across momentum, that would point at the field rather than the momentum, because a field map scaled wrongly by a fixed fraction produces a fixed relative over-bend at every momentum.

Test two: the width against Highland. For the random part I compare the measured width with what multiple Coulomb scattering should give. The standard parameterisation is Highland's [8],

$$\theta_0 = \frac{13.6\text{ MeV}}{\beta c p} z_{\text{ch}} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right], \quad (32)$$

where θ_0 is the root-mean-square scattering angle projected onto one plane, p is the momentum, βc the velocity, which is c to an excellent approximation for every track here since the slowest is 1.5 GeV, $z_{\text{ch}} = 1$ is the charge in units of the elementary charge, and x/X_0 is the thickness traversed in **radiation lengths**. A radiation length X_0 is the material thickness in which an electron loses all but $1/e$ of its energy to bremsstrahlung, and it is the natural unit in which scattering is quoted; for air it is about 304 m. A particle scattering continuously over a length L ends up displaced sideways by

$$\sigma_{\text{displacement}} \approx \frac{\theta_0 L}{\sqrt{3}}, \quad (33)$$

the $\sqrt{3}$ arising because deflections earned near the end of the path have almost no lever arm left to act over. The only material between the two planes that can be accounted for from first principles is the air: $L = 5177.8\text{ mm}$ against $X_0 = 304\text{ m}$ gives $x/X_0 = 0.01703$, that is 1.70 per cent of a radiation length. I compute the prediction at each band's median momentum, compare it with the measured width, and invert Eq. (32) numerically for the thickness the measured width implies.

Table 8: The simulated truth minus the field-only reference at each particle’s own first fibre-tracker plane, on all 14,482 crossings, in the paper’s bands. Columns: the band population split by charge; the signed median of Δx for each charge, which flips sign; the median of $\Delta x/b$ for each charge, which does not; the momentum defect that ratio implies, computed as the pooled median ratio times the band’s median momentum; and the charge-corrected 68 per cent half-width of Δx . Positions in micrometres. The row marked “window” is the 10–50 GeV loss window, which overlaps two bands and is not part of the partition. **This table is the original study’s Table 4.2 re-binned into the paper’s bands;** its content is the same measurement on the same crossings. Source: `results/tab_decomposition_bands.csv` and `results/paper_numbers.json` (`reference_vs_truth.rebinned.decomposition`).

band [GeV]	n (q^+/q^-)	median Δx [μm], q^+/q^-	median $\Delta x/b$ [10^{-3}], q^+/q^-	implied Δp [MeV]	68% half-width of Δx [μm]
< 3	1684 (880/804)	−8641/ + 8130	+6.90/ + 6.53	+17.6	6119
3–8	6606 (3442/3164)	−2277/ + 2151	+3.62/ + 3.50	+17.3	2676
8–20	4265 (2171/2094)	−257/ + 245	+1.08/ + 0.974	+12.2	815
20–50	1650 (801/849)	+14/ − 7	−0.149/ − 0.0712	−3.1	351
> 50	277 (145/132)	+51/ − 58	−1.22/ − 1.42	−86.8	160
10–50 (window)	4713 (2366/2347)	−71/ + 74	+0.443/ + 0.470	+7.5	582
all	14,482 (7439/7043)	−1032/ + 873	+2.65/ + 2.37	+17.0	2262

Because the two charges have systematically different centres, pooling them carelessly would let the systematic inflate the width. I therefore move each charge onto its own median first, then pool, then take the 68th percentile of the absolute deviation from that centre. That is the statistic called “68 per cent half-width” below.

The decomposition, band by band and charge by charge. Table 8 is the main measurement, on all 14,482 crossings, re-binned into this paper’s five bands.

Read with test one in mind, the table says three things at once.

First, **the signed median flips sign with the charge and the ratio to the bend does not.** In the 3–8 GeV band a positive particle is found 2277 μm to one side of the reference and a negative particle 2151 μm to the other, while both give $\Delta x/b \approx +3.6 \times 10^{-3}$. That is the signature of a momentum error and of nothing else. Scattering cannot produce it, because scattering has no preferred side. Note also how small the pooled signed median is once the two charges are added together, −205 μm in that band against $\pm 2200 \mu\text{m}$ per charge: a pooled analysis would have seen almost nothing.

Second, **below 20 GeV the implied defect is a roughly constant number of MeV and not a constant fraction.** It is +17.6, +17.3 and +12.2 MeV in the three lowest bands and +7.5 MeV in the loss window, while the relative ratio $\Delta x/b$ falls by a factor of six over the same range. A fixed absolute momentum loss is what ionisation energy loss looks like and is not what a field-scale error looks like. The simulated particle really is slower than the number handed to the integrator, by of order 17 MeV, because the start state carries the particle’s *production* momentum: the state harvesting joins the momentum from the particle table, and the truth dump never wrote the hit’s own local momentum, which Geant4 records and which my dump simply did not ask for. What the 17 MeV contains is the energy the particle lost in the vertex detector and the Upstream Tracker before reaching the start plane, plus the energy it loses in the crossing itself, which the integrator also does not model. The two cannot be separated without re-dumping the truth, and that is the first of the two loose ends in item 6 of Section 8.

Third, **above 20 GeV the sign reverses and the ratio grows.** The implied defect becomes −3.1 MeV in the 20–50 GeV band and −86.8 MeV above 50 GeV, a relative under-bend of 0.01 and 0.13 per cent. A defect that grows with momentum is not energy loss. I state this as unexplained and name the candidates without choosing: a difference between the map I integrate in and the map and current the simulation was run with under its conditions tag, a scale difference of a tenth of a per cent being exactly the right size; or the accuracy of Geant4’s

Table 9: The charge-corrected 68 per cent half-width of Δx against the Highland prediction for the air between the two planes, on all 14,482 crossings, in the paper’s bands. The air thickness is $x/X_0 = 5177.8\text{mm}/304\text{ m} = 0.01703$ everywhere. The last column inverts Eq. (32) for the thickness the measured width implies, which is a measurement of the simulation’s material budget and not a confirmation of anything, since I have not looked that budget up independently. Source: `results/tab_highland_air.csv`.

band [GeV]	n	median p [GeV]	θ_0 , air [μrad]	air, Eq. (32) [μm]	measured [μm]	ratio	implied x/X_0
< 3	1684	2.63	571	1706	6119	3.59	0.179
3–8	6606	4.84	310	926	2676	2.89	0.120
8–20	4265	11.95	125.6	375	815	2.17	0.071
20–50	1650	27.28	55.0	164	351	2.14	0.069
> 50	277	65.49	22.9	68	160	2.34	0.081
10–50 (loss window)	4713	16.40	91.5	273	582	2.13	0.068

Table 10: The over-bend by particle type, measured at 5–25 GeV with the bend cut applied. This table is quoted in the original study’s own selection and binning, because no re-binned version exists. The ordering pion, kaon, proton is the one an energy-loss explanation predicts; the muons carry no statistical weight. Source: `single_network_chain_discrete_approach/reference_vs_truth/results/decomposition_by_pid.csv`, via `results/paper_numbers.json` (`reference_vs_truth.verbatim.decomposition_by_pid`).

particle	n	median p [GeV]	median $\Delta x/b$	median $ \Delta x $ [μm]	implied Δp [MeV]
pion	6139	9.25	$+1.65 \times 10^{-3}$	811	+15.3
kaon	909	10.44	$+1.48 \times 10^{-3}$	698	+15.4
proton	893	9.82	$+1.23 \times 10^{-3}$	395	+12.1
muon	42	7.40	$+1.11 \times 10^{-3}$	811	+8.2

own field stepper. The two bands hold 1650 and 277 crossings across all three splits, and only 170 and 30 of those are in the test split, so the high-momentum statement in particular carries little weight. In the original study’s finer binning the same quantity reads -8.1 MeV at 25–50 GeV, -77.5 MeV at 50–100 GeV and -240.4 MeV at 100–200 GeV, over 1016, 240 and 37 crossings, which shows the growth more clearly than the paper’s coarser bands can.

The random part. Table 9 puts the last column of Table 8 against Eq. (32) for air alone.

Above 8 GeV the measured width is a stable 2.1 to 2.3 times the air-only prediction, and the thickness it implies is a stable 6.8 to 8.1 per cent of a radiation length against the 1.70 per cent that air alone provides. The simulation therefore puts about four times more scattering material between the two planes than the air does, which is entirely plausible: the magnet region is not empty, and the last Upstream Tracker and first fibre-tracker stations have structure of their own. Below 8 GeV the ratio climbs to 2.9 and 3.6. Highland’s parameterisation describes the Gaussian core and degrades in exactly that regime, where the non-Gaussian single-scattering tail is large and where a 68 per cent half-width begins to sample it, so I do not read the two softest rows as evidence of more material.

The particle-type cross-check. If the systematic really is energy loss, then at the same momentum the over-bend should order itself by particle mass, because ionisation loss depends on velocity. Table 10 is that test.

The ordering is the predicted one and the effect is modest, about 25 per cent in the ratio between pion and proton. I record it as consistent with an energy-loss explanation and do not push it further; only a rebuilt truth dump carrying the local momentum can make this test sharp.

Table 11: The reference against itself and against the schemes, beside the two comparators that bracket the problem. All entries are endpoint errors in micrometres; the exact-scheme and comparator rows are radial medians on the 1452 test tracks and the rest are as noted in the n column. The three *label against the particle's next hit* rows, including the > 5 GeV and < 2 GeV rows, are quoted in the original study's own selection and binning, because no re-binned version exists. Source: `single_network_chain_discrete_approach/reference_vs_truth/results/rk6_self_consistency.csv`, via `results/paper_numbers.json` (`reference_vs_truth.verbatim.rk6_self_consistency`).

quantity	value [μm]	n
closure of the integrator, forward then back, median	2.3×10^{-6}	2000
closure of the integrator, worst case	22.4	2000
step convergence, 5 mm against 1 mm, median	2.8×10^{-5}	200
step convergence, worst case	2.8	200
label against the particle's next hit, short legs, median	11.9	100,660
the same, above 5 GeV	3.8	100,660
the same, below 2 GeV	38.8	100,660
exact scheme, $N = 64$, $q = 2$, $\Delta z = 80.9$ mm	0.147	1452
exact scheme, $N = 128$, $q = 8$, $\Delta z = 40.5$ mm	0.0115	1452
exact scheme, $N = 256$, $q = 16$, $\Delta z = 20.2$ mm	1.05×10^{-3}	1452
straight line, no magnet at all	450,542	1452
the simulated truth against the reference	1896	1452

The slopes say the same thing. Everything above is about position. The slope residuals behave identically, which is what a single physical cause requires. Over all 14,482 crossings with both charges pooled, the median absolute slope residual is 3.68×10^{-4} in t_x and 2.20×10^{-4} in t_y ; below 3 GeV it is 3.31×10^{-3} and 7.37×10^{-4} , and above 50 GeV it is 3.11×10^{-5} and 1.92×10^{-5} . Two features are worth naming. At low momentum the horizontal residual is about four times the vertical one in position and four to five times in slope, because the magnet bends in x and the momentum systematic therefore appears there and not in y . Above 20 GeV, where the systematic has largely gone, the two components converge, which is what a purely random cause requires: at 20–50 GeV the median absolute residuals are $204 \mu\text{m}$ in x against $219 \mu\text{m}$ in y and 6.19×10^{-5} in t_x against 5.69×10^{-5} in t_y .

What the reference does agree with. Table 11 collects the self-consistency numbers in one place, so that the size of the gap to the simulation can be read against the size of everything the reference *can* be checked against.

The reading is stark. Sub-micrometre closure, picometre step convergence and sub-micrometre agreement with the exact collocation scheme leave no room for integration error anywhere near the millimetre scale. The reference solves its equation essentially exactly. The equation is what is incomplete, and the $1736 \mu\text{m}$ median gap to the simulation, $1896 \mu\text{m}$ in the radial metric on the test split, is that incompleteness and nothing else. Figure 8 separates that gap into the charge-odd systematic and the charge-blind width, band by band.

What this means for the programme. The gap has three parts and a remainder, and they have different status.

1. **The start momentum is the production momentum**, not the momentum the particle had at the Upstream Tracker. This is a defect in my own data pipeline and not physics, and it is fixable cheaply: it shows as a charge-antisymmetric over-bend worth $+17.6$, $+17.3$ and $+12.2$ MeV in the three bands below 20 GeV.
2. **Energy loss along the crossing itself.** The equation holds q/p constant; a real particle loses energy continuously. This is deterministic and therefore addable to the equation,

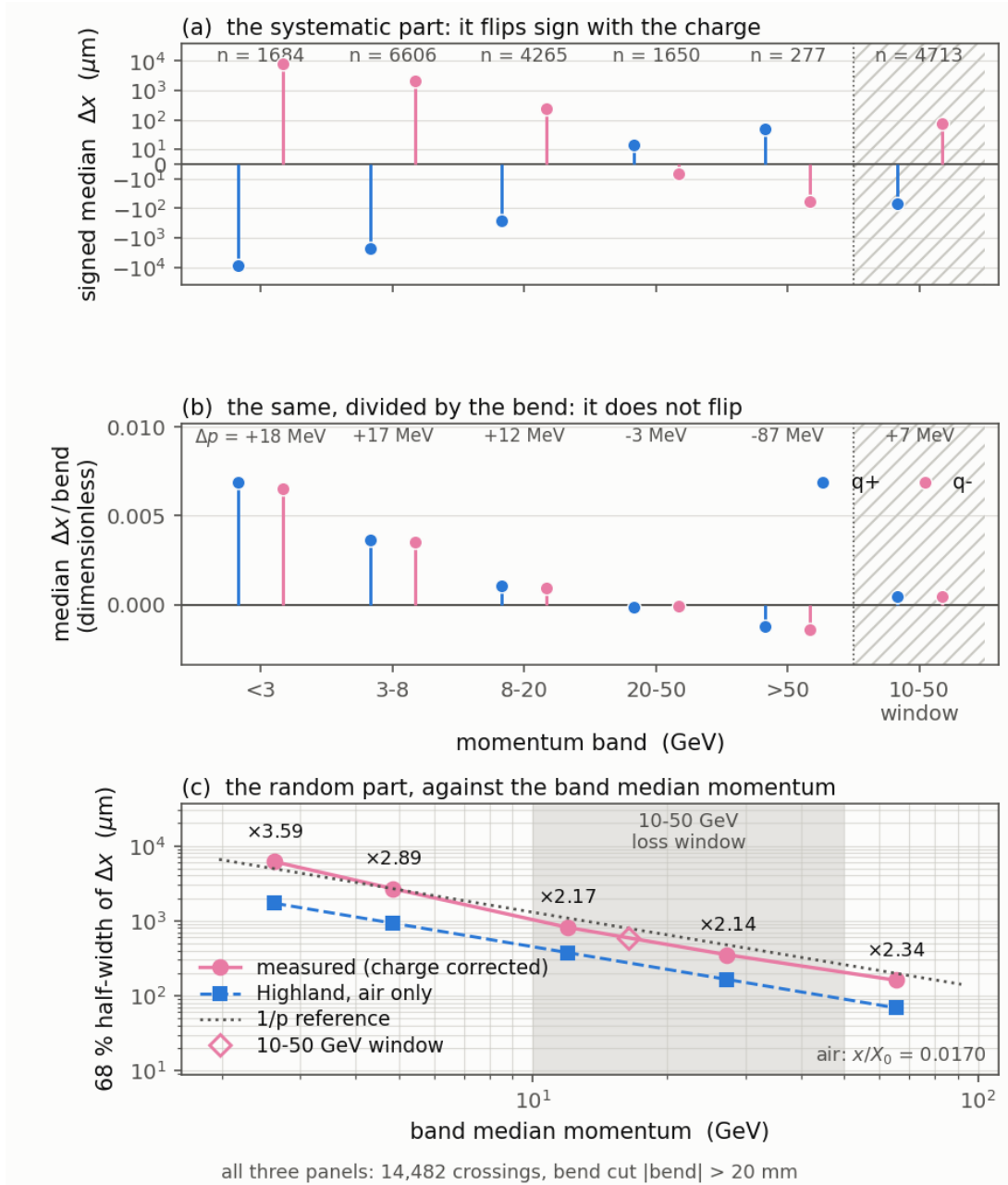


Figure 8: What separates the field-only reference from the real particle, on all 14,482 crossings. Panel (a): the systematic part, the signed median of Δx per momentum band drawn separately for positive and negative tracks, on a symmetric logarithmic axis so that the four orders of magnitude between the softest and the stiffest band are all visible; below 20 GeV the two charges sit on opposite sides of zero, which is what a charge-odd effect looks like. Panel (b): the same quantity divided by each track's own bend. The ratio is dimensionless and is plotted at its own scale, multiplied by nothing. It does not flip sign with the charge, which is the signature of a momentum defect, and the defect it implies is printed above each band to the nearest MeV: +18, +17 and +12 MeV below 20 GeV, then -3 and -87 MeV in the two stiffest bands and +7 MeV for the loss window. Panel (c): the random part, the charge-corrected 68 per cent half-width of Δx against the band's median momentum on logarithmic axes, beside Highland's prediction for the air between the two planes and a pure $1/p$ line anchored on the 3-8 GeV measurement; the measured width is a factor 2.14 to 3.59 wider than air alone over the five bands, and that ratio is printed at each of those five points. Source: `scripts/fig06_material_gap.py`, `results/fig06_material_gap.csv`.

which is what LHCb’s production extrapolator does in a separate material layer. Part of the 17 MeV belongs here rather than to the start plane, and only a rebuilt dataset will say how it splits.

3. **Multiple Coulomb scattering.** Random, charge-blind, falling as $1/p$, measured here at $582\,\mu\text{m}$ of half-width in the 10–50 GeV loss window and about 2.1 times the air-only estimate. This is stochastic and therefore irreducible for any deterministic reference. A deterministic extrapolator can predict its variance and propagate it as a covariance, which is exactly what a Kalman filter does with it, but it can never predict the individual deflection. No extra Runge–Kutta order, no extra collocation stage and no extra network capacity can touch it.
4. Separately from these three, the **momentum-growing residual above 20 GeV**, which I have not explained.

Nothing in the network studies moves because of any of this: the networks and the reference share the same equation and the same q/p , so every network-against-reference number in Section 5 is unaffected. What this subsection fixes is the level at which the programme stops being about integration and starts being about physics the equation was never given: about 1.7 mm in the median over this sample, $532\,\mu\text{m}$ in the loss window, and $153\,\mu\text{m}$ above 50 GeV.

5 Results

This section reports the experiment. It is long because there are two losses, three step lengths and six trained networks, and because the interesting part of the answer is not the headline number but the way the improvement is distributed. I take the pieces in the order in which they were measured. Section 5.1 lays out the sixteen networks trained under the pooled loss of Eq. (22) over a grid of step counts and stage counts, which is what fixed the three working points the rest of the paper uses. Section 5.2 is the pre-flight: where each of the two objectives actually spends itself, measured on a trained network before any reweighted network existed, together with the scan that chose the clamp. Section 5.3 says which of the six runs had finished training and which had not, because that determines what the comparison is allowed to claim. Section 5.4 is the comparison itself, three networks under two losses, and the test of the criterion that was fixed in advance. Section 5.5 takes the endpoint error apart by state component and by momentum band, Section 5.6 looks closely at the region around 5 GeV, and Section 5.7 decomposes what is left into per-step errors and their accumulation. Section 5.8 asks what shorter steps bought, Section 5.9 sets every number against the simulated truth rather than against the reference, and Section 5.10 reports what all of it cost to train and what a trained network costs to run.

Every number in this section is measured on the 1,452 test tracks unless it says otherwise, and every error is against the sixth-order reference of Section 2.7 carried to the first SciFi plane unless it says otherwise. The statistics are those defined in Section 2.8 and are not redefined here. The three networks that carry the comparison are named by their step count, their stage count and their step length — $N = 64$, $q = 2$, $\Delta z = 80.9\,\text{mm}$; $N = 128$, $q = 8$, $\Delta z = 40.5\,\text{mm}$; and $N = 256$, $q = 16$, $\Delta z = 20.2\,\text{mm}$ — and after this paragraph the step count and the stage count alone identify them.

5.1 The pooled loss over the grid of step counts and stage counts

The first question is whether chaining one network to itself works at all, and at what step length and stage count it works best under the objective the construction comes with; Table 12 is the sixteen-run answer.

Table 12: The sixteen networks trained under the pooled loss of Eq. (22), one per combination of step count N and stage count q , scored on the 1,452 test tracks. *Endpoint* is the chain of N applications from the track’s real Upstream Tracker state, radial metric at z_1 . *Ceiling* is the same q -stage scheme at the same step length, chained the same N times and solved directly by a root-finder with no network anywhere, from Table 6. *Single step* is one application of the same network from a reference state on an interior plane, median over all planes and tracks. **This table, and only this table, uses the stopped snapshot of 18 September 2026;** none of these runs had plateaued when it was taken (Section 5.3), so the ordering within the grid is a snapshot and not a ranking. The three combinations carried forward are marked $\star$. Source: `results/tab_pooled_grid.csv` and `results/tab_exact_ceiling.csv`.

N	q	Δz [mm]	endpoint med. [μm]	endpoint p95 [μm]	ceiling [μm]	single step [μm]	restarts
2	2	2588.9	9,250	39,347	9,270.9	1,430.8	109
2	4	2588.9	283.8	2,011	195.40	92.46	159
2	8	2588.9	188.1	1,428	7.472	62.53	130
2	16	2588.9	223.6	1,683	5.167	72.81	106
64	2	80.90	166.3	1,154	0.1465	0.4151	760 $\star$
64	4	80.90	172.4	1,051	0.4783	0.5568	714
64	8	80.90	204.2	1,249	0.1030	0.6477	657
64	16	80.90	194.2	1,207	0.0719	0.8103	546
128	2	40.45	207.0	1,206	0.5107	0.1509	760
128	4	40.45	195.6	1,246	0.1159	0.2567	713
128	8	40.45	158.6	1,090	0.0115	0.3086	599 $\star$
128	16	40.45	246.2	1,410	0.000176	0.3739	383
256	2	20.23	185.3	1,319	0.2284	0.0906	758
256	4	20.23	195.6	1,275	0.0295	0.0854	709
256	8	20.23	193.6	1,204	0.000426	0.1276	615
256	16	20.23	136.8	1,082	0.00105	0.0944	550 $\star$

Read it this way. The four rows at $N = 2$ are the control, and they reproduce the first paper’s central demonstration on a completely different geometry. At $N = 2$, $q = 2$ the chained network lands at $9,250\mu\text{m}$ against a ceiling of $9,271\mu\text{m}$: the two agree to 0.2 per cent. The network is not failing there. It is solving the equations it was given, and a two-stage Gauss–Legendre scheme taking two steps of 2589mm through the magnet is itself 9mm wrong. At $q = 4$ the ceiling drops to $195\mu\text{m}$ and the network sits at $284\mu\text{m}$, a factor 1.45 above it. From $q = 8$ onwards the ceiling falls through $7.5\mu\text{m}$ to $5.2\mu\text{m}$ while the network stops at 188 and $224\mu\text{m}$: the discretisation has stopped being the limitation and the optimiser has become it. That is the same crossing the first paper reported on its single frozen leg, and it happens here at the same stage count.

The twelve rows at $N \geq 64$ are the experiment proper, and the first thing to say about them is that the ceiling is irrelevant. The exact scheme chained 64, 128 or 256 times lands between $0.000176\mu\text{m}$ and $0.51\mu\text{m}$ of the reference, so every one of those twelve networks sits between two and six orders of magnitude above the answer its own objective is pointing at. Nothing in this part of the grid is limited by the numerical scheme; everything is limited by how well the network solves it. That is the single most useful fact in the table, because it means the rest of the paper is about optimisation and about the objective, not about discretisation error.

The second thing to say is that the twelve endpoint medians do not order themselves. They run from $136.8\mu\text{m}$ at $N = 256$, $q = 16$ to $246.2\mu\text{m}$ at $N = 128$, $q = 16$, a total spread of a factor 1.80, with no monotone trend in either N or q and with the best and the worst run sharing a stage count. The single-step column behaves quite differently: it falls smoothly with the step, from 0.42 – $0.81\mu\text{m}$ at $N = 64$ through 0.15 – $0.37\mu\text{m}$ at $N = 128$ to 0.085 – $0.13\mu\text{m}$ at $N = 256$. One application of the network gets steadily better as the step gets shorter, and the chain of N of them does not. That gap is the subject of Sections 5.7 and 5.8, and it is the reason the paper does not simply recommend the shortest step available.

It is worth one sentence on how small the single-step numbers are in context. Ignoring the

magnet and continuing along the incoming slopes over one step of 80.90 mm costs $95.2\ \mu\text{m}$; over 40.45 mm it costs $23.8\ \mu\text{m}$; over 20.23 mm it costs $5.96\ \mu\text{m}$. The single-step errors above are therefore between 0.44 and 0.85 per cent of the straight line at $N = 64$ and between 1.4 and 2.1 per cent at $N = 256$. A single application of any of these networks is a good step. It is the chaining that costs.

Three combinations were carried forward into the comparison, marked $\star$ in the table: $N = 64$ with $q = 2$, $N = 128$ with $q = 8$ and $N = 256$ with $q = 16$. The choice pairs each halving of the step with a doubling of the stage count, so that the three points differ in step length while keeping the per-step arithmetic growing in a controlled way, and the middle point is the grid's best run at that step length. The three were fixed before any reweighted network was trained. Because the runs in Table 12 were all stopped short, every one of the three was afterwards extended to its own plateau before being used as a counterpart, and it is those extended runs, not these, that appear in every later table.

5.2 Where each loss puts its weight

Before any reweighted network was trained, both objectives were evaluated on the same trained network and the same states, so that the difference between them could be described rather than assumed; this section reports that measurement, the ablations that attribute it to the three factors of Eq. (23), and the scan that chose the clamp.

The measurement is the one already quoted in Section 2.5 and is set out in full in Table 13. Eight thousand (track, plane) states were drawn evenly over the 64 start planes of the extended $N = 64$, $q = 2$ network, the residuals of Eq. (13) were formed once, and each state's share of the total loss was accumulated under five weighting modes.² Two of the five modes are the objectives the networks were actually trained under. The other three switch one factor of Eq. (23) off by replacing the quantity that varies with its own average, so that the overall size of the loss is unchanged and only its distribution moves. Those three are diagnostic only. **No network was ever trained under them**, and they appear here solely to say which factor does which part of the work.

Read it this way, one column at a time. The pooled loss spends 89.8 per cent of itself on tracks of 3–8 GeV, which are 45.8 per cent of the states, and 1.23 per cent on the 10–50 GeV window, which is 32.0 per cent of them. Its 20–50 GeV share is three hundredths of a per cent and its > 50 GeV share is nine thousandths. In z it is equally lopsided: the first quarter of the crossing, where an error has the longest lever arm and costs the most, carries 0.65 per cent, and the second quarter carries 89.4. And the concentration within all of that is extreme: the worst one per cent of states carry 98.3 per cent of the entire loss. This is not a defect of the implementation. It is what a normalisation by a pooled per-component spread does when the residuals themselves span two orders of magnitude across the momentum range.

The cost-weighted loss moves both distributions. The 3–8 GeV share falls from 89.8 to 38.3 per cent, the window share rises from 1.23 to 47.0, 20–50 GeV goes from 0.031 to 17.1 and > 50 GeV from 0.009 to 6.6. In z the first quarter rises from 0.65 to 23.3 per cent and the last

²An earlier record of this pre-flight exists and was taken on the stopped checkpoint of 18 September 2026 rather than on the extended network. The quantities that record stored are carried in the companion file of Figure 9 as a separate column: it gives, for the pooled loss, a rank correlation of 0.690 overall and 0.559 in the window against 0.684 and 0.532 here, a top-one-per-cent share of 97.9 against 98.3 per cent, and first- and last-quarter shares of 0.556 and 6.15 against 0.650 and 6.22 per cent; for the cost-weighted loss it gives 0.561 and 0.889, 83.8 per cent, and 21.2 and 1.75 per cent. Re-running the pre-flight on that checkpoint reproduces the stored record exactly, to zero deviation in every row, which is the check that the replayed random draw and the method here are the ones the original gate used. That record's momentum binning was the older 1/2/5/10/25/200 GeV edge set, so per-band shares in this paper's bands exist for the extended network only, and it also reported a larger in-window share for the cost-weighted loss than the extended network gives: 57.82 per cent against the 47.01 of Table 13, with the pooled loss at 1.509 per cent there against 1.23 here (`results/tab_preflight_snapshot_record.csv`). The weights are identical in the two; only the residuals they weigh have moved, because the network has been trained further.

Table 13: The pre-flight: the share of the loss carried by each momentum band and by each quarter of the crossing, on 8,000 (track, plane) states drawn evenly over the 64 start planes of the extended pooled-loss $N = 64$, $q = 2$, $\Delta z = 80.9$ mm network. All figures are percentages of the whole loss. The first two columns are the two trained objectives; the last three switch one factor of Eq. (23) off and are **diagnostic only, never trained**. The 10–50 GeV loss window overlaps two of the five bands and is reported as an extra row. The bottom three rows are the rank correlation of a state’s share with what that state’s error actually costs at z_1 — a diagnostic computed against the reference, which enters neither loss — and the share carried by the worst one per cent of states. Source: `results/tab_preflight_shares.csv` and `results/paper_numbers.json` (preflight).

	states [%]	pooled loss [%]	cost-weighted loss [%]	lever arm off [%]	track factor off [%]	window off [%]
<i>by momentum band</i>						
< 3 GeV ($n = 907$)	11.34	8.83	4.16	2.85	6.02	6.48
3–8 ($n = 3,665$)	45.81	89.79	38.32	34.71	83.11	74.55
8–20 ($n = 2,360$)	29.50	1.35	33.84	38.21	9.81	10.22
20–50 ($n = 904$)	11.30	0.031	17.05	17.06	0.87	6.01
> 50 ($n = 164$)	2.05	0.009	6.63	7.16	0.19	2.74
10–50 (loss window) ($n = 2,559$)	31.99	1.23	47.01	51.04	8.95	15.04
<i>by quarter of the crossing in z</i>						
quarter 1 (leaves z_0)	25	0.65	23.35	9.89	7.70	9.59
quarter 2	25	89.43	56.91	44.56	85.11	82.51
quarter 3	25	3.69	18.36	28.01	6.11	6.97
quarter 4 (arrives at z_1)	25	6.22	1.38	17.54	1.08	0.93
<i>how well the loss ranks states by what they cost</i>						
rank corr. with true cost	—	+0.684	+0.583	+0.260	+0.853	+0.766
the same, within 10–50 GeV	—	+0.532	+0.896	+0.554	+0.953	+0.888
share carried by the worst 1 %	1	98.30	80.85	78.48	94.25	91.13

quarter falls from 6.2 to 1.4, which is the lever arm doing exactly what Eq. (24) says it should: weight is moved towards the steps whose errors have the furthest to travel. The one-per-cent concentration falls from 98.3 to 80.8 per cent, so the objective is also markedly less dominated by a handful of states than the one it replaces — though 80.8 per cent is still a concentrated objective, and I say so rather than pretending otherwise.

The three ablation columns say which factor is responsible for which move. Switching the lever arm off leaves the momentum redistribution almost intact — the window share is 51.0 per cent, slightly *higher* than the full weight’s 47.0 — but destroys the z redistribution, taking the first quarter to 9.9 per cent and the last quarter to 17.5. So the lever arm is the whole of the z behaviour and none of the momentum behaviour. Switching the per-track factor off sends the 3–8 GeV share back to 83.1 per cent and the window share down to 8.9, which is most of the way back to the pooled loss: the per-track factor a_n , not the momentum window, is what removes the momentum skew. Switching the window off leaves 74.6 per cent in 3–8 GeV and 15.0 in the window — better than pooled but far short of the full weight. The reading is that the two momentum factors are not redundant: dividing by the track’s own bend does the bulk of the work and the window then finishes it, and neither alone reaches 47 per cent.

The last three rows need reading carefully, because they do not all point the same way. Within the 10–50 GeV window, the cost-weighted loss ranks states by what they truly cost at z_1 far better than the pooled loss does: +0.896 against +0.532. Overall, it ranks them slightly *worse*: +0.583 against +0.684. That is not a failure of the weight, it is the weight working. A state’s true endpoint cost is largest on the soft tracks, which the objective deliberately discounts; an objective that has been told to care about 10–50 GeV will necessarily agree less well with a cost ranking that is dominated by tracks below 3 GeV. The number to read is the in-window one, and it is the number the weight was designed to move.

Read the clamp scan this way. Unclamped, the 164 states above 50 GeV — two per cent of the sample — take 21.7 per cent of the loss, which is the failure mode Section 2.6 predicted

Table 14: The clamp scan. The factor a_n of Eq. (25) is clamped to $[1/c, c]$ of its median over the batch; the table varies c over $\{\text{off}, 2, 3, 5, 10\}$ on the same 8,000 states. All shares are percentages of the whole cost-weighted loss. The runs used $c = 5$, in bold. Source: `results/tab_clamp_scan.csv`.

clamp c	< 3 [%]	3–8 [%]	8–20 [%]	20–50 [%]	> 50 [%]	10–50 (window) [%]	rank corr. in window	worst 1 % [%]
n	907	3,665	2,360	904	164	2,559	—	—
off	0.92	28.15	29.02	20.24	21.67	45.93	+0.880	76.39
2	11.33	73.70	13.20	1.31	0.46	12.82	+0.953	94.27
3	8.61	59.59	24.98	5.04	1.78	27.12	+0.937	89.33
5	4.16	38.32	33.84	17.05	6.63	47.01	+0.896	80.85
10	0.96	28.69	29.58	20.63	20.15	46.81	+0.880	76.33

Table 15: The six runs against the plateau rule of Section 2.8: the median validation endpoint error of the last ten rounds no more than 5 per cent below the median of the ten before, holding at each of the last three rounds. *Flat?* is the verdict at the run’s last round. *First held* is the earliest round at which the rule fired, which for a run that is not flat at its last round means it fired and later stopped firing. *Last ten vs previous ten* is the change between those two ten-round medians, negative meaning still improving. The *validation headline* is the median over the last ten rounds with the round-to-round spread beside it, on the 1,463 validation tracks, max metric. The three pooled-loss runs are the extended runs, carrying respectively 500, 1,000 and 500 restarts and 20, 40 and 20 rounds beyond the snapshot of Table 12. Source: `results/tab_headline_pairs.csv` and `results/tab_extended_vs_snapshot.csv`.

network	loss	rounds	restarts	flat?	first held	last ten vs previous ten	validation headline [μm]
$N = 64, q = 2$	pooled	50	1,260	yes	38	−4.19 %	126.6 ± 17.5 %
$N = 64, q = 2$	cost-weighted	40	1,000	yes	35	+2.82 %	89.3 ± 11.7 %
$N = 128, q = 8$	pooled	63	1,599	no	28	−7.24 %	128.7 ± 12.4 %
$N = 128, q = 8$	cost-weighted	40	1,000	yes	35	+0.50 %	107.0 ± 10.8 %
$N = 256, q = 16$	pooled	42	1,050	no	34	−10.94 %	149.5 ± 9.8 %
$N = 256, q = 16$	cost-weighted	60	1,807	yes	60	+0.55 %	86.4 ± 25.9 %

from the $|q/p|^2$ scaling. Clamping at $c = 10$ barely changes anything, 20.1 per cent. Clamping hard, at $c = 2$, over-corrects in the opposite direction: a_n is flattened to the point where the objective reverts towards its pooled behaviour, 73.7 per cent of the loss returns to 3–8 GeV, the window share collapses to 12.8 per cent and the one-per-cent concentration rises to 94.3. The in-window rank correlation is highest there, +0.953, but that is precisely the point at which the in-window share is smallest, so ranking the window well while spending almost nothing on it is not what was wanted. The criterion was the window share subject to the tails being held, and $c = 5$ is where that is met: 47.0 per cent inside the window, the maximum over the scan, with > 50 GeV held to 6.6 per cent. Leaving the clamp off, or setting it at 10, gives a window share of 45.9 and 46.8 per cent and a slightly lower one-per-cent concentration, 76.4 and 76.3 against 80.8, but lets the > 50 GeV band take 21.7 and 20.1 per cent of the objective, which is the thing the clamp exists to prevent. The clamp was fixed at five on that basis and not revisited. Figure 9 draws the whole of this section.

5.3 Convergence to plateau

The comparison that follows is only meaningful if both sides of it had finished training, so this section applies the plateau rule of Section 2.8 to all six runs and states, for each, whether it had.

Read it this way. All three cost-weighted runs are flat at their last round. Their ten-round medians moved by +2.8, +0.5 and +0.5 per cent over the preceding ten, which is to say they moved by nothing, and each fired the rule several rounds before it was stopped — at rounds 35, 35 and 60 out of 40, 40 and 60. Of the three pooled-loss counterparts only one is flat. The $N = 64, q = 2$ run is: it fired at round 38 of 50 and its last ten rounds sit 4.2 per cent below

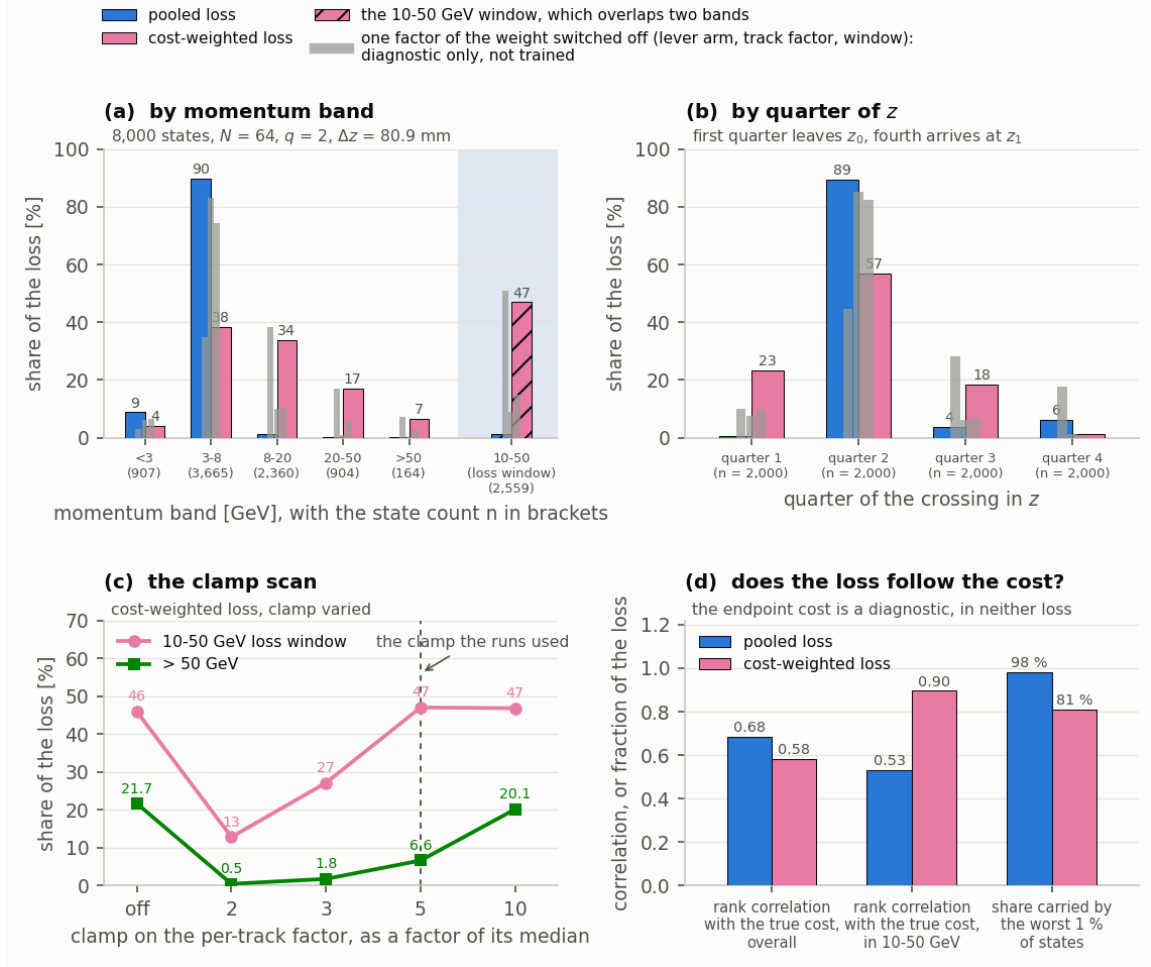


Figure 9: Where each loss spends itself, on the 8,000 pre-flight states drawn evenly over the 64 start planes of the extended pooled-loss $N = 64$, $q = 2$, $\Delta z = 80.9$ mm network. (a) The share of the loss carried by each momentum band under the pooled loss and under the cost-weighted loss, with the 10–50 GeV window drawn apart on a shaded background because it overlaps two of the five bands, and the state count n printed under each label; the three thin grey bars behind each pair are the same shares with the lever arm, the per-track factor or the momentum window switched off, which are diagnostics and were never trained. (b) The same by quarter of the crossing in z , the first quarter being the one that leaves z_0 . (c) The clamp scan of Table 14: the share carried by the window and by the > 50 GeV band as the clamp on the per-track factor is tightened, with the value the runs used marked. (d) How well each objective ranks states by what their error actually costs at the endpoint, overall and within the window, beside the share of the whole loss carried by the worst one per cent of states; the cost is a diagnostic and enters neither objective. Source: `scripts/fig08.loss.shares.py`.

the ten before, inside the rule’s five per cent. **The other two are not.** The $N = 128$, $q = 8$ run was still improving at 7.2 per cent per ten rounds when it stopped at round 63, and the $N = 256$, $q = 16$ run at 10.9 per cent per ten rounds when it stopped at round 42 — and that is after both had already been extended well past the 18 September snapshot, by 1,000 and 500 restarts respectively. Both fired the rule once, at rounds 28 and 34, and then resumed falling, which is exactly the behaviour the rule’s one-sidedness is there to catch.

The consequence has to be stated before the comparison and not after it. Two of the three pooled-loss rows in every later table are **lower bounds on how good that network would eventually be**, not converged numbers. If those two runs were trained to a real plateau their endpoint errors would fall further, and the margin the cost-weighted loss shows over them at $N = 128$ and $N = 256$ would shrink by an unknown amount. Extrapolating the observed rate is not a substitute for running them, and I do not do it. The conclusions of this paper therefore

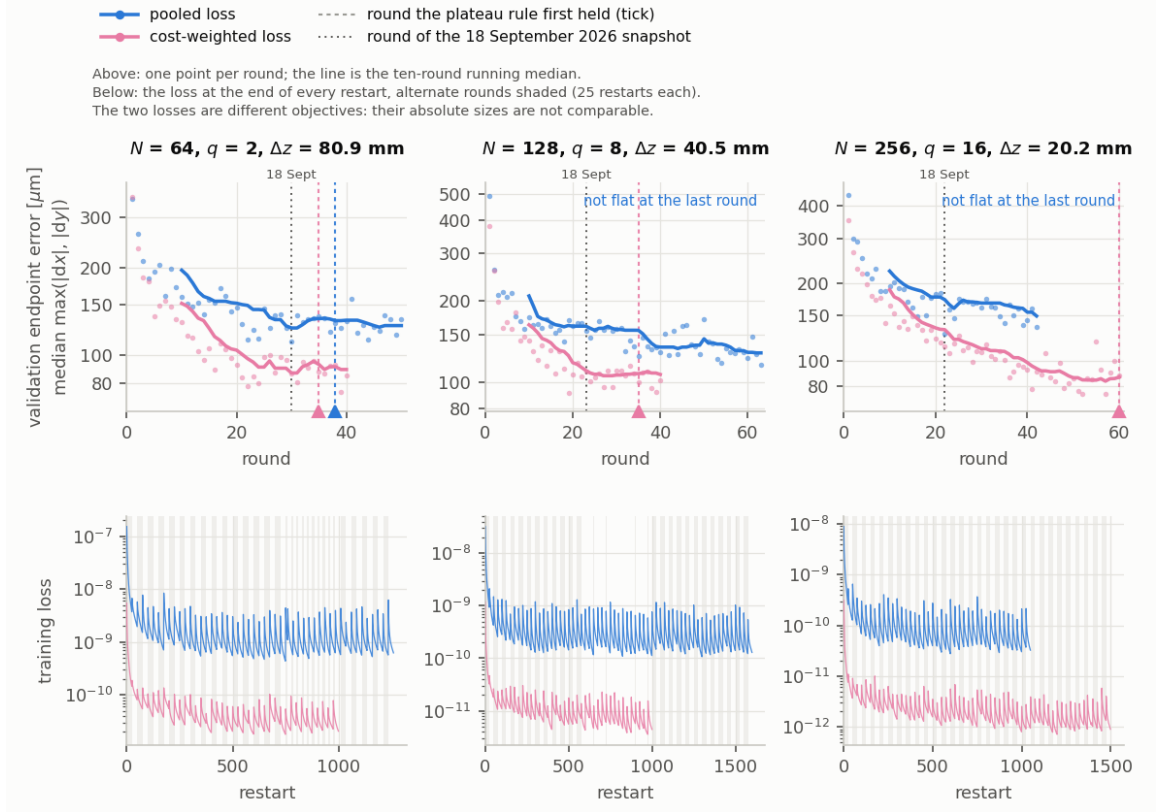


Figure 10: How the six runs converged, one column per network. Top row: the validation endpoint error — the median over the 1,463 validation tracks of $\max(|\Delta x|, |\Delta y|)$ at z_1 — round by round as points, with the ten-round running median over them as a line, for the pooled-loss run and the cost-weighted-loss run together. On each curve for which the plateau rule holds at its last round, a tick marks the round at which the rule first held; the two pooled-loss curves that do not satisfy the rule at their last round carry no tick and are annotated “not flat”. The dotted vertical line on each pooled-loss curve is the round of the 18 September 2026 snapshot, so everything to the right of it is the extension this paper uses. The axis is the ordinal round, which is what the tables count. Bottom row: the training loss at the end of every restart on a logarithmic axis, with alternate rounds shaded so that the round boundaries and the redrawn batch of 32,000 collocation states at each are visible. The two losses are different objectives and their absolute sizes are not comparable; only the shape of each curve is. Source: `scripts/fig09_convergence.py`.

rest on the $N = 64, q = 2$ pair, which is the pre-registered one and is the only pair in which both sides are flat; the other two pairs are reported in full, with this caveat attached to every row, and are read as supporting rather than as independent evidence.

One further number in the table deserves attention because it affects how the headline should be read. The validation headline carries a round-to-round spread, and that spread is not small: 17.5, 11.7, 12.4, 10.8, 9.8 and 25.9 per cent over the six runs. The last of those, the $N = 256, q = 16$ cost-weighted run, is the widest in the set, and its plateau verdict rests on a rule applied to a noisier signal than the others. Any difference between two runs that is comparable with these numbers is not a difference at all, which is why the criterion of Section 2.8 was written as a band to be cleared rather than as an inequality between two medians.

Figure 10 shows the histories.

5.4 The headline: three networks, two losses

This is the comparison the paper exists to make: the same network, at the same three step lengths, trained twice and scored identically, against the criterion that was fixed before the runs were submitted.

Read it this way. Take the pre-registered pair first. Under the pooled loss the $N = 64, q = 2$

Table 16: The headline comparison. Three networks, each trained under the pooled loss of Eq. (22) and under the cost-weighted loss of Eq. (23), scored on the same 1,452 test tracks against the reference at z_1 . Positions are radial medians in micrometres; the per-step slope errors are medians over all steps and all test tracks of the local deviation of one application, and are dimensionless. *In window* is the 10–50 GeV loss window. **The pooled-loss runs are the runs extended past the 18 September snapshot, and two of the three — $N = 128, q = 8$ and $N = 256, q = 16$ — were still improving at their last round** (Table 15); their rows are therefore not converged and the differences in those two pairs are upper bounds. Source: `results/tab_headline_pairs.csv`, `results/tab_reweighted_runs.csv` and `results/tab_anatomy.csv`.

network	loss	median [μm]	p95 [μm]	> 1 mm [%]	in window [μm]	< 3 GeV [μm]	per-step t_x / t_y [10^{-7}]	wall [h]
$N = 64, q = 2$	pooled	145.7	898	4.41	108.1	353.7	13.0 / 7.2	31.2
	cost-weighted	88.8	1,399	7.16	24.8	615.1	7.3 / 9.7	21.6
$N = 128, q = 8$	pooled	122.2	928	4.41	70.5	333.7	5.5 / 3.2	54.7
	cost-weighted	104.6	1,611	9.57	27.3	900.2	3.8 / 5.0	32.8
$N = 256, q = 16$	pooled	147.7	1,032	5.23	89.6	373.5	3.0 / 1.8	63.2
	cost-weighted	92.2	1,230	6.54	34.6	556.7	1.9 / 2.4	100.4

network reaches a median endpoint error of $145.7 \mu\text{m}$; under the cost-weighted loss, $88.8 \mu\text{m}$. That is a fall of 39.1 per cent. Inside the 10–50 GeV window the fall is from 108.1 to $24.8 \mu\text{m}$, a factor 4.37. Both of those are improvements, both are large, and neither is the whole story. Over the same tracks the 95th percentile rises from 898 to $1,399 \mu\text{m}$, a factor 1.56; the fraction of tracks ending more than a millimetre from the reference rises from 4.41 to 7.16 per cent, from 64 tracks to 104; and the median for tracks below 3 GeV rises from 353.7 to $615.1 \mu\text{m}$, a factor 1.74. The cost-weighted loss made the typical track better, made the window it was aimed at dramatically better, and made the soft tracks and the tail worse. That is a trade and not a free improvement, and it is the trade the objective was designed to make.

The same shape appears at the other two step lengths, with the caveat of Section 5.3 attached. At $N = 128, q = 8$ the median falls from 122.2 to $104.6 \mu\text{m}$, only 14.4 per cent, while the window falls from 70.5 to $27.3 \mu\text{m}$, a factor 2.58; the p95 rises by a factor 1.74 and the sub-3 GeV median by a factor 2.70, the largest degradation anywhere in the table. At $N = 256, q = 16$ the median falls from 147.7 to $92.2 \mu\text{m}$, 37.6 per cent, and the window from 89.6 to $34.6 \mu\text{m}$, a factor 2.59, with the p95 rising by only a factor 1.19. The in-window gain is the one consistent feature: a factor between 2.6 and 4.4 at all three step lengths.

The per-step slope errors are worth reading across rather than down. Within each pair the cost-weighted loss improves the x slope — the bending plane — and worsens the y slope: 13.0 to 7.3 and 7.2 to 9.7, in units of 10^{-7} , at $N = 64$; 5.5 to 3.8 and 3.2 to 5.0 at $N = 128$; 3.0 to 1.9 and 1.8 to 2.4 at $N = 256$. Section 5.7 takes that apart. Down the table, both slopes fall by roughly a factor two for each halving of Δz , which is what a well-behaved step should do, and the endpoint medians do not follow them at all; Section 5.8 is about that.

Read the band table this way. There is a crossover and it sits between 3–8 and 8–20 GeV. Below it the cost-weighted network is worse in five of the six comparisons, the exception being 3–8 GeV at $N = 256$, where the median falls from 200.4 to $177.7 \mu\text{m}$, a reduction of 11.3 per cent; above it, it is better in all nine. At $N = 64$ the ratios run 0.58, 0.91, 2.59, 9.01 and 7.62 across the five bands from soft to hard — that is, 1.7 times worse below 3 GeV and nine times better at 20–50 GeV. At $N = 128$ they run 0.37, 0.72, 1.83, 3.31, 5.33, and at $N = 256$, 0.67, 1.13, 2.29, 3.89, 4.64. The > 50 GeV column holds only 30 test tracks and is quoted here for completeness; its medians move by factors of 5 to 8 and a band of that size cannot support a quantitative claim, so I read it only as agreeing in direction with the 20–50 GeV band beside it. The 95th percentiles tell the same story more sharply: above the crossover the tail improves along with the median, by a factor of up to 5.9 at 20–50 GeV; below it the tail is where the whole cost is paid, and at $N = 128$ the sub-3 GeV p95 more than doubles, to 3.67 mm.

Table 17: The same six runs, radial endpoint median per momentum band, with the 95th percentile beside it. The five bands partition the test set; the 10–50 GeV loss window overlaps two of them and is reported as an extra row. n is the number of test tracks in the band. The > 50 GeV row holds 30 tracks and is read cautiously throughout this paper. The pooled-loss rows for $N = 128$, $q = 8$ and $N = 256$, $q = 16$ are not converged (Table 15). Source: `results/tab_headline_bands.csv`.

band [GeV]	n	$N = 64, q = 2$		$N = 128, q = 8$		$N = 256, q = 16$	
		pooled	cost-w.	pooled	cost-w.	pooled	cost-w.
<i>median, μm</i>							
< 3	164	353.7	615.1	333.7	900.2	373.5	556.7
3–8	655	160.4	175.5	153.7	214.1	200.4	177.7
8–20	433	86.3	33.4	79.2	43.4	106.5	46.6
20–50	170	145.5	16.2	70.2	21.2	78.6	20.2
> 50	30	170.7	22.4	168.0	31.5	216.0	46.6
10–50 (loss window)	487	108.1	24.8	70.5	27.3	89.6	34.6
<i>95th percentile, μm</i>							
< 3	164	1,390	2,768	1,537	3,668	1,857	2,866
3–8	655	1,035	1,513	1,106	1,626	1,247	1,213
8–20	433	368	210	362	273	455	246
20–50	170	373	64	308	67	301	76
> 50	30	1,111	124	521	77	742	111
10–50 (loss window)	487	353	130	306	158	389	142

The pre-registered criterion. Section 2.8 fixed what success would have to look like, and the test is as follows. The first condition was that the $N = 64$, $q = 2$ network’s test radial error clear a ± 13 per cent run-to-run band downward against its pooled-loss counterpart. It falls by 39.1 per cent, from 145.7 to $88.8\mu\text{m}$. The band was set at ± 13 per cent in advance; the counterpart’s actual round-to-round spread over its last ten rounds, measured after the fact, is ± 17.5 per cent (Table 15), and the fall clears that too. **The first condition holds.** The second condition was that the median per-step slope error fall below 1.6×10^{-6} . That threshold is the earlier record’s “0.0016”, which was quoted in units of 10^{-3} of slope and is 1.6×10^{-6} as a dimensionless number; it was the $N = 64$, $q = 2$ network’s own value at the time the criterion was written. The cost-weighted network reaches 7.34×10^{-7} in t_x , a factor 2.18 below it, and 2.25×10^{-7} within the loss window. **The second condition holds.** For completeness, the extended pooled-loss counterpart has itself since fallen to 1.30×10^{-6} , below the threshold as well, so the second condition is the weaker of the two: the first is what distinguishes the two losses.

A partial result had been defined in advance as an in-band improvement without an overall one, and a null result as nothing clearing the band. Neither applies. The outcome is the full one, with the pre-registered price attached: the improvement is real, it is concentrated where the weight was put, and it is paid for below 3 GeV and in the tail. Figures 11 and 12 show both halves.

5.5 Per component, per band

A radial median hides which of the two transverse directions the error is in, and that turns out to be the most informative thing about this result, so this section splits the endpoint error into its four predicted components.

Read the two tables together. The whole of the gain is in x , and x is the bending plane: the magnet’s field is close to vertical, so it deflects the particle horizontally, and the correction the surrogate has to reproduce lives almost entirely in x . Over all tracks the median $|\Delta x|$ falls from 110.7 to $37.5\mu\text{m}$, a factor 2.95, while the median $|\Delta y|$ goes from 52.3 to $55.2\mu\text{m}$, which is 5.6 per cent *worse*. Inside the loss window the two fall together — 86.9 to $11.6\mu\text{m}$ in x and 39.6 to $13.9\mu\text{m}$ in y — but the ordering has swapped: under the pooled loss the x error is 2.2 times the

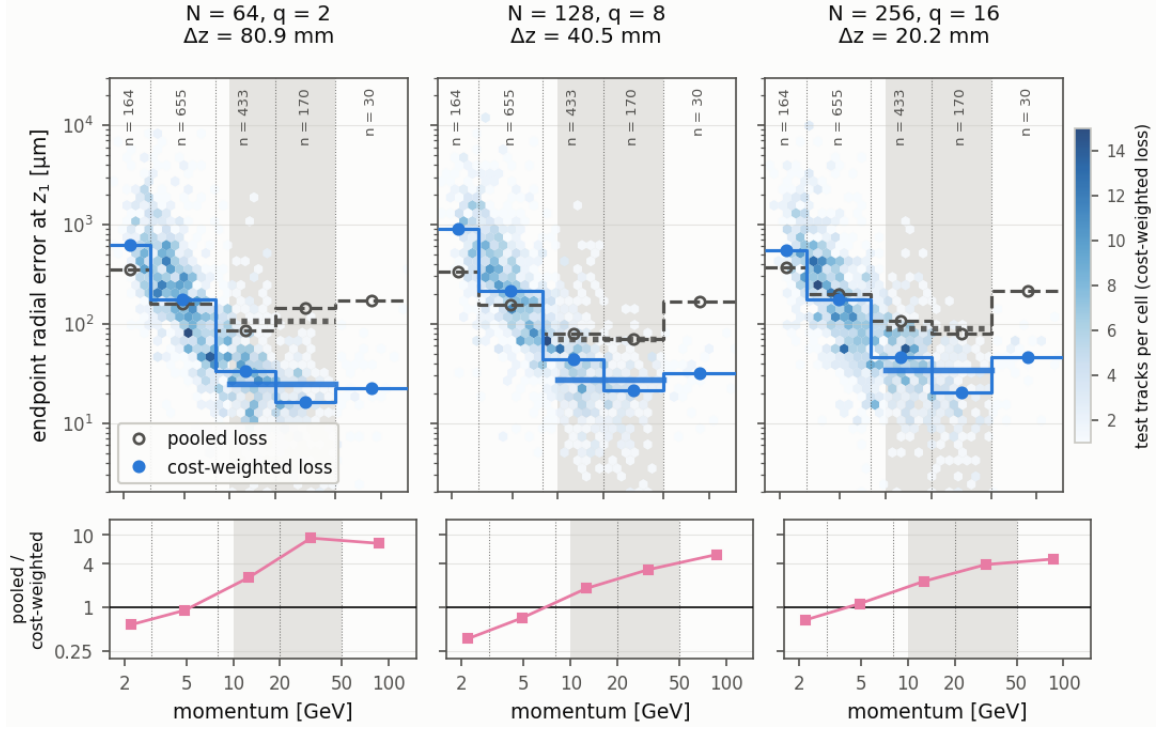


Figure 11: The endpoint error against momentum, one column per network. Top row: the radial deviation of every test track from the reference at $z_1 = 7826.0$ mm, in micrometres on a logarithmic axis, against the track’s momentum, also logarithmic. The density cloud is the cost-weighted-loss run; over it the per-band median of both losses is drawn as a step line with a marker at each band centre. The four band edges are thin vertical lines, the number of test tracks in each band is printed along the top, and the 10–50 GeV loss window is the shaded span. Bottom row: the ratio of the two per-band medians, pooled over cost-weighted, on a logarithmic axis with unity marked; the curve crosses unity between 3–8 and 8–20 GeV in all three networks. No slopes appear in this figure. Source: `scripts/fig10_headline_vs_p.py`.

Table 18: The position components of the endpoint error for the $N = 64, q = 2, \Delta z = 80.9$ mm network under both losses, per momentum band. *med* is the median of $|\Delta|$, *sgn* the median of the signed deviation and *hw* the 68 per cent half-width $(q_{84} - q_{16})/2$ of the signed deviation, all in micrometres. A signed median close to the median magnitude means a systematic offset; one much smaller means scatter. The > 50 GeV row holds 30 tracks. Source: `results/tab_headline_components.csv`.

band [GeV]	<i>n</i>	pooled loss						cost-weighted loss					
		Δx			Δy			Δx			Δy		
		med	sgn	hw	med	sgn	hw	med	sgn	hw	med	sgn	hw
all	1452	110.7	−39.4	164.7	52.3	−15.5	86.4	37.5	−3.6	97.6	55.2	+2.9	139.8
< 3	164	249.7	+7.6	382.4	183.4	−25.6	275.3	323.3	+139.1	479.0	393.1	+119.6	725.6
3–8	655	118.8	−29.1	192.7	63.5	−10.4	102.7	84.3	−5.6	148.0	118.3	+7.2	233.3
8–20	433	62.9	−38.6	90.6	40.8	−18.8	53.3	14.4	−1.6	24.5	24.3	+1.1	41.7
20–50	170	134.1	−98.6	135.8	37.5	−27.5	47.2	10.4	−8.0	13.9	6.2	+0.2	12.3
> 50	30	145.6	−60.4	236.1	23.9	+3.4	32.7	16.6	−12.6	25.2	10.4	+5.1	14.2
10–50	487	86.9	−54.4	105.9	39.6	−22.8	49.5	11.6	−6.3	17.5	13.9	+2.0	24.2

y error in the window, and under the cost-weighted loss the y error is the larger of the two by a factor 1.20. What limits the reweighted network in its own window is now the non-bending plane.

The signed medians are the sharper statement. In the window the pooled-loss network sits at $−54.4 \mu\text{m}$ in x against a median magnitude of 86.9: two thirds of its typical error is a common offset in one direction, which is a systematic under-bend and not scatter. The cost-weighted network sits at $−6.3 \mu\text{m}$ against 11.6. The absolute offset has fallen by a factor 8.7, in step

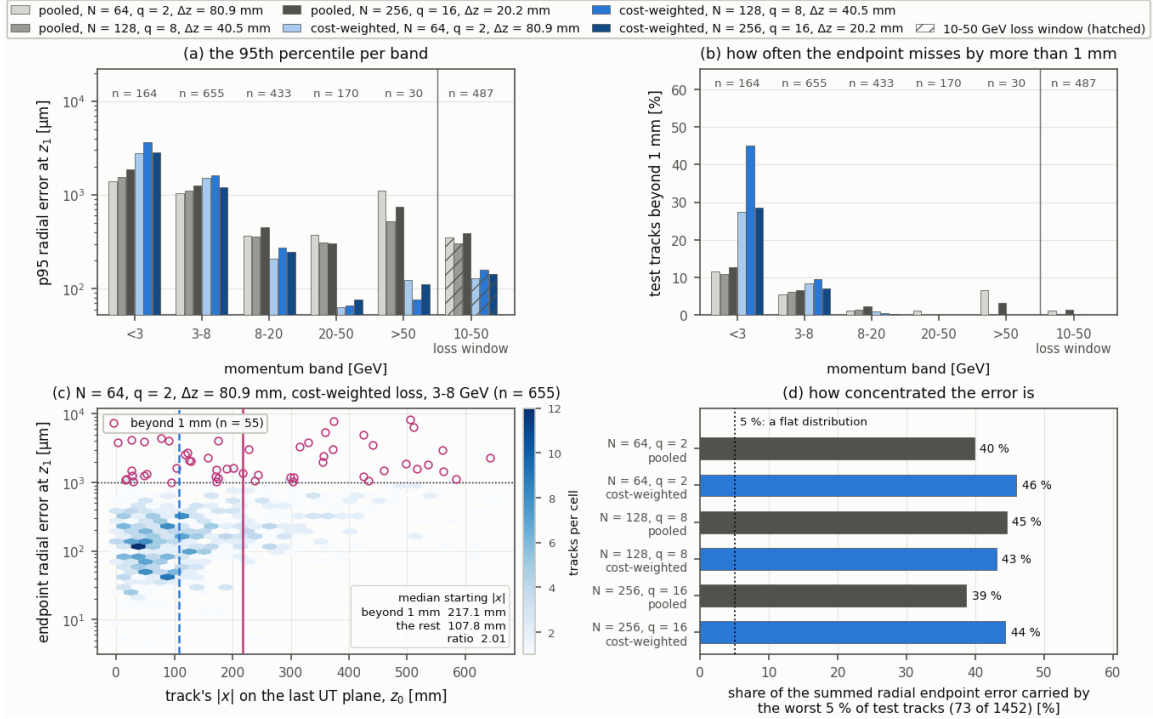


Figure 12: The tail. (a) The 95th percentile of the radial endpoint deviation per momentum band for the six runs, logarithmic, with the three pooled-loss networks in grey darkening with N beside the same three under the cost-weighted loss in blue; the 10–50 GeV loss window is the hatched group at the right and n is printed per band. (b) The fraction of test tracks whose radial endpoint error exceeds one millimetre, per band, on a linear axis. (c) Where the tail tracks start: the radial endpoint error against the track’s $|x|$ on the last Upstream Tracker plane, for the $N = 64, q = 2, \Delta z = 80.9$ mm network under the cost-weighted loss in the 3–8 GeV band; the tracks beyond one millimetre are circled and the median starting $|x|$ of the tail and of the rest are drawn and annotated. (d) How concentrated the error is: the share of the summed radial endpoint error carried by the worst five per cent of test tracks, 73 of 1,452, for all six runs, against the five per cent a flat distribution would give. Source: `scripts/fig15.tail.py`.

Table 19: The slope components of the same endpoint error, same network, same bands. t_x and t_y are dx/dz and dy/dz and are dimensionless; every entry is a pure number and nothing here is scaled. *med* is the median of $|\Delta|$, *sgn* the median of the signed deviation, *hw* the 68 per cent half-width, all $\times 10^{-5}$. Source: `results/tab.headline.components.csv`.

band [GeV]	n	pooled loss						cost-weighted loss					
		Δt_x			Δt_y			Δt_x			Δt_y		
		med	sgn	hw	med	sgn	hw	med	sgn	hw	med	sgn	hw
all	1452	3.68	−1.62	5.48	2.14	−0.30	3.52	3.23	−1.08	7.65	3.89	+0.79	10.02
< 3	164	9.06	−0.61	14.18	5.60	+0.99	8.54	25.39	−1.81	41.40	34.67	+3.54	67.81
3–8	655	3.48	−1.50	5.89	2.68	−0.16	4.30	6.98	−3.59	12.20	8.98	+3.00	17.16
8–20	433	2.82	−1.66	3.49	1.59	−0.32	2.30	1.37	−0.68	1.88	1.91	+0.74	2.60
20–50	170	4.03	−2.99	4.50	1.69	−1.25	1.94	0.85	−0.64	1.10	0.85	+0.19	1.20
> 50	30	5.57	−0.11	9.01	1.18	−0.08	1.49	0.99	−0.40	1.62	0.49	+0.30	0.57
10–50	487	3.42	−2.08	3.89	1.55	−0.74	2.14	1.06	−0.75	1.35	1.26	+0.43	1.88

with the error itself. I want to be careful about how far that generalises. As a fraction of the median magnitude the offset is 0.63 before and 0.54 after at $N = 64$, which is barely a change; at $N = 128$ it goes from $+17.9$ against $53.9\ \mu\text{m}$ to $+0.15$ against $13.7\ \mu\text{m}$, a fraction of 0.33 falling to 0.011, and at $N = 256$ from $+30.3$ against $68.9\ \mu\text{m}$ to -2.7 against $12.3\ \mu\text{m}$, 0.44 falling to 0.22. The sign of the pooled-loss offset is also not common to the three: it is negative at $N = 64$ and positive at $N = 128$ and $N = 256$. So the defensible claim is that the cost-weighted loss reduces the in-window offset in absolute terms in all three pairs and reduces it relative to the error's own size in two of the three; it is not that it removes a universal under-bend.

The slope table repeats the position table one derivative down, which is what Section 5.7 would lead one to expect. In the window the median $|\Delta t_x|$ falls from 3.42×10^{-5} to 1.06×10^{-5} and $|\Delta t_y|$ from 1.55×10^{-5} to 1.26×10^{-5} , so again both improve and again y becomes the larger. Below 3 GeV both slopes get much worse, 9.06 to 25.4 and 5.60 to 34.7 in units of 10^{-5} , which is the same trade seen in the positions and is where the soft-track degradation of Table 17 comes from. The 68 per cent half-widths are consistently about 1.5 times the median magnitude throughout, which is what a roughly symmetric distribution with a modest tail gives, and they move with the medians rather than independently.

The other two networks behave the same way and are summarised rather than tabulated in full. At $N = 128$, $q = 8$ the median $|\Delta x|$ over all tracks falls from 81.7 to $51.3\ \mu\text{m}$ and $|\Delta y|$ rises from 59.2 to $61.2\ \mu\text{m}$; in the window x falls from 53.9 to $13.7\ \mu\text{m}$ and y from 34.7 to $16.9\ \mu\text{m}$. At $N = 256$, $q = 16$ the overall $|\Delta x|$ falls from 103.4 to $37.6\ \mu\text{m}$ while $|\Delta y|$ falls only from 69.7 to $61.3\ \mu\text{m}$; in the window x falls from 68.9 to $12.3\ \mu\text{m}$ and y from 40.4 to $27.0\ \mu\text{m}$. In all three pairs the x error improves by between 1.6 and 7.5 times and the y error by between 0.95 and 2.9, and in all three the cost-weighted network's in-window error is dominated by y . That is a structural statement about the result and it is the one the discussion returns to. Figure 13 draws all six runs and all four components together.

5.6 Around 5 GeV

The region around 5 GeV was asked for specifically, it sits below the crossover of Section 5.4, and it is where the cost-weighted loss is paying rather than earning, so it is worth a section of its own rather than a row in a band table.

Read it this way. In the 4–6 GeV band the honest single number for the core of the distribution is the 68 per cent half-width, not the median magnitude and certainly not the RMS. Under the cost-weighted loss the half-width in x is $146\ \mu\text{m}$ and the RMS is $618\ \mu\text{m}$, a factor 4.2 apart, and the reason is visible in Table 21: 17 of the 286 tracks end more than a millimetre away and those few set the RMS entirely. Under the pooled loss the same two numbers are 203 and $426\ \mu\text{m}$, a factor 2.1 apart with 14 tracks beyond a millimetre. The half-width in x is therefore the one component where the cost-weighted network is better in this band, 146 against $203\ \mu\text{m}$, and it is better because the reweighting did reach down this far in the bending plane. Everything else here is worse.

The degradation is concentrated in y and it is a shift, not a widening only. In the 4–6 GeV band the signed median in y goes from $-9.2\ \mu\text{m}$ under the pooled loss to $+28.1\ \mu\text{m}$ under the cost-weighted one, and the half-width from 96.6 to $223.3\ \mu\text{m}$: the distribution has both moved and more than doubled in width. The same happens in t_y , where the signed median goes from -2.9×10^{-6} to $+3.5 \times 10^{-5}$ and the half-width from 3.97×10^{-5} to 1.61×10^{-4} . In the wider 3–8 GeV band the pattern is identical with slightly smaller numbers. Whatever the reweighting bought in the window, it took the soft tracks' vertical plane with it, and it took it by a factor four in the slope's half-width.

The tail columns say something the momentum bands cannot. The tracks that end beyond a millimetre are not randomly placed: in the 3–8 GeV band under the cost-weighted loss their median starting $|x|$ on the last Upstream Tracker plane is 217.1 mm against 107.8 mm for the rest of the band, a ratio of 2.01; under the pooled loss the same ratio is 1.46, and in the narrower

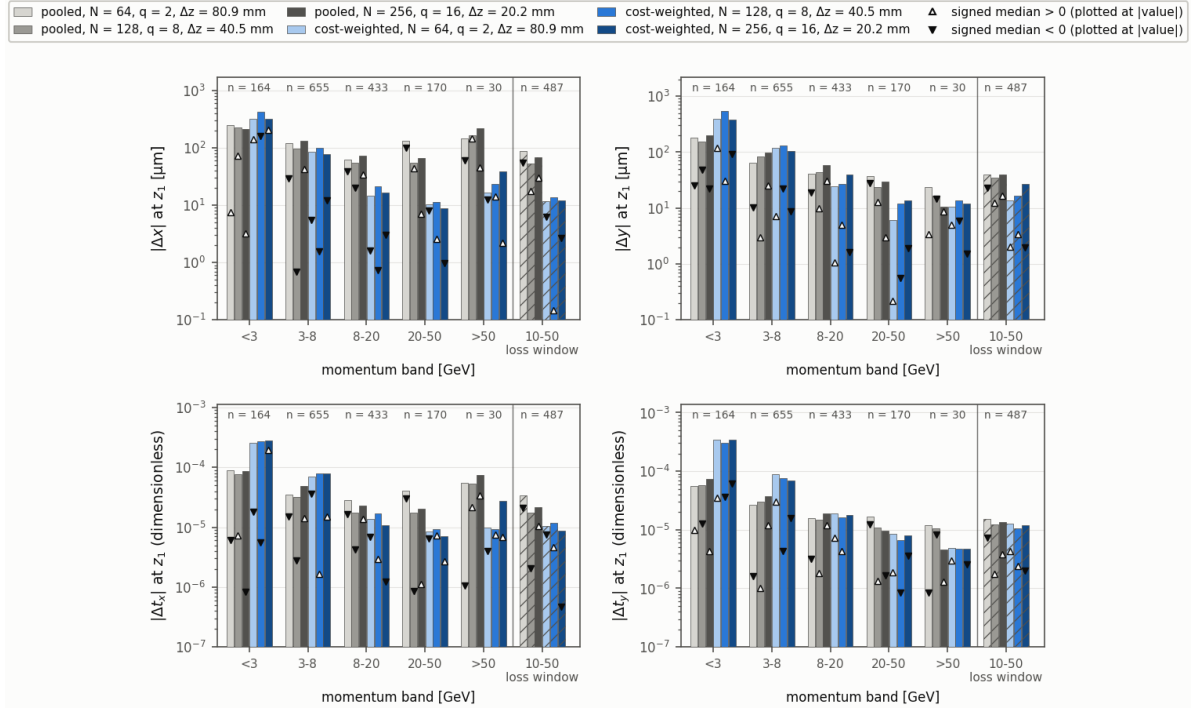


Figure 13: The four state components of the endpoint error at z_1 , band by band, for all six runs: Δx and Δy in micrometres and the dimensionless slope deviations Δt_x and Δt_y , all on logarithmic axes so that the slope decades read as scientific notation. In each panel the grouped bars give the median |deviation| of the three networks under the pooled loss, in grey darkening with N , beside the same three under the cost-weighted loss in blue. The five bands partition the test set and the 10–50 GeV loss window is the hatched group at the right. The number of test tracks is printed above each group. On top of every bar a small triangle marks the signed median plotted at the same magnitude, pointing up when it is positive and down when it is negative, so that the size of the systematic part can be compared with the size of the error directly. Source: `scripts/fig11_components_by_band.py`.

Table 20: The $N = 64$, $q = 2$, $\Delta z = 80.9$ mm network under both losses in the 3–8 GeV band and in the 4–6 GeV band, per component. Position entries are micrometres; slope entries are dimensionless and are quoted $\times 10^{-5}$. *sgn* is the median of the signed deviation, *hw68* the 68 per cent half-width of it, *RMS* the root mean square of it and *med* the median of its magnitude. The RMS is quoted beside the half-width because the size of the disagreement between them is itself the statement about the tail. Source: `results/tab_near_5gev.csv`.

band	loss	component	n	sgn	hw68	RMS	med
3–8 GeV	pooled	Δx [μm]	655	−29.1	192.7	708.2	118.8
		Δy [μm]	655	−10.4	102.7	739.8	63.5
		Δt_x [10^{-5}]	655	−1.50	5.89	20.75	3.48
		Δt_y [10^{-5}]	655	−0.16	4.30	25.07	2.68
	cost-weighted	Δx [μm]	655	−5.6	148.0	634.9	84.3
		Δy [μm]	655	+7.2	233.3	716.0	118.3
		Δt_x [10^{-5}]	655	−3.59	12.20	32.65	6.98
		Δt_y [10^{-5}]	655	+3.00	17.16	52.08	8.98
	4–6 GeV	Δx [μm]	286	−35.5	203.4	426.1	125.1
		Δy [μm]	286	−9.2	96.6	518.2	64.0
		Δt_x [10^{-5}]	286	−2.02	5.48	13.09	3.55
		Δt_y [10^{-5}]	286	−0.29	3.97	21.65	2.66
	cost-weighted	Δx [μm]	286	−26.5	146.2	618.1	92.8
		Δy [μm]	286	+28.1	223.3	425.4	130.0
		Δt_x [10^{-5}]	286	−5.36	10.27	28.37	7.76
		Δt_y [10^{-5}]	286	+3.49	16.11	32.86	9.55

Table 21: The same two bands summarised, and where the tail tracks start. *radial median* and *p95* are of $\sqrt{\Delta x^2 + \Delta y^2}$ at z_1 ; *beyond 1 mm* counts the tracks in the band whose radial endpoint error exceeds one millimetre; the last three columns are the median $|x|$ of those tracks on the last Upstream Tracker plane z_0 , the same for the rest of the band, and their ratio. Source: `results/tab_near_5gev.csv`.

band	loss	n	radial med. [μm]	p95 [μm]	beyond 1 mm	start $ x $ tail [mm]	start $ x $ rest [mm]	ratio
3–8 GeV	pooled	655	160.4	1,035	36	159.6	109.6	1.46
	cost-weighted	655	175.5	1,513	55	217.1	107.8	2.01
4–6 GeV	pooled	286	159.6	890	14	160.8	115.6	1.39
	cost-weighted	286	187.1	1,106	17	188.8	113.6	1.66

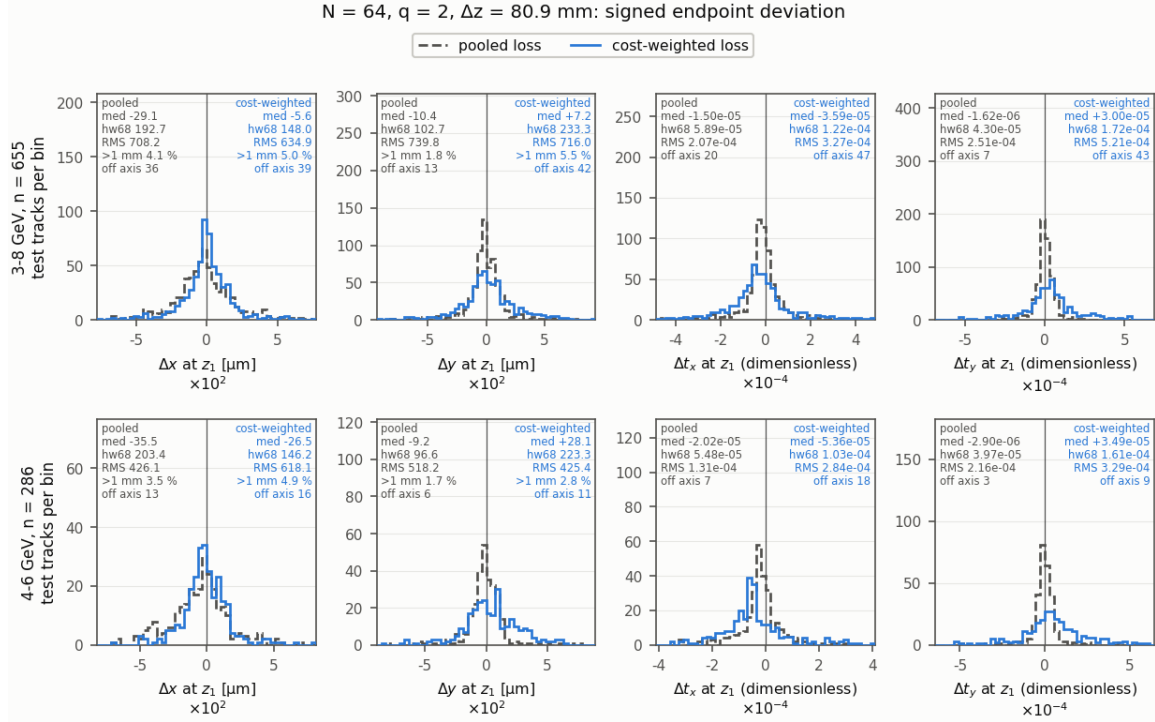


Figure 14: The signed endpoint deviation of the $N = 64, q = 2, \Delta z = 80.9 \text{ mm}$ network from the reference at z_1 , under both losses, as step histograms on a common binning. The four columns are Δx and Δy in micrometres and the dimensionless Δt_x and Δt_y , whose axes therefore read in scientific notation. Top row: the 3–8 GeV band, $n = 655$. Bottom row: the 4–6 GeV band, $n = 286$. Each panel annotates, for both losses, the signed median, the 68 per cent half-width and the RMS; the two position columns also print the fraction of tracks whose deviation in that component exceeds one millimetre, and every panel prints how many tracks fall outside the drawn range, since the bin range is finite. Source: `scripts/fig12_signed_3to8.py`.

4–6 GeV band the ratios are 1.66 and 1.39. The tail is made of soft tracks that start far from the beam line, and it becomes more so under the cost-weighted loss. A momentum window cannot address that, because the distinguishing variable is not momentum. Panel (c) of Figure 12 draws it, and Figure 14 shows the full signed distributions these numbers summarise.

5.7 What the remaining error is made of

Everything so far describes the error at the far side of the magnet. This section asks where it came from: whether the endpoint error is accumulated per-step error and nothing else, how much the individual steps agree with one another, and which plane carries what is left.

The measurement is made by comparing each step's output with a reference integration started from the same state that step was given, over the same Δz , so that the quantity is the

Table 22: The per-step error of all six runs. Each step’s output is compared with a reference integration started from the same state the network was given, over the same Δz ; the entries are medians over all N steps and all 1,452 test tracks. Slopes are dimensionless and are quoted $\times 10^{-7}$; positions are micrometres. *radial* is the radial per-step position error. The three momentum columns restrict the same statistic to tracks in that band. Source: `results/tab_anatomy.csv`.

network	loss	all tracks			3–8 GeV		8–20 GeV		10–50
		$ \Delta t_x $ [10^{-7}]	$ \Delta t_y $ [10^{-7}]	radial [μm]	$ \Delta t_x $ [10^{-7}]	$ \Delta t_y $ [10^{-7}]	$ \Delta t_x $ [10^{-7}]	$ \Delta t_y $ [10^{-7}]	$ \Delta t_x $ [10^{-7}]
$N = 64, q = 2$	pooled	13.02	7.16	0.333	14.94	9.97	8.56	4.93	9.49
	cost-weighted	7.34	9.74	0.433	14.70	21.79	2.96	4.24	2.25
$N = 128, q = 8$	pooled	5.49	3.15	0.172	6.69	4.85	3.41	1.98	3.65
	cost-weighted	3.77	4.98	0.221	7.64	10.76	1.55	2.24	1.18
$N = 256, q = 16$	pooled	2.99	1.83	0.089	3.65	2.79	1.83	1.23	1.82
	cost-weighted	1.87	2.38	0.058	3.63	4.94	0.73	1.20	0.57

Table 23: How the per-step errors accumulate. *x part* and *y part* are the medians of $|\Delta x|$ and $|\Delta y|$ at z_1 , repeated here so that the two planes can be compared against the coherences. *coherence* is, per track, the magnitude of the sum of the per-step slope errors divided by the sum of their magnitudes, with the value $1/\sqrt{N}$ that independent steps would give in the column beside it. *explained* is the ratio of the lever-arm model of Section 2.8 — each step’s own position error plus its slope error times the distance it still has to travel — to the measured radial endpoint error, once with both slopes and once with the *x* slope alone. Source: `results/tab_anatomy.csv`.

network	loss	<i>x part</i> [μm]	<i>y part</i> [μm]	coherence t_x	coherence t_y	independent steps	explained both slopes	explained <i>x</i> slope only
$N = 64, q = 2$	pooled	107.7	51.9	0.357	0.397	0.125	0.984	0.733
	cost-weighted	36.4	54.8	0.491	0.518	0.125	0.977	0.377
$N = 128, q = 8$	pooled	79.0	59.1	0.312	0.432	0.0884	0.983	0.634
	cost-weighted	52.4	59.9	0.527	0.459	0.0884	0.989	0.448
$N = 256, q = 16$	pooled	98.9	69.3	0.411	0.453	0.0625	0.989	0.673
	cost-weighted	36.7	61.6	0.454	0.474	0.0625	0.994	0.363

local error of one application in the middle of a chain rather than a difference from the true trajectory.

Read it this way, and start with the last two columns of Table 23, because they license everything else. Modelling the endpoint error as the sum over steps of each step’s own position error plus that step’s slope error multiplied by the distance it still has to travel reproduces the measured radial endpoint error to between 97.7 and 99.4 per cent in all six runs. The decomposition is complete. The endpoint error of a chained network is accumulated per-step error carried by the lever arm and there is nothing else in it — no separate effect of feeding the network its own output, no instability, nothing that is not already in the per-step numbers. That is a strong statement and it is measured rather than assumed, and it is what makes the lever arm the right variable to weight by.

Next, the per-step slopes. In the loss window the cost-weighted network’s *x* slope error is 2.25×10^{-7} against the pooled network’s 9.49×10^{-7} , a factor 4.21; at $N = 128$ the same comparison is 1.18 against 3.65, a factor 3.08, and at $N = 256$, 0.57 against 1.82, a factor 3.21. Over all tracks the improvement is smaller, a factor 1.77, 1.46 and 1.60, because the soft tracks that dominate an unweighted median were deliberately de-weighted. And in t_y the cost-weighted network is *worse* over all tracks in all three pairs — 9.74 against 7.16, 4.98 against 3.15, 2.38 against 1.83, in units of 10^{-7} — while being better in the 8–20 GeV band. The 3–8 GeV columns show where that comes from: there the cost-weighted t_y error is 2.2, 2.2 and 1.8 times the pooled one. The *y* degradation of Section 5.5 is already present at the level of a single step.

Table 24: What shortening the step buys. *per-step radial* and *per-step* $|\Delta t_x|$ are the local errors of one application of the network in the middle of its own chain, medians over all steps and test tracks. *single step* is one application started from a reference state on an interior plane, median over all planes and tracks; it agrees with the per-step radial column to the digits shown, although the two are computed from different starting states. *endpoint* and *in window* are the radial medians of the full N -application chain, repeated from Table 16. *chain / step* is the last column of the reference divided by the single-step column. Source: `results/tab_anatomy.csv`, `results/tab_along_z.csv` and `results/tab_single_step_vs_z.csv`.

network	loss	Δz [mm]	per-step radial [μm]	per-step $ \Delta t_x $	single step [μm]	endpoint [μm]	in window [μm]
$N = 64, q = 2$	pooled	80.90	0.333	1.30×10^{-6}	0.333	145.7	108.1
	cost-weighted	80.90	0.433	7.34×10^{-7}	0.433	88.8	24.8
$N = 128, q = 8$	pooled	40.45	0.172	5.49×10^{-7}	0.172	122.2	70.5
	cost-weighted	40.45	0.221	3.77×10^{-7}	0.221	104.6	27.3
$N = 256, q = 16$	pooled	20.23	0.089	2.99×10^{-7}	0.089	147.7	89.6
	cost-weighted	20.23	0.058	1.87×10^{-7}	0.058	92.2	34.6

Now the coherences, which explain why the endpoint did not improve as much as the steps did. If the N steps of a chain erred independently the magnitude of the sum of their slope errors would be about $1/\sqrt{N}$ of the sum of their magnitudes — 0.125 at $N = 64$, 0.0884 at $N = 128$, 0.0625 at $N = 256$. The measured values are between 0.31 and 0.53, so the steps are between 2.9 and 7.6 times more aligned than chance in every run. The steps are not making independent mistakes; they are making the same mistake repeatedly, which is what one should expect of a single deterministic map applied to a smoothly varying sequence of states. Worse, the cost-weighted networks are *more* coherent than their counterparts in all six of the matched comparisons: 0.491 and 0.518 against 0.357 and 0.397 at $N = 64$, 0.527 and 0.459 against 0.312 and 0.432 at $N = 128$, 0.454 and 0.474 against 0.411 and 0.453 at $N = 256$. Smaller per-step errors that all point the same way accumulate as efficiently as larger random ones would. That is the mechanism by which a factor 4.2 in the in-window per-step slope became a factor 4.4 in the in-window endpoint, while a factor 1.8 in the overall per-step x slope became a factor 1.6 in the overall endpoint, and it is the clearest single reason why shortening the step does not help as much as it ought to.

Finally, the two plane columns and the x -slope-only model. Under the pooled loss the x part of the endpoint error is larger than the y part in all three networks — 107.7 against 51.9, 79.0 against 59.1, 98.9 against 69.3 μm — and a decomposition using the x slope alone recovers 63 to 73 per cent of the endpoint error. Under the cost-weighted loss the x part is the smaller in two of the three — 36.4 against 54.8 and 36.7 against 61.6 μm , with $N = 128$ close to equal at 52.4 against 59.9 — and the x -slope-only model recovers only 36 to 45 per cent. A one-sided decomposition was adequate for the pooled-loss networks and is not adequate for the reweighted ones, because what limits them has moved into the other plane. Figure 15 shows the step-by-step measurement for the pre-registered pair.

5.8 Shorter steps

The design in this paper has a free parameter that the first paper did not have — how finely to cut the crossing — and the obvious expectation is that a shorter step is an easier problem and should give a better crossing; this section measures that and finds it is not so.

Read it this way. Halving the step halves the per-step error, near enough. Under the pooled loss the per-step radial error goes $0.333 \rightarrow 0.172 \rightarrow 0.089 \mu\text{m}$, ratios of 1.93 and 1.93 against the 2.00 that exact proportionality to Δz would give; the per-step x slope error goes $1.30 \rightarrow 0.549 \rightarrow 0.299$ in units of 10^{-6} , ratios of 2.37 and 1.84. Under the cost-weighted loss the slope goes $7.34 \rightarrow 3.77 \rightarrow 1.87$ in units of 10^{-7} , ratios of 1.95 and 2.02, which is as close to proportional as one could ask. Each network is doing its own job better as the job gets easier.

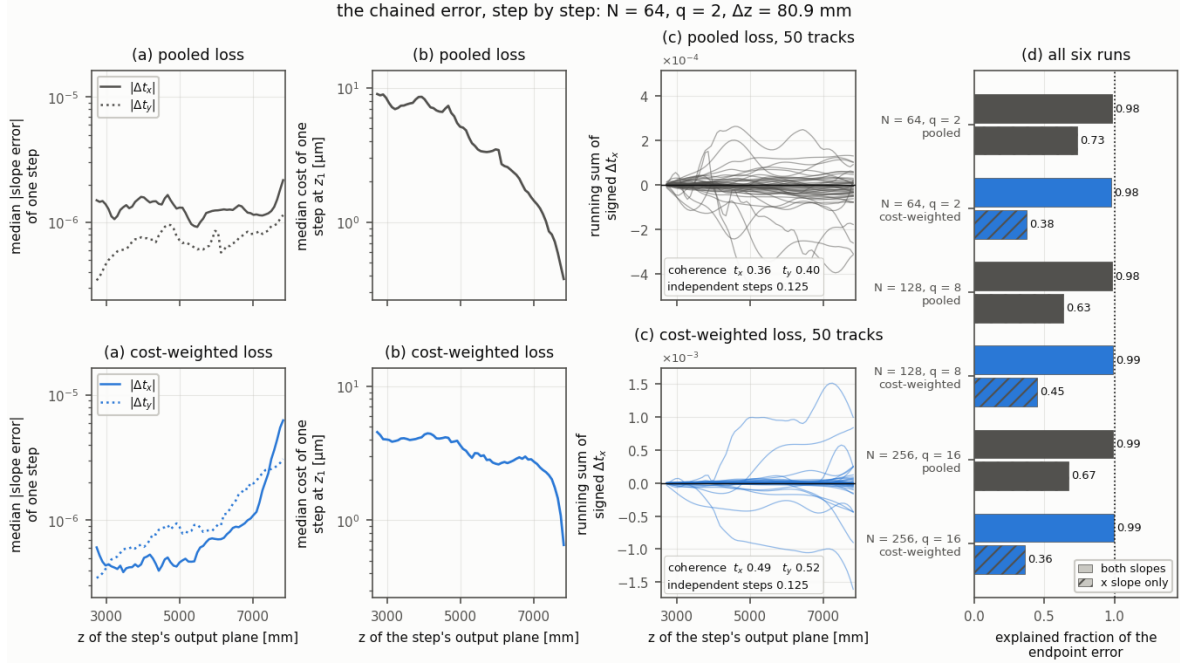


Figure 15: The anatomy of the chained error for the two $N = 64, q = 2, \Delta z = 80.9 \text{ mm}$ networks, the pooled loss on the top row and the cost-weighted loss on the bottom. Each step's output is compared with a reference integration started from the same state the network was given, over the same step. (a) The median $|\Delta t_x|$ and $|\Delta t_y|$ of a single step against the z of that step's output plane, dimensionless on a logarithmic axis. (b) What each step costs at the endpoint: the median over test tracks of that step's position error plus its slope error times the distance still to run, in micrometres — an early step is expensive because its slope error is carried a long way. (c) The running sum of the signed per-step slope error along the track for 50 test tracks; the lines drift in one direction rather than wandering, which is what the coherence annotated in the panel measures, quoted against the 0.125 that independent steps would give. The two rows carry different vertical scales because the drifts differ by about a factor four. (d) The explained fraction of the endpoint error for all six runs, with both slopes and with the x slope alone. Source: `scripts/fig14_anatomy.py`.

The endpoint does not follow. Under the pooled loss it goes $145.7 \rightarrow 122.2 \rightarrow 147.7 \mu\text{m}$ and under the cost-weighted loss $88.8 \rightarrow 104.6 \rightarrow 92.2 \mu\text{m}$: neither sequence is monotone and neither moves by more than a factor 1.21 across a factor four in step length. Inside the loss window it is worse than that — the cost-weighted sequence is $24.8 \rightarrow 27.3 \rightarrow 34.6 \mu\text{m}$, which gets steadily *worse* as the step shortens, so the network with the smallest per-step error has the largest in-window endpoint error of the three. The arithmetic is not mysterious: halving the step halves each step's error and doubles the number of steps, so a perfectly incoherent chain would improve only as $\sqrt{\Delta z}$ and a perfectly coherent one would not improve at all. The coherences of Table 23 are between 0.31 and 0.53 rather than at their independent values of 0.13, 0.088 and 0.063, which puts the chains much closer to the coherent limit than to the incoherent one, and the measured endpoints are flat, as that limit says they should be.

The ratio of the two errors makes the same point in one number. The chain's endpoint error is 438 times its own single-step error at $N = 64$ under the pooled loss and 205 times under the cost-weighted one; 710 and 474 times at $N = 128$; and 1,653 and 1,587 times at $N = 256$. A chain of N independent steps would show roughly $\sqrt{N}$, which is 8, 11 and 16, and a chain in which every step's slope error is carried the full remaining length would show something of order N times the lever arm in units of the step — which is where these numbers live. Figure 16 puts the two measurements side by side on a common axis, and the gap between the panels is the whole of this section.

The practical conclusion is that shortening the step is not the lever. What would be the lever is making successive steps disagree with one another, since the accumulation and not the

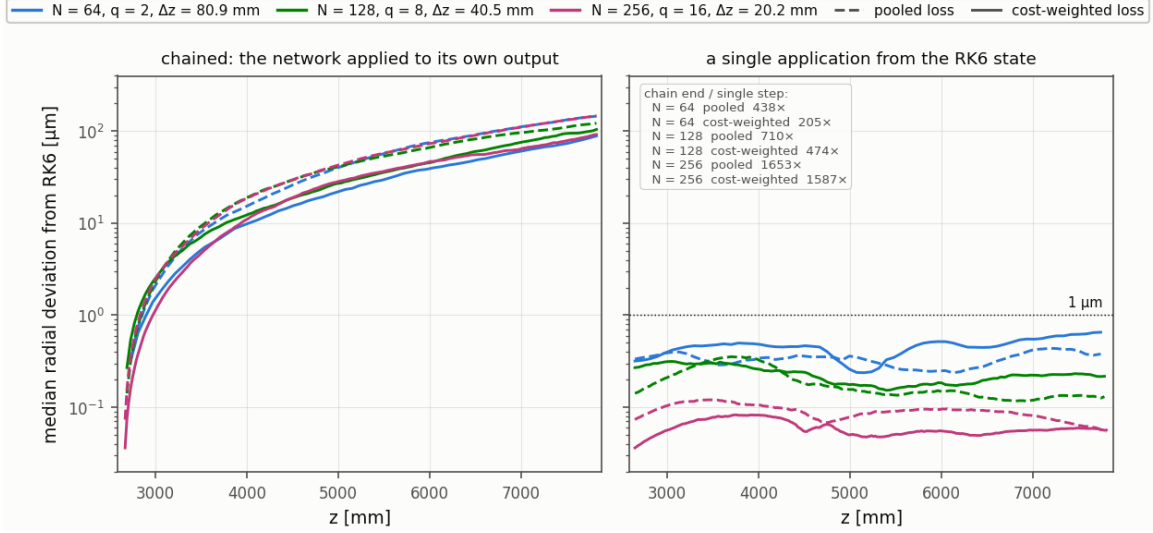


Figure 16: The error along the crossing: one step against the whole chain, for the six runs. Left: the median radial deviation from the reference at every plane of the chain, against z from $z_0 = 2648.2$ mm to $z_1 = 7826.0$ mm, micrometres on a logarithmic axis; the colour names the network and the pooled-loss run is dashed while the cost-weighted one is solid. Plane zero is the track’s own starting state and is exactly zero, so the curves begin at the first output plane. Right: the same six networks applied *once*, from the reference state on every start plane and compared with the reference one plane later, against the start plane’s z on the same scale. The two panels share their vertical range on purpose: a single application of any of these networks is well under one micrometre while the chain of N of them ends near a hundred, and the ratio per run is printed in the right panel. No slopes appear in this figure. Source: `scripts/fig13_along_z.py`.

individual step is what is binding, and that is a property of the map rather than of any single residual.

5.9 Against the Geant4-true state

Every number so far is a distance from a field-only reference, which answers how well the network solves the problem it was given and not whether that problem is the particle’s real trajectory; this section puts all three reference points of Section 2.7 on the same axis so that the distinction can be read off.

Read it this way, one column pair at a time. The *nn-true* and *RK6-true* columns are, to the eye, the same numbers. Over all test tracks the network’s median distance from the simulated truth is between 97.4 and 102.0 per cent of the reference’s own distance from it, across all six runs; band by band, for the three cost-weighted networks, the largest departure from unity anywhere in the table outside the thirty-track > 50 GeV row is 4.1 per cent. The network and the reference are, at this resolution, the same object as far as the real particle is concerned. That is not a coincidence and it is not a measure of quality: it is the statement that the gap to the truth is the material the particle crosses — scattering and energy loss — which neither the reference nor the network models, and which is an order of magnitude larger than the difference between them.

The *ratio* column quantifies that. For the three cost-weighted networks the reference is between 12.5 and 25.2 times further from the truth than the network is from the reference, in every band from < 3 to 20–50 GeV and in the loss window. Over all tracks the same ratio is 21.0, 17.7 and 20.1. For the pooled-loss networks it is more variable, from 2.7 at 20–50 GeV to 31.7 below 3 GeV and 12.6 to 15.4 overall, because their distance from the reference is both larger and more strongly momentum-dependent. The honest reading of the whole table is the one Section 2.7 warned about: a network can be two to six orders of magnitude above its own numerical ceiling (Table 12) and still be twenty times below the material term, and both

Table 25: The three reference points, per momentum band, for the three cost-weighted networks. *nn-RK6* is the network against the field-only reference at z_1 , *nn-true* the same network’s endpoint carried with the reference to the particle’s own first SciFi plane and compared with its simulated state there, and *RK6-true* the reference itself against that same state. Positions are the max metric $\max(|\Delta x|, |\Delta y|)$ in micrometres, slopes the max metric over the two slope components and dimensionless. *RK6-true* does not depend on the network and is printed once per band. *ratio* is *RK6-true* divided by *nn-RK6* in position: how many times further the reference is from the truth than the network is from the reference. The > 50 GeV band holds 30 tracks. Source: `results/tab_against_true_state.csv`.

band [GeV]	network	position, median [μm]				slope, median		ratio
		n	nn-RK6	nn-true	RK6-true	nn-RK6	nn-true	
< 3	$N = 64, q = 2$	164	528.4	9,338	9,403	4.73×10^{-4}	3.61×10^{-3}	17.8
	$N = 128, q = 8$	164	753.6	9,785		4.21×10^{-4}	3.57×10^{-3}	12.5
	$N = 256, q = 16$	164	511.3	9,365		5.01×10^{-4}	3.48×10^{-3}	18.4
3–8	$N = 64, q = 2$	655	159.4	2,701	2,699	1.21×10^{-4}	8.71×10^{-4}	16.9
	$N = 128, q = 8$	655	198.1	2,730		1.26×10^{-4}	8.46×10^{-4}	13.6
	$N = 256, q = 16$	655	157.0	2,663		1.35×10^{-4}	9.22×10^{-4}	17.2
8–20	$N = 64, q = 2$	433	30.9	736.4	734.3	2.49×10^{-5}	2.33×10^{-4}	23.8
	$N = 128, q = 8$	433	40.4	758.2		2.55×10^{-5}	2.22×10^{-4}	18.2
	$N = 256, q = 16$	433	45.1	745.7		2.29×10^{-5}	2.19×10^{-4}	16.3
20–50	$N = 64, q = 2$	170	15.7	368.8	369.2	1.22×10^{-5}	1.11×10^{-4}	23.5
	$N = 128, q = 8$	170	19.8	360.1		1.19×10^{-5}	1.05×10^{-4}	18.6
	$N = 256, q = 16$	170	18.2	369.4		1.10×10^{-5}	1.06×10^{-4}	20.3
> 50	$N = 64, q = 2$	30	20.7	181.3	182.8	1.18×10^{-5}	4.67×10^{-5}	8.8
	$N = 128, q = 8$	30	29.3	185.5		1.17×10^{-5}	5.14×10^{-5}	6.2
	$N = 256, q = 16$	30	44.4	161.8		2.76×10^{-5}	3.20×10^{-5}	4.1
10–50 (window)	$N = 64, q = 2$	487	21.8	555.7	548.9	1.80×10^{-5}	1.54×10^{-4}	25.2
	$N = 128, q = 8$	487	24.7	555.0		1.74×10^{-5}	1.52×10^{-4}	22.3
	$N = 256, q = 16$	487	32.2	552.7		1.62×10^{-5}	1.50×10^{-4}	17.0

statements are true at once. The optimiser is the binding constraint on the surrogate, and the surrogate is not yet the binding constraint on a track fit that has to model the material anyway.

Two cautions belong here. First, the > 50 GeV row holds 30 test tracks; its ratios are the smallest in the table, 8.8, 6.2 and 4.1, and at $N = 256, q = 16$ the network’s distance from the truth is 88.5 per cent of the reference’s, which is the only entry that departs from unity by more than a few per cent. With thirty tracks that is one or two tracks’ worth of median and should be read as such. Second, this comparison cannot be used to score the networks against one another. The network was never trained on the simulated state, cannot reproduce it by construction, and the quantity that varies between the six runs — their distance from the reference — is precisely the small number being swamped. Figure 17 shows the arrangement.

5.10 Cost

The last thing to report is what all of this cost: what each network took to train, and what a trained network costs to carry a track across the magnet.

Read it this way. The two losses cost the same to train. At every step length the median wall per restart differs between the two by less than half a per cent — 77.2 against 77.7 seconds, 117.9 against 118.3, and 219.8 against 184.1, the last being the one affected by the duplicate job discussed below. The reweighting changes the weights inside the loss and nothing else, so this is what one would expect, and it is worth recording because a cheaper objective that cost more to optimise would be a different result. The total walls differ only because the runs were given different numbers of restarts. The per-restart cost roughly follows the arithmetic of a step: N applications of a network emitting $4(q + 1)$ numbers, with q field lookups each, gives $N \times q$ field evaluations per crossing, which is 128, 1024 and 4096 at the three working points, and the

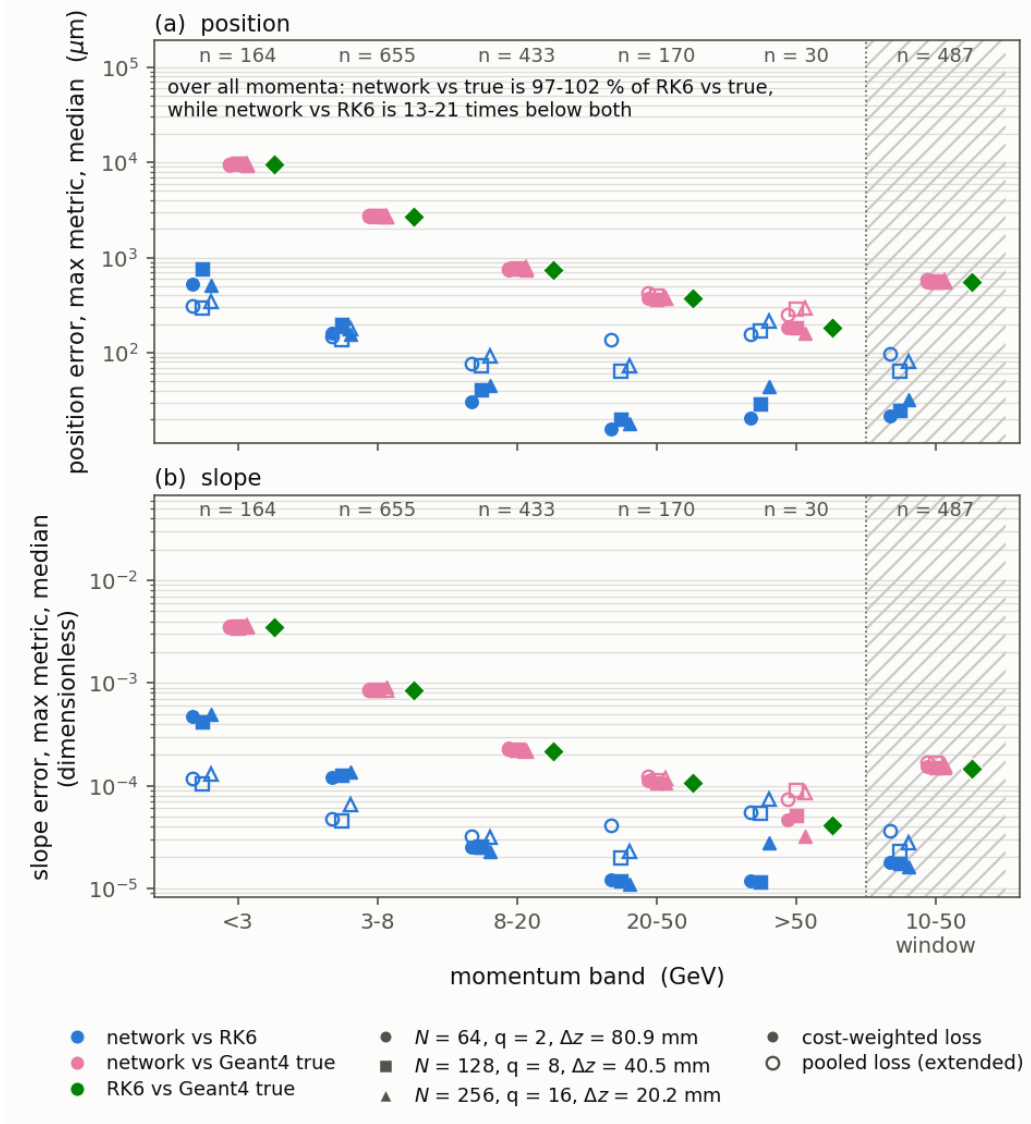


Figure 17: The three reference points for the same crossing, band by band. (a) The position error as the max metric $\max(|\Delta x|, |\Delta y|)$, median per band, in micrometres on a logarithmic axis, for the network against the field-only reference, the network against the particle’s simulated state on its own first SciFi plane, and the reference against that same state. (b) The same for the slope, $\max(|\Delta t_x|, |\Delta t_y|)$, dimensionless, so the axis reads in scientific notation. Filled markers are the three cost-weighted networks and hollow markers the three pooled-loss networks extended to their own plateau, with the marker shape naming the network; the reference-against-truth point does not depend on the network and is drawn once per band. The five momentum bands are drawn as a partition and the 10–50 GeV loss window, which overlaps two of them, is drawn apart on a hatched background; every band prints its n . The message is the vertical arrangement: network-against-truth and reference-against-truth sit on top of one another in every band, while for the three cost-weighted networks network-against-reference is more than an order of magnitude below both in every band from < 3 to 20–50 GeV and in the loss window. The thirty-track > 50 GeV band is the exception: there the pooled-loss networks’ hollow markers sit level with the reference-against-truth point, at ratios of 1.16, 1.09 and 0.85, and the cost-weighted networks are a factor 4.1 to 8.8 below rather than an order of magnitude. Source: `scripts/fig05_three_references.py`.

Table 26: What the six runs cost. *restarts* is the number of optimiser restarts recorded, with the number of distinct restart indices beside it where the two differ; *rounds* is the number of distinct rounds, with the number of recorded round rows beside it where the two differ. Every round is 25 restarts and redraws 32,000 collocation states, and every restart ran to the same iteration cap, a median of 200 iterations, with no early stop anywhere. *wall* is the total recorded training wall; *per restart* its median over the de-duplicated history. *carry* is the measured cost of taking one test track across the magnet with the trained network, the full chain of N applications. Source: `results/tab_compute.csv` and `results/paper_numbers.json` (compute).

network	loss	params	restarts	rounds	per restart [s]	wall [h]	carry [μ s/track]
$N = 64, q = 2$	pooled	18,956	1,260	50	77.2	31.2	653
	cost-weighted	18,956	1,000	40	77.7	21.6	657
$N = 128, q = 8$	pooled	22,052	1,599	63	117.9	54.7	1,353
	cost-weighted	22,052	1,000	40	118.3	32.8	1,322
$N = 256, q = 16$	pooled	26,180	1,050	42	219.8	63.2	2,964
	cost-weighted	26,180	1,807 (1,500)	60 (74)	184.1	100.4	3,655

measured walls are in the ratio 1 : 1.53 : 2.85 rather than 1 : 8 : 32, because the batch is large and much of the per-restart cost is not the field.

The duplicate-job note has to be stated plainly because it affects one row. The $N = 256$, $q = 16$ cost-weighted run was trained by two farm jobs at once from restart 1,198 onwards. Its history file therefore holds 302 repeated restart numbers and its round file 13 repeated round numbers: 1,807 history rows for 1,500 distinct restarts, and 74 round rows for 60 distinct rounds. Every reader of those files in this paper de-duplicates them by keeping the last row written per number, which is the rule the analysis folder already used, and the restart and round counts quoted everywhere in this paper are the de-duplicated ones. The wall is the one quantity where the two bookkeepings genuinely disagree: the run’s own record says 100.4 hours, which counts the duplicate job’s work, while the de-duplicated history sums to 81.1 hours. The table quotes the recorded 100.4 hours, since that is what the farm actually spent, and the reader should know that about 19 hours of it was work done twice. No other run in the set has a duplicated row.

The carry cost is a measurement of this implementation and not a deployment figure, and I say so rather than letting it be read as one. It is 653 to 657 μ s per track at $N = 64$, 1,322 to 1,353 at $N = 128$ and 2,964 to 3,655 at $N = 256$, measured in the same framework the networks were trained in, on one thread, with no fusion, no reduced precision and no attempt at optimisation. For scale, the sixth-order reference integrating the same crossing at 0.1 mm steps costs about 0.12 seconds per track when amortised over a large batch, so the $N = 64$ network is already about 180 times faster than the reference it is being scored against — but the reference is a double-precision research integrator taking fifty thousand steps, not the production extrapolator, and the comparison that matters for the trigger is against the latter and has not been made here. What the table does establish is the scaling: the carry cost doubles with each halving of the step, as it must, so the shortest-step network costs four and a half to five and a half times the longest-step one and delivers, by Section 5.8, no better a crossing. That is a second and independent reason not to take the step length below $\Delta z = 80.9$ mm on this evidence.

6 The low-momentum window (reserved)

Reserved. This section is held for the study in which the momentum window of Eq. (26) is moved down onto the 3–8 GeV band. Those runs are **in training at the time of writing**, so no number from them is reported here. When they land, this section will report them in exactly the form Sections 5.4 to 5.6 use for the 10–50 GeV runs: the headline comparison of the same chained networks under the moved window against the same pooled-loss counterparts, as Table 16; the radial endpoint median and 95th percentile in the paper’s five momentum bands with the moved window as the extra row, as Table 17; and the per-component breakdown, as Tables 18 and 19, with the region around 5 GeV treated as in Section 5.6. The same three reference points of Section 2.7, the same plateau rule and the same test tracks apply, so the effect of moving the window can be read off directly against Section 5.

7 Discussion

Section 5 reported the experiment; this section reads it against the two questions of Section 1.4. I take the improvement first (Section 7.1), then the price (Section 7.2), then everything that limits how far either half can be pushed (Section 7.3). No new measurement appears here; every number has already been given a table and a source above.

7.1 What the reweighting did

The experiment had a pre-registered outcome and it is the full one. Section 2.8 fixed two conditions in advance, and Section 5.4 tested them on the $N = 64$, $q = 2$, $\Delta z = 80.9$ mm pair. The first was that the test radial endpoint error clear a ± 13 per cent run-to-run band downward: it falls from 145.7 to $88.8 \mu\text{m}$, by 39.1 per cent, which clears that band and also clears the ± 17.5 per cent round-to-round spread the counterpart turned out to have. The second was that the median per-step slope error fall below 1.6×10^{-6} : the cost-weighted network reaches 7.34×10^{-7} in t_x , a factor 2.18 below the threshold. Both hold.

The second condition is the weaker of the two and I say so rather than counting it twice. The extended pooled-loss counterpart has itself fallen to 1.30×10^{-6} since the threshold was written, which is also below 1.6×10^{-6} , so that condition no longer separates the two losses. What separates them is the first condition and the size of the margin on it.

The gain is concentrated where the weight was put. Inside the 10–50 GeV loss window the $N = 64$ median endpoint error falls from 108.1 to $24.8 \mu\text{m}$, a factor 4.37 . That is the one feature common to all three step lengths: the same window falls by a factor 2.58 at $N = 128$ and 2.59 at $N = 256$, so the in-window gain is a factor between 2.6 and 4.4 everywhere it was measured. It is visible one derivative down as well. In the window the median per-step x -slope error falls from 9.49×10^{-7} to 2.25×10^{-7} , a factor 4.21 , against factors of 3.08 and 3.21 at the two shorter steps. The endpoint gain and the per-step gain are the same size, which is what Section 5.7 says they must be: modelling the endpoint error as accumulated per-step error carried by the lever arm reproduces the measured radial endpoint error to between 97.7 and 99.4 per cent in all six runs, so there is nothing between the step and the endpoint for the two numbers to disagree about.

The pre-flight of Section 5.2 says why this happened, and it was measured before any reweighted network existed. The share of the loss carried by the loss window rises from 1.23 to 47.0 per cent, the share carried by the 3–8 GeV band falls from 89.8 to 38.3 , the share carried by the first quarter of the crossing — where an error has the longest lever arm — rises from

0.65 to 23.3 per cent, and within the window the rank correlation between a state’s share of the loss and what that state’s error actually costs at z_1 rises from +0.532 to +0.896. The objective was asked to look somewhere else and it did, and the endpoint error followed it there.

The whole of the gain is in x . Over all test tracks the median $|\Delta x|$ at the first SciFi plane falls from 110.7 to 37.5 μm , a factor 2.95, while the median $|\Delta y|$ moves from 52.3 to 55.2 μm , which is 5.6 per cent worse. The magnet’s field is close to vertical and therefore bends the track horizontally, so x is where the correction the surrogate has to reproduce lives, and x is what the reweighting improved. Inside the window both components do fall — 86.9 to 11.6 μm in x and 39.6 to 13.9 μm in y — but their ordering swaps: under the pooled loss the x error is 2.2 times the y error there, and under the cost-weighted loss the y error is the larger by a factor 1.20. What limits the reweighted network inside its own window is now the non-bending plane.

The in-window offset falls in every pair. The signed median is the sharper statement, because a signed median close to the median magnitude is a common offset in one direction rather than scatter. In the window the pooled-loss $N = 64$ network sits at $-54.4 \mu\text{m}$ in x against a median magnitude of 86.9 μm , and the cost-weighted one at $-6.3 \mu\text{m}$ against 11.6 μm : the absolute offset falls by a factor 8.7. The same happens at the other two step lengths, from +17.9 to +0.15 μm at $N = 128$ and from +30.3 to $-2.7 \mu\text{m}$ at $N = 256$. So the absolute in-window offset falls in all three pairs. As a fraction of the error’s own size it falls in two of the three, from 0.33 to 0.011 at $N = 128$ and from 0.44 to 0.22 at $N = 256$, while at $N = 64$ it is 0.63 before and 0.54 after, which is barely a change. And the sign of the pooled-loss offset is not common to the three: it is negative at $N = 64$ and positive at $N = 128$ and $N = 256$. The defensible claim is therefore the one Section 5.5 makes and no more: the reweighting reduces the in-window offset in absolute terms everywhere and relative to the error in two of three pairs. It is not that it removes a systematic under-bend, because there is no single under-bend across the three to remove.

None of this cost the label-free property. Both factors of Eq. (23) are functions of the network’s own inputs and of the measured field map: the lever arm is geometry, and the track factor needs $|q/p|$, which is the fifth component of the input, and the field integral $\bar{I}$. No reference trajectory, from an integrator or from a simulation, enters the weight, so the property that makes the construction worth having is intact. The weight was also checked not to move the minimum before any network was trained: collocation states solved exactly by a root-finder give between 6×10^{-30} and 2×10^{-29} of the straight line’s loss under every weighting mode, which is machine zero. The reweighting changes the shape of the surface around the minimum and not where the minimum is.

7.2 What it did not do

It is worse below the crossover. The crossover sits between the 3–8 and the 8–20 GeV bands, and below it the cost-weighted network is worse in five of the six comparisons. The exception is the 3–8 GeV band at $N = 256$, where the median falls from 200.4 to 177.7 μm , a reduction of 11.3 per cent. At $N = 64$ the median below 3 GeV rises from 353.7 to 615.1 μm , a factor 1.74; the 95th percentile over all tracks rises from 898 to 1399 μm , a factor 1.56; and the fraction of tracks ending more than a millimetre from the reference rises from 4.41 to 7.16 per cent, which is 64 tracks becoming 104. At $N = 128$ the degradation below 3 GeV is the largest anywhere in the paper, a factor 2.70 in the median and a 95th percentile that more than doubles to 3.67 mm. This is a trade and it was the trade the objective was designed to make, since the window was written into the loss deliberately; but a surrogate that has to serve every track in the event does not get to ignore the half of the population that sits below 8 GeV, and the price is therefore real rather than presentational.

The vertical plane did not move. Over all tracks the median $|\Delta y|$ is $52.3\mu\text{m}$ under the pooled loss and $55.2\mu\text{m}$ under the cost-weighted one at $N = 64$; at $N = 128$ it goes from 59.2 to $61.2\mu\text{m}$ and at $N = 256$ from 69.7 to $61.3\mu\text{m}$. Across the three pairs the x error improves by between 1.6 and 7.5 times and the y error by between 0.95 and 2.9. The effect is already present in a single step: the per-step $|\Delta t_y|$ over all tracks is worse under the cost-weighted loss in all three pairs, 9.74 against 7.16, 4.98 against 3.15 and 2.38 against 1.83, in units of 10^{-7} , and in the 3–8 GeV band the cost-weighted t_y error is 2.2, 2.2 and 1.8 times the pooled one. Near 5 GeV the vertical distribution has both shifted and widened: in the 4–6 GeV band the signed median in y goes from -9.2 to $+28.1\mu\text{m}$ and the 68 per cent half-width from 96.6 to $223.3\mu\text{m}$, and in t_y the half-width goes from 3.97×10^{-5} to 1.61×10^{-4} . Equation (23) weights the two position components equally, in millimetres, because the endpoint measure is radial, so nothing in the objective pushed on y ; what the reweighting did was remove the term that had been holding x down, and y was left where it was or slightly worse.

What is left is more coherent, not less. The per-step errors of a chain can add or cancel, and Section 2.8 measures which by dividing the magnitude of the sum of the per-step slope errors by the sum of their magnitudes. Independent steps would give $1/\sqrt{N}$, that is 0.125, 0.0884 and 0.0625 at the three step counts. The measured values run from 0.31 to 0.53, so every chain in this paper is between 2.9 and 7.6 times more aligned than chance — and the cost-weighted networks are the more aligned of each pair in all six of the matched comparisons, 0.491 and 0.518 against 0.357 and 0.397 at $N = 64$, 0.527 and 0.459 against 0.312 and 0.432 at $N = 128$, and 0.454 and 0.474 against 0.411 and 0.453 at $N = 256$. Smaller per-step errors that all point the same way accumulate as efficiently as larger random ones, so part of the per-step gain was handed back at the endpoint. The same shift shows in the decomposition: a model built from the x slope alone recovers 63 to 73 per cent of the endpoint error under the pooled loss and only 36 to 45 per cent under the cost-weighted one. The error that remains is a coherent drift and it is no longer confined to one plane.

The heavy tail is geometric, not a matter of momentum. The tracks that end more than a millimetre from the reference are not scattered through the population. In the 3–8 GeV band under the cost-weighted loss their median starting $|x|$ on the last Upstream Tracker plane is 217.1 mm against 107.8 mm for the rest of the band, a ratio of 2.01; under the pooled loss the same ratio is 1.46, and in the narrower 4–6 GeV band the two ratios are 1.66 and 1.39. The tail is made of soft tracks that start far from the beam line, and the reweighting made it more so. A momentum window cannot reach this, because the variable that distinguishes the tail is not momentum. That is a concrete statement about what the next weight would have to contain, and it is in Section 8.

Shortening the step did not help. The design has a free parameter the first study did not have, and the obvious expectation is that a shorter step is an easier problem. Each step does get easier: under the pooled loss the per-step radial error goes $0.333 \rightarrow 0.172 \rightarrow 0.089\mu\text{m}$ as Δz is halved twice, ratios of 1.93 and 1.93, and under the cost-weighted loss the per-step x -slope error goes $7.34 \rightarrow 3.77 \rightarrow 1.87$ in units of 10^{-7} , ratios of 1.95 and 2.02. The crossing does not follow. The pooled-loss endpoint goes $145.7 \rightarrow 122.2 \rightarrow 147.7\mu\text{m}$ and the cost-weighted one $88.8 \rightarrow 104.6 \rightarrow 92.2\mu\text{m}$: neither is monotone and neither moves by more than a factor 1.21 across a factor four in step length. Inside the loss window the cost-weighted sequence is $24.8 \rightarrow 27.3 \rightarrow 34.6\mu\text{m}$, which gets steadily worse, so the network with the smallest per-step error has the largest in-window endpoint error of the three. The coherences above are the reason: a perfectly incoherent chain would improve as $\sqrt{\Delta z}$ and a perfectly coherent one not at all, and these chains sit much nearer the coherent limit. The chain’s endpoint error is 438 and 205 times its own single-step error at $N = 64$, 710 and 474 at $N = 128$ and 1653 and 1587 at $N = 256$,

against the 8, 11 and 16 that $\sqrt{N}$ independent steps would give. Since the measured carry cost doubles with each halving of the step, the shortest-step network costs four and a half to five and a half times the longest-step one and delivers no better a crossing. On this evidence there is no reason to go below $\Delta z = 80.9$ mm.

7.3 Caveats, stated plainly

- **Two of the three pooled-loss runs had not finished training.** Applying the plateau rule of Section 2.8 to the six runs’ own round logs, all three cost-weighted runs are flat at their last round and only one of the three pooled-loss runs is. The $N = 128$, $q = 8$ pooled run was still improving at 7.24 per cent per ten rounds when it stopped at round 63, and the $N = 256$, $q = 16$ pooled run at 10.94 per cent per ten rounds at round 42, after both had already been extended well past the stopped snapshot. Their rows are lower bounds on how good those networks would eventually be, so the margins at those two step lengths are upper bounds on the effect of the loss. **The conclusions of this paper therefore rest on the $N = 64$, $q = 2$ pair**, which is the pre-registered one and the only pair in which both sides are flat; the other two pairs are reported in full and are read as supporting evidence rather than as independent evidence.
- **One seed.** Every run in this paper uses seed 0 (Table 4). There is no seed ensemble anywhere, so the only measure of run-to-run variation available is the round-to-round spread of the validation headline, which is 17.5, 11.7, 12.4, 10.8, 9.8 and 25.9 per cent over the six runs. That is why the pre-registered criterion was written as a band to be cleared rather than as an inequality between two medians, and it is why no difference of order ten per cent anywhere in this paper should be read as a difference.
- **The stiffest band is thirty tracks.** The > 50 GeV band holds 30 test tracks. Its medians move by factors of 5 to 8 between the two losses, among the largest in Table 17 — whose largest factor is the 9.01 of the 20–50 GeV band at $N = 64$ — and a band of that size cannot support a quantitative claim. It is read here only as agreeing in direction with the 20–50 GeV band beside it.
- **The three ablations were never trained.** The lever-arm-off, track-factor-off and window-off columns of Table 13 are pre-flight diagnostics: one factor of Eq. (23) is replaced by its own average on a network that was trained under a different objective entirely. They say how each factor redistributes a fixed set of residuals. They do not say what a network trained under any of them would do, and no such network exists.
- **The output parameterisation was chosen for the earlier fixed-step study and inherited here.** Every network in this paper writes its outputs as the straight-line extrapolation of the input plus a per-track-scaled correction (Eq. (17)), so it learns the correction to the straight line rather than the stage states that the network of Raissi, Perdikaris and Karniadakis emits. The form was introduced for a network serving step lengths from 70 mm to 8.6 m, chosen explicitly for the earlier fixed-step study, and inherited by this paper’s single-network chain without re-examination. Its effect here is unmeasured: no network in this paper was trained to emit the stage states directly, so nothing in Section 5 says whether any of its numbers depends on the choice, in either direction.
- **The pre-flight was measured on the extended network, not on the checkpoint the original gate used.** The stored record of this measurement was taken on the stopped checkpoint of 18 September 2026, and the numbers in Table 13 are the same measurement on the extended network. Re-running the pre-flight on that checkpoint reproduces the stored record exactly, to zero deviation in every row, so the method and the replayed random draw are the ones the original gate used; but the two sets of numbers differ by

0.2 to 8.5 per cent, because the weights are identical and the residuals they weigh are not. The largest of those differences is the cost-weighted loss’s share of the 10–50 GeV window, and the value in this paper is the smaller one.

- **The reference is field-only.** Every network number in this paper is a distance from a sixth-order integration of the equation of motion in the measured field, and Section 4.4 measures what that equation leaves out: a median of $1736\ \mu\text{m}$ between the field-only reference and the simulated particle over the whole set, $532\ \mu\text{m}$ in the loss window and $153\ \mu\text{m}$ above 50 GeV. Part of that gap is a defect in my own pipeline, a start momentum that is the particle’s production momentum and implies an over-bend worth $+17.6$, $+17.3$ and $+12.2$ MeV in the three bands below 20 GeV; part is energy loss along the crossing, which the equation does not model; part is multiple scattering, which is stochastic and irreducible; and the momentum-growing residual above 20 GeV is unexplained. None of this moves a single network-against-reference number, because the network and the reference share the same equation and the same q/p . What it does mean is that the levels in Section 5.9 are the levels that decide whether any of this yet matters physically.
- **The carry cost is a framework number, not a deployment number.** The 653 to $657\ \mu\text{s}$ per track at $N = 64$, and the figures at the two shorter steps, were measured in the same framework the networks were trained in, on one thread, in double precision, with no fusion and no attempt at optimisation. The comparison that matters for the trigger is against the production extrapolator in the deployment precision on the deployment hardware, and it has not been made here. What the measurement does establish is the scaling with N , which is the part the design choice turns on.
- **Two of the sixteen exact ceilings are not fresh solves.** The exact collocation scheme was solved for this paper at $N = 2$ and $N = 256$ and stored per component; the $N = 64$ and $N = 128$ cells of Table 6 are read from the earlier chained comparator table, which stores the radial median alone. That is why the max-metric block of that table is empty in those two rows. It does not affect the reading, because the point those cells carry is that the ceiling is orders of magnitude below the networks and not what its third digit is.

8 Outlook

Seven things follow from the result and from how it was built, in the order in which they are worth doing.

1. The rework: emit the stage states directly. Every network in this paper learns the correction to the straight line, a form inherited from a network that served steps of very different lengths and never re-examined here (Section 2.4). The direct test is to retrain the same chain with the network emitting the stage states and the endpoint directly, as in the source construction, with the same two losses and everything else identical, starting with $N = 64$, $q = 2$. If the chain reaches the same endpoint error, the form was not load-bearing here and the design is closer to its source than it looked; if it does not, the correction to the straight line is part of why the chain works, and that belongs in the description of the method rather than in its implementation.

2. The window moved down onto 3–8 GeV. The clearest test of the claim made here is to move the loss window and see whether the trade moves with it. Those runs are built, they have passed the same gates as the runs reported above, and they are in training as this is written; Section 6 is reserved for them and states what it will contain. The expected outcome is a trade in the opposite direction: the 3–8 GeV band should improve by something like the factor the

10–50 GeV window improved by here, and the 10–50 GeV band and the tail should pay for it. If that is what happens, the weight is behaving as a knob rather than as an improvement, which is the more useful thing to know; if the soft band does not improve, then the soft tracks are hard for a reason the weight cannot reach, and the tail measurement above says where to look next.

3. Train the three ablations. The three diagnostic columns of Table 13 say how the lever arm, the per-track factor and the momentum window each redistribute a fixed set of residuals, and they say it cleanly: the lever arm is the whole of the z redistribution and none of the momentum redistribution, the per-track factor does the bulk of the momentum work and the window finishes it. What they cannot say is which factor produced the endpoint gain, because no network was trained under any of them. Three runs at $N = 64$, $q = 2$, each with one factor switched off and everything else identical, would settle that in the same way the pre-registered pair settled the full weight, and at the same cost as the pair.

4. Supervise the step Jacobian. The binding constraint is not the size of a single step’s error but the fact that successive steps make the same error. The chains here are between 2.9 and 7.6 times more aligned than independent steps would be, and the cost-weighted chains are the more aligned of each pair, so the accumulation absorbed part of the per-step gain. Nothing in either objective refers to more than one step at a time, which is why the alignment is free to be whatever the map happens to produce. A term that reads the derivative of one step’s output with respect to its input — the step Jacobian — and asks it to match the Jacobian the collocation equations imply would constrain how an error made at step k is carried into step $k + 1$, and it would still be label-free, since the collocation equations and the field map are all it needs. This is the item with the most room in it: the decomposition of Section 5.7 says the endpoint error is accumulated per-step error and nothing else, so anything that breaks the alignment goes straight to the endpoint.

5. Weight on the starting geometry, not only on the momentum. The tail is made of soft tracks that start far from the beam line: a median starting $|x|$ of 217.1 mm against 107.8 mm for the rest of the same band. Both weights in this paper are blind to that, because both are built from momentum and from position in z . A factor in a_n that rises with the track’s starting $|x|$, or with the straight-line path’s distance from the axis, would be as cheap and as label-free as the two factors already there, and it would act on the one population that neither loss currently reaches. The tail is also the quantity a real-time system schedules against, so it is worth more than its share of the median suggests.

6. Settle the reference. Section 4.4 leaves two loose ends and both are cheap to close. The first is the start momentum: the state harvesting joins the momentum from the particle table, which is the particle’s production momentum, and the truth dump never wrote the hit’s own local momentum, which the simulation records. Re-dumping the truth with the local momentum would separate the energy the particle lost before the start plane from the energy it loses in the crossing, which is the difference between a pipeline defect and a piece of physics the equation is missing; at present the two are summed into one +17 MeV. The second is the material itself. The measured scattering width is a stable 2.1 to 2.3 times the air-only prediction above 8 GeV and implies 6.8 to 8.1 per cent of a radiation length against the 1.70 per cent air provides, which is a measurement of the simulation’s material budget and not a confirmation of it, because I have not looked that budget up independently. Doing so would say whether the remaining factor is material that is known to be there. Neither item changes a network number, and both change what the numbers mean.

7. The cost comparison that has not been made. The argument for a surrogate is that its arithmetic is fixed and known before the track arrives, and this paper supports that argument with an operation count and a framework measurement, not with a deployment measurement. The $N = 64$ network carries a track across the magnet in 653 to 657 μs in double precision on one thread with no optimisation, about 180 times faster than the double-precision research integrator it is scored against; but that integrator takes fifty thousand steps and is not the production extrapolator. The comparison that decides anything is against the production routine, in reduced precision, on the hardware the trigger runs on [4], and until it is made the cost argument remains a promise.

9 Conclusion

One small neural network, trained to take a single fixed step in z and then applied to its own output as many times as it takes to cross the LHCb magnet, solves a problem the first study of this programme left open [1]. There, a single network asked to cross the magnet in one 5178 mm step, inside a network that also served every other kind of step, sat a factor 22 above the numerical scheme it was being trained to solve, and that one leg dominated everything chained to it. Cutting the same crossing into 64 steps of 80.9 mm and training one network to take a step of that length, wherever in the crossing it starts, removes the discretisation from the problem entirely: the exact collocation scheme chained the same 64 times sits 0.147 μm from the reference, so every trained network in this paper is limited by its own optimisation and by the accumulation along the chain, and not at all by the scheme. That is the fact everything else rests on, and it was measured rather than assumed.

The experiment this paper reports is a change of objective on that design. Under the loss the construction comes with, the network reaches a median endpoint error of 145.7 μm over 1452 held-out tracks, and the pre-flight shows where that loss spends itself: 89.8 per cent on tracks of 3–8 GeV, 1.23 per cent on the 10–50 GeV momentum region of interest, and 0.65 per cent on the quarter of the crossing where an error has the furthest to travel. Replacing it with a loss that measures every residual by the displacement it would cause at the far side of the magnet, as a fraction of that track’s own total bend and with the region of interest weighted up, moves the endpoint error to 88.8 μm and the in-window error from 108.1 to 24.8 μm , a factor 4.37. The median per-step slope error in t_x falls from 1.30×10^{-6} to 7.34×10^{-7} . Both pre-registered conditions are met, although the second is weak, because the pooled-loss counterpart has since fallen below that threshold as well. The new weight is built from the network’s own inputs and the measured field map alone, so the objective is still label-free in the precise sense of this paper: no reference trajectory enters it anywhere. Measured against the particle’s simulated state rather than against the field-only reference, the network and the reference are indistinguishable — the network’s median distance from the simulated truth is between 97.4 and 102.0 per cent of the reference’s own distance from it across all six runs — because both omit the material the particle crosses, and that omission is, for the three cost-weighted networks, between 12.5 and 25.2 times larger than the difference between them in every band from < 3 to 20–50 GeV and in the loss window. In the thirty-track > 50 GeV band the pooled networks instead approach the reference-to-truth gap, the ratio falling to 0.85.

The negatives are as specific as the gain. The improvement is a trade, not a free lunch: below the crossover between 3–8 and 8–20 GeV the reweighted network is worse in five of six comparisons, the median below 3 GeV rises from 353.7 to 615.1 μm , the 95th percentile over all tracks rises from 898 to 1399 μm , and the fraction of tracks ending more than a millimetre away rises from 4.41 to 7.16 per cent. The whole of the gain is in the bending plane; the vertical plane does not move, and inside the window it becomes the larger of the two errors. What is left is more coherent than what it replaced, in all six matched comparisons, which is why a factor of four in the per-step slope error did not become a larger factor at the endpoint. Shortening

the step from 80.9 to 20.2 mm halves each step’s error and does not improve the crossing at all, while costing four and a half to five and a half times as much to carry a track. And the heavy tail is made of soft tracks that start far from the beam line, which no momentum window can reach.

Two of the three pooled-loss runs had not finished training when they were stopped, so the conclusions rest on the one pair in which both sides had; there is one seed; the stiffest band holds thirty tracks; the network’s outputs are written as a correction to the straight line, a form chosen for the earlier fixed-step study and inherited here, whose effect in this design has not been measured; and the deployment cost comparison has not been made. What the paper establishes is narrower than the headline and more useful: on this design the endpoint error is accumulated per-step error carried by the lever arm and nothing else, to between 97.7 and 99.4 per cent, the objective decides where that error sits in momentum and in z , and the accumulation — not the size of a single step — is what now limits the crossing.

Data availability

The code, the committed result tables and the figures are in the repository `LHCb.Extrapolation.Project` at <https://github.com/GeorgeWilliam1999/LHCb-Extrapolation>. Everything the experiments of this paper produced — the track set builder, the six training runs, the comparators, the analysis folders and the reference-against-truth study, that is the whole `single_network_chain_discrete_approach/` tree — is committed as `b84693f7` on `main`.

Two things are not in that commit. The paper folder `Self_chained_paper/`, which holds this source, its scripts, its committed result files and its figures, is untracked at the time of writing and will be committed by George. The figure supplement `single_network_chain_discrete_approach/data_supplement/` was committed afterwards, as `aac1692f`.

Every figure in this paper is regenerated by a script in `Self_chained_paper/scripts/` from committed result files only; no figure is copied in from an experiment folder. The pipeline is two commands from inside `Self_chained_paper/`: first

```
PYTHONNOUSERSITE=1 /data/bfys/gscriven/conda/envs/TE/bin/python
scripts/numbers.py
```

which reads the experiment folders read-only and writes `results/paper_numbers.json`, the `results/tab_*.csv` tables and the `results/cache/` arrays; then each of `scripts/fig01_crossing.py` to `scripts/fig17_loss_weights_schematic.py` in turn, each of which writes one PNG into `figures/` and one companion `results/fig*.csv` carrying the numbers it drew. Model weights and the track set itself are not committed; they regenerate from the recorded seed, the committed metadata and the scripts. Appendix A gives the commits, the data set, the cluster identifiers and the maps from every figure and every table to the file it was computed from.

A Provenance

This appendix is the audit trail. It is the only part of the paper in which experiment folder names, block letters and cluster identifiers appear; the body names networks by their step count N , their stage count q and their step length Δz , and nothing else.

Repository and commits

Repository	<code>LHCb.Extrapolation.Project</code> ,	https://github.com/
GeorgeWilliam1999/LHCb-Extrapolation ,	branch	<code>main</code> .
		Local checkout

/data/bfys/gscriven/LHCb_Extrapolation_Project.

- **b84693f7** (22 September 2026), “Single-network chain across the magnet: Blocks E, F, G; reference-vs-truth study; write-up plans”. This is the commit every number in this paper traces to. It carries the whole `single_network_chain_discrete_approach/` tree: the track set builder and its gates (`Block_E_single_network_chain/E0_Track_dataset/`), the sixteen pooled-loss runs and their extensions (`E1_Network_grid/`), the exact-collocation comparators (`E2_Comparators/`), the pooled-loss analysis (`E3_Analysis/`), the weighting module and its pre-flight (`Block_F_reweighted_loss/F0_Weighting/`), the three cost-weighted runs (`F1_Training/`), their analysis (`F2_Analysis/`), the low-momentum-window study that Section 6 is reserved for (`Block_G_low_momentum_window/`), the reference-against-truth study (`reference_vs_truth/`) and the shared package (`_shared/`).
- **aac1692f** (22 September 2026), “Data supplement: seven figures of the crossing drawn to scale”. Adds `single_network_chain_discrete_approach/data_supplement/`, which is not used by this paper.
- **Untracked at the time of writing**, and to be committed by George: `Self_chained_paper/` (this source, its scripts, its `results/` tables and its `figures/`), `Block_F_reweighted_loss/F3_Analysis/email/`, and the low-momentum-window job list and first results under `Block_G_low_momentum_window/G1_Training/`.
- **Superseded callouts**. Earlier records of this programme quote “local 1040ee9, origin a062bd8” as the state of the work and ask for a push before figures would render. That pair refers to the preceding multi-network study and is stale for this paper: 1040ee96 is the commit immediately before **b84693f7** and carries the earlier design, in which a separate network was trained for each step. Nothing in this paper is read from it.

The experiment identifiers used in the day-to-day plan map to the paper as follows. Block E is the single-network chain: E0 the track set, E1 the sixteen pooled-loss runs of Table 12 and their extensions, E2 the exact collocation solves of Table 6, E3 the pooled-loss analysis. Block F is the cost-weighted loss: F0 the weight and the pre-flight of Tables 13 and 14, F1 the three runs, F2 the analysis behind Tables 16 to 26, F4 the write-up. Block G is the low-momentum window reserved in Section 6. The folder `reference_vs_truth/` is Section 4.4, and `Block_C_step_size_and_stages/C1_Fine_reference/` of the preceding study `multi_network_chain_discrete_approach/` is the reference measurement of Section 4.3.

Data set and field maps

Source sample: LHCb TestFileDB entry `expected_2024_minbias_xdigi`, production 00212966, XDIGI format, detector description `ddb-20231017`, conditions `sim-20231017-vc-mu100`, data type 2024. Five of the entry’s eight files, 19 GB, are mirrored on CVMFS, and 200 events were read from them; no grid access was used. The conditions tag is **magnet-up**.

Track set: `single_network_chain_discrete_approach/Block_E_single_network_chain/E0_Track_dataset/results/tracks.npz`, with `results/tracks.meta.json` and `build.tracks.log` beside it. 14,482 crossings after the ten-step cascade of Table 1, split by particle with seed 20260718 into 11,567 training, 1463 validation and 1452 test. Each crossing is stored on 257 planes, $z_k = z_0 + kL/256$ with $z_0 = 2648.2$ mm, $z_1 = 7826.0$ mm, $L = 5177.8$ mm and a plane spacing of 20.22578125 mm; the step lengths that divide it are $\Delta z = 2588.9, 80.903125, 40.4515625$ and 20.22578125 mm at $N = 2, 64, 128$ and 256. Each track’s own planes lie within 60 mm of the nominal ones, z_{pre} from 2593.15 to 2663.26 mm and z_{post} from 7818.75 to 7833.45 mm.

Field maps. Every trajectory in this paper is integrated on `/cvmfs/lhcb.cern.ch/lib/lhcb/DBASE/FieldMap/v8r1/cdf/field.v8r1.up.bin`, md5

9e49ddc4313b589f273540e7e0bb513b, trilinear on a 100 mm cell and therefore C^0 in its derivative. The cross-magnet labels stored in the harvested training file were integrated with the magnet-down map `field.v8r1.down.bin`, md5 `af284c6954d2273c637a5e766b82b58e`; those labels are not used anywhere in this paper, as the box in Section 3.1 states.

Reference integrator: Butcher’s seven-stage explicit Runge–Kutta method of order six at a fixed step of 0.1 mm, in double precision, implemented in `single_network_chain_discrete_approach/_shared/reference.py`. The stored truth arrays `<split>_truth` of `tracks.npz` are that integration on the 257-plane grid.

Compute

All training ran on the Nikhef HTCondor farm, one job per run, single-threaded and in double precision. The run folders are `Block_E_single_network_chain/E1_Network_grid/results/N064.q02`, `N128.q08` and `N256.q16` for the three extended pooled-loss runs, and `Block_F_reweighted_loss/F1_Training/results/full/N064.q02`, `N128.q08` and `N256.q16` for the three cost-weighted runs. Every round is 25 restarts and redraws 32,000 collocation states; every restart ran its full 200 iterations.

- **Pooled-loss grid**, the sixteen runs of Table 12: cluster **5805460** (17 September 2026, 16 jobs).
- **Pooled-loss extensions**, which produced the three extended runs used in every comparison: cluster **5809659**, then **5815090** and **5816352** (18 September 2026 onwards). The three run folders carry 1260 restarts in 50 rounds, 1599 in 63 and 1050 in 42; their own `record.json` files do not record a cluster identifier, so these are taken from `E1_Network_grid/README.md`.
- **Exact-collocation comparators**: cluster **5805461** (17 September 2026).
- **Cost-weighted runs**: cluster **5809660** (18 September 2026, 3 jobs), giving 1000 restarts in 40 rounds at $N = 64$ and at $N = 128$. The $N = 256$, $q = 16$ run was reopened as cluster **5823724** and acquired a duplicate job, cluster **5833416**, from restart 1198 onwards; it holds 1807 history rows for 1500 distinct restarts and 74 round rows for 60 distinct rounds. Every reader in this paper de-duplicates by keeping the last row written per number. As with the pooled runs, the per-run `record.json` does not carry a cluster identifier; these come from `Block_F_reweighted_loss/README.md` and `F4_Writeup/page.md`.

Median wall per restart, over the de-duplicated history: 77.19 and 77.74 seconds at $N = 64$, 117.9 and 118.3 at $N = 128$, 219.8 and 184.1 at $N = 256$, pooled and cost-weighted respectively. Recorded training wall: 31.2, 21.6, 54.7, 32.8, 63.2 and 100.4 hours. The last of those counts the duplicate job; the de-duplicated history sums to 81.1 hours.

Figure-to-file map

Every figure is drawn by the script named, which writes both the PNG in `figures/` and a companion CSV in `results/` carrying the numbers it drew. PN abbreviates `results/paper_numbers.json` and S abbreviates `single_network_chain_discrete_approach/`. All scripts open every input read-only.

Table-to-file map

Paths without a prefix are relative to `Self_chained_paper/`; those with S are relative to `single_network_chain_discrete_approach/`. Every `results/tab_*.csv` is written by `scripts/numbers.py`.

figure	file in figures/	script	inputs
4	fig_crossing.png	fig01_crossing.py	PN dataset.geometry, dataset.field_map; the v8r1 up map; S/.../tracks.npz
5	fig_sample.png	fig02_sample.py	PN dataset.splits; S/.../tracks.npz
6	fig_ceiling.png	fig03_ceiling.py	results/tab_exact_ceiling.csv; PN pooled_grid_snapshot
7	fig_rk6_reference.png	fig04_rk6_reference.py	C1.Fine_reference/ results/reference_convergence.csv, _meta.json, tableau_checks.json
17	fig_three_references.png	fig05_three_references.py	PN against_true_state; results/tab_against_true_state.csv
8	fig_material_gap.png	fig06_material_gap.py	PN reference_vs_truth.rebinned_decomposition, .highland.air_rebinned
3	fig_weights.png	fig07_weights.py	PN constants, worked_example; F0.Weighting/weighted_loss.py; tracks.npz
9	fig_loss_shares.png	fig08_loss_shares.py	PN preflight (modes, clamp_scan, the 18 September record)
10	fig_convergence.png	fig09_convergence.py	the six runs' rounds.csv and history.csv; tab_extended_vs_snapshot.csv; PN headline
11	fig_headline_vs_p.png	fig10_headline_vs_p.py	the six runs' chain_states.npz; tracks.npz; PN headline
13	fig_components_by_band.png	fig11_components_by_band.py	PN headline (per-band medians and signed medians)
14	fig_signed_3to8.png	fig12_signed_3to8.py	the two $N = 64$, $q = 2$ runs' chain_states.npz; tracks.npz; PN near_5gev
16	fig_along_z.png	fig13_along_z.py	PN along_z; results/cache/single_step*.npz
15	fig_anatomy.png	fig14_anatomy.py	results/cache/ anatomy_pooled_N064_q02.npz and anatomy_reweighted_N064_q02.npz; PN anatomy
12	fig_tail.png	fig15_tail.py	PN headline, near_5gev; the six runs' chain_states.npz; tracks.npz
1	fig_loss_schematic.png	fig16_loss_schematic.py	S/.../tracks.npz (test track 275); the pooled $N = 64$, $q = 2$ run's scale.json and network.pt through chain_network.load_network; _shared/irk.py, model.py, reference.py; PN constants
2	fig_loss_weights_schematic.png	fig17_loss_weights_schematic.py	PN constants, worked_example, anatomy; F0.Weighting/weighted_loss.py; tracks.npz; the pooled $N = 64$, $q = 2$ run's scale.json

The same two maps are repeated in README.md next to this source, together with the list of bibliography fields still to verify and the record of the disagreements found between the earlier write-ups and the committed result files while this paper was being written.

table	what it holds	source
1	the cut cascade	results/tab_dataset_cuts.csv
2	the splits in the five bands	results/tab_dataset_bands.csv
3	the material floor	results/tab_material_floor.csv
4	what is held fixed across the runs	results/tab_compute.csv; PN compute; S/Block_F_reweighted_loss/F4_Writeup/page.md
5	the Gauss–Legendre tableau checks	results/tab_tableau.csv
6	the exact ceiling and the snapshot networks	results/tab_exact_ceiling.csv; results/tab_pooled_grid.csv
7	the reference step ladder	results/tab_rk6_convergence.csv
8	truth minus reference, by band and charge	results/tab_decomposition_bands.csv; PN reference_vs_truth_rebinned_decomposition
9	the width against the Highland prediction	results/tab_highland_air.csv
10	the over-bend by particle type	S/reference_vs_truth/results/decomposition_by_pid.csv, via PN
11	the reference against itself	S/reference_vs_truth/results/rk6_self_consistency.csv, via PN
12	the sixteen pooled-loss networks	results/tab_pooled_grid.csv; results/tab_exact_ceiling.csv
13	the pre-flight shares	results/tab_preflight_shares.csv; PN preflight
14	the clamp scan	results/tab_clamp_scan.csv
15	the six runs against the plateau rule	results/tab_headline_pairs.csv; results/tab_extended_vs_snapshot.csv
16	the headline comparison	results/tab_headline_pairs.csv; results/tab_reweighted_runs.csv; results/tab_anatomy.csv
17	the same six runs per band	results/tab_headline_bands.csv
18	the position components	results/tab_headline_components.csv
19	the slope components	results/tab_headline_components.csv
20	the two bands around 5 GeV, per component	results/tab_near_5gev.csv
21	the same two bands, and where the tail starts	results/tab_near_5gev.csv
22	the per-step error of the six runs	results/tab_anatomy.csv
23	coherence and the explained fraction	results/tab_anatomy.csv
24	what shortening the step buys	results/tab_anatomy.csv; results/tab_along_z.csv; results/tab_single_step_vs_z.csv
25	the three reference points	results/tab_against_true_state.csv
26	what the six runs cost	results/tab_compute.csv; PN compute

References

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